**Module Code & Module Title**

**CS6PO5NT- Project**

**Assessment Weightage & Type**

**Final Year Interim Report (25%)**

**Credit-30**

**Year and Semester**

**2025-26 Autumn**


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**Assignment Submission Date:30/12/2025**

**Assignment Due Date:30/12/2025**

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# Title: AI-Powered Crop Disease Detection and Recommendation System

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# 1. Introduction

## 1.1. Project Description

The object of work of this Final Year Project is the design and development of an AI-Powered Crop Disease Detection and Recommendation System which is planned to modernize the archaic agricultural activities by the means of intelligent automation. Many economies rely on agriculture as a major sector but the crop diseases are leading to a lot of losses in yield as a result of late diagnosis, misdiagnosis, and inability to get professional advice particularly in rural and remote regions. The proposed project is going to solve these issues by offering farmers with a technology-based method of offering early disease detection and disease treatment services.

The proposed system allows farmers to take or post photos of the leaves of the crops using a simple web or mobile interface. These images are then subjected to computer vision algorithms and deep learning architectures or models namely Convolutional Neural Networks (CNNs). CNNs perform very well in image-based processes because they are able to automatically identify features that are important to the task including color change, spot pattern, texture irregularity and leaf deformity amongst others that suggests the existence of the disease. The model that is trained is used to match the uploaded image with a database of labeled crop disease images in order to categorize the disease correctly or healthy crops.

As soon as the disease is detected, the system creates real-time diagnostic outcomes and individualized treatment prescriptions. These can be recommendations based on proper application of pesticides, organic treatments, nutrient modification and preventive measures to prevent further spread of the disease. It focuses on sustainable agricultural practice by encouraging environment-friendly solutions and reducing the massive use of chemicals thus ensuring the health of soil and the environment.

Besides diagnosis and recommendations, the system has the capability of storing the historical data in terms of disease occurrences, type of crops and the seasonal patterns. The data can be considered to find trends that would enable the farmers make data-driven decisions regarding future planting periods and disease management techniques. Agricultural researchers and policy makers can also find such insights useful in the patterns of diseases in regions.

Comprehensively, this project is evidence of how artificial intelligence can be applied in the farming industry, ensuring that farmers have a cheap, scaled, and accessible solution. The system is able to mitigate crop losses, increase productivity, and food security by incorporating deep learning, image analysis, and expert knowledge. The project eventually helps in the development of smart and sustainable agriculture where farmers are able to access information that is timely, accurate and takes action.

## 1.2. Problem Domain

This project problem domain lies in the imperative issues of agricultural sector caused by crop diseases that are a significant threat to agricultural productivity, livelihoods of farmers, as well as food security in the world. Fungus, bacteria, viruses, and pests infecting crops are transmissible and can cause massive losses in yields and cost of production unless detected at the initial stage. The small-scale and subsistence farmers are especially susceptible because they generally cannot access prompt diagnostic equipment and the professional farm services.

Historically, farmers visually identify the disease through agricultural inspection or use of the services of agricultural experts. This method is time-consuming, subjective, and inaccurate in nature since the symptoms of disease may change with different types of crops, different stages of growth and the environmental conditions. Furthermore, most rural and isolated agricultural societies do not have access to professional agronomists, laboratories, or extension services thus, professional advice is expensive and unfeasible. This often causes farmers to use guesswork or late measures, which may increase disease outbreaks and cause too much or improper use of pesticides.

The domain of this problem is at the cross-point of agriculture, artificial intelligence, and information systems, where the need of automated and scalable disease detection solutions that are accurate and free of mistakes is increasing. Recent developments in the field of computer vision and deep learning, specifically, Convolutional Neural Networks (CNNs), allow impressive visual pattern analysis of images of crop leaves and the detection of diseases with high accuracy.

Overall, the problem domain highlights the urgent need for intelligent, data-driven agricultural solutions that reduce dependency on manual expertise, enable early disease detection, support sustainable farming practices and ultimately enhance food security and rural economic stability.

### 1.2.1. Local Scenario

Agriculture is a major economic activity in Nepal and other agricultural economies with a significant number of people living on it. Most farmers engage in small-scale or subsistence farming and they greatly depend on traditional knowledge and experience to take care of crop health. Identification of crop diseases in these areas relies on manual visual identification which many times cannot be adequate to identify any disease with similar symptoms. This has resulted in poor diagnostics thus more crops are lost through ineffective treatment.

Professional agricultural services, including those of plant pathologists, agronomists or government extension officers are inaccessible, especially in remote and rural areas. Farmers also tend to rely on informal recommendation by fellow farmers, local suppliers or community networks and they may not be accurate, or scientifically proven. Moreover, the time spent on consulting the specialists and transport limitations and financial constraints often lead to the inability to diagnose the disease in time and spread it quickly and lead to the consequences of its spread on crops.

The other major issue in the local environment is incompatible internet connectivity. The network coverage, bandwidth or unstable access to the digital services is the norm in many rural locations in Nepal. This limits the ability of farmers to utilize data intensive farming systems or real-time online consultancy. Thus, any technological solution that is supposed to work in this environment must be light, portable, and able to operate within low-connectivity conditions, e.g., by compressing images, means of caching the data offline or by synchronizing with the offline data.

Moreover, the digital literacy of farmers is rather different, which is why simplicity and ease of use should be one of the key design factors. Adoption can be greatly increased with a system that is easy to use, as it has clear and visual feedback and localized recommendations. Through these localized problems, an artificial intelligence-based crop disease detection system can provide accessible, timely, and practical assistance to Nepalese farmers and reduce their crop losses to improve their productivity and the sustainability of the agricultural industry despite infrastructural constraints.

### 1.2.2. Global Scenario

Crop diseases are one of the primary threats to agricultural productivity and food security at the global scale as they bring about significant losses in the yields of crops annually in the developed and developing countries. Pathogens such as fungi, bacteria, viruses, and pests which cause plant diseases lower the quality and quantity of crops and lead to losses to farmers and elevated food prices across the globe. The agricultural research organizations reveal that a good portion of the total crop output in the world is every year wasted on plant diseases that could have been prevented or detected late.

This problem has been increased by the effects of climate change. The increase in temperatures, alteration of rainfall patterns and the increase in humidity have provided the conducive environment with which crop diseases can spread very fast to a new geographical location. Previously, region-specific pathogens have been seen to attack crops in different continents, which complicate and predict the disease burden. The changing global environment requires more adaptive and quicker disease detection systems.

In spite of the fact the developed diagnostic methods like laboratory testing, hyperspectral imaging, and expert-based agricultural advisory systems are in use in some places, they are usually costly, infrastructure-based, and need special skills to work. This means that they can be used mostly in large-scale commercial farms or research institutions. Smallholder farmers who form a great percentage of the global agricultural labor force are not often able to access these advanced tools either because of cost, technical complexity and geographical location.

The recent years have seen the rise of AI-based image classification through deep learning model as a potential solution in the world to detect crop diseases. Such systems are able to scan photographs of plant leaves and diagnose accurately diseases in real time with devices that are readily available like smartphones. Their potential is not yet widely used by smallholder farmers as they face some barriers including low levels of digital literacy, insufficient access to the internet, absence of localized disease data and insufficient awareness/confidence in AI-based solutions.

Therefore, the global scenario highlights the need for affordable, accessible and farmer-centric AI solutions that can be adapted to diverse agricultural environments. Bridging the gap between technological innovation and practical adoption is essential to ensure that AI-powered disease detection systems deliver real-world impact at a global scale.

## 1.3. Project as a solution

The suggested AI-Powered Crop Disease Detection and Recommendation System is a viable and scalable answer to the problems of the traditional process of crop disease diagnosis. The project will offer affordable diagnostic platform to help farmers identify crop diseases within a short time and accurately without involving experts all the time by combining artificial intelligence, computer vision, and information systems.

The system enables farmers to post images of leaves of crops using an easy to use web or mobile platform, meaning that the solution can be used even by people with minimal technical expertise. After an image is provided, it is fed into one of the trained Convolutional Neural Network (CNN) models, which has been trained to identify disease trends based on a vast collection of labeled image crops. This model examines the visual features like discoloration, texture, spots and lesions to determine the most probable disease to the crop. This automated system saves on a lot of time in diagnosing and the risk of human error is minimized.

Once the disease prediction is created, the platform gives customized treatment advice to help farmers act instantly and make an informed decision. These suggestions can involve organic solutions, chemical pesticide advice, manure and nutrient proposals and precautions to minimize future disease incidences. The system provides a variety of treatment options, which promotes sustainable and responsible farming systems, enabling the farmers to select methods that can best fit their resources and environmental requirements.

Besides the real-time diagnosing, the system will archive past prediction records such as type of crop, disease and the time the disease occurred. This information would also be processed to create visual analytics and reports on diseases trends, so farmers can understand what problems are recurring, what are the disease patterns during the seasons and how effective the treatments are. These insights can be used in the decision making that is data-driven and long-term crop management.

i. Core Solution Capabilities

- Disease detection through CNN based deep learning model.
- Output of disease within seconds following image uploads.
- Fertilizer, pesticide and organic treatment suggestions.
- Disease trends and frequency patterns visualized with the use of charts.
- All predictions have a history, which is guaranteed by cloud-based storage.
- Is a Progressive Web App (PWA) that can work on mobile and low-bandwidth environments?
- Admin dashboard to add, manage or update disease datasets.

The best thing about this system is that it is not only able to diagnose but also guide, monitor and educate. It also does not end at the identification farmers are made structured recommendations to treatment that are in line with sustainable agricultural practices. Past performance is also recorded in the system, which makes farmers to keep track of the performance and observe the patterns of recurrence over time.

ii. How This System Solves Existing Challenges

*Table 1 : Table of Traditional Problem and AI-Based Solution*

| S.N | Traditional Problem | AI-Based Solution |
|---|---|---|
| i. | Slow and inaccurate manual identification. | Instant disease prediction powered by CNN. |
| ii. | Lack of expert access in rural areas. | Accessible from any device with internet. |
| iii. | Confusion between similar disease symptoms. | High accuracy pattern recognition model. |
| iv. | No documentation for case history. | Auto saved prediction logs and reports. |
| v. | Overuse of pesticides. | Balanced treatment and prevention suggestions. |

iii. Direct Benefits to Farmers

- Faster diagnosis → early treatment → reduced crop loss
- Increased yield and revenue due to timely intervention
- Lower dependence on physical experts and middlemen
- Better planning for future farming cycles through history analytics
- Improved soil and crop health with informed chemical usage

Overall, this is a project that converts conventional farming to smart farming, which is a combination of technology and field viability. The AI-powered system can serve as a digital agricultural assistant through which with a touch of a button, farmers can obtain accurate, empowering and more direct and timely information, able to work round the clock, scale easily and enable farmers to make reliable decisions. This system has the potential to make agriculture efficient, intelligent and resilient with the ability to diversify its crops, types of diseases and health monitoring.

## 1.4. Aims and Objectives

The primary aim of this Final Year Project is to create a full-fledged end-to-end AI-based crop disease detection and advisory system that will be able to interpret the images of plant leaves correctly and provide a diagnosis and a treatment prescription in real time. The system aims at assisting the agricultural sector by decreasing reliance on the manual diagnosis, reducing the financial loss as a result of disease outbreaks, and facilitating the farmer in making better farming decisions with the help of the digital solution. This project will fill the gap between the early detection of diseases and taking corrective measures on time by incorporating Artificial Intelligence, deep learning, REST API communication, and dashboard analytics based on MERN.

In its essence, the project is aimed at enhancing agricultural productivity, sustainability and accuracy of decisions. The traditional agriculture mostly relies on experience-based recognition that is liable to misunderstanding and slowness. Compared to it, this AI system facilitates the prediction of the disease through visual identification by comparison with a learned set of symptoms so that even unprofessional users can identify the problem in a short period of time and with high accuracy. With the additions of history tracking, notification, and visual analytics, this project is not only a detection platform, but an entire agricultural intelligence platform.

### 1.4.1. Aim of the Project

The main aim of the project is to create and innovate an AI-powered web application that can detect the disease of crops based on uploaded photos of leaves, give correct treatment information and give analytical data to help farmers avoid crop loss and enhance agricultural production.

### 1.4.2. Objectives of the Project

The objectives of the project are given below:

**i. Technical Development Objectives**

- To train a CNN-based deep learning model using an agricultural dataset for disease classification.
- To implement a Flask API microservice that processes uploaded images and returns prediction results.
- To develop a MERN-based web interface where users can upload images and view results.
- To store prediction logs, user records, and disease data in a scalable MongoDB database.
- To integrate analytical graphs using Chart.js/Recharts for trend visualization.
- To develop a Progressive Web Application (PWA)-based UI for mobile accessibility.
- To support history tracking, notification alerts, and offline caching for usability.

**ii. Functional Output Objectives**

- To automatically detect diseases with high accuracy and confidence percentage.
- To provide remedy solutions including:
- Organic remedies
- Fungicides & pesticides
- Fertilizer recommendation & dosage
- To store past predictions for monitoring disease patterns over time.
- To allow admin users to update disease categories, retrain the AI model & manage users.
- To generate region-wise disease frequency reports and usage stats for agricultural study.

**iii. End-User Benefit Objectives**

- Minimize crop damage by diagnosing diseases at early stages.
- Reduce farmers' dependency on external consultancy for basic diagnosis.
- Increase crop production through fast intervention and guided treatment.
- Improve resource management by reducing wasted pesticide usage.
- Enable farmers to digitally track plant health progression across seasons.

### 1.4.3. Why This Objective Matters

With the achievement of these goals, this project will directly affect the smart agriculture transformation which will enable technology to substitute the uninformed decision-making process with the data analysis. Farmers are not obliged to wait days waiting the inspection of the experts but they get the results immediately and make sure that the treatment is provided at the appropriate time, thus avoiding the spread of diseases and loss of money. This system can be scaled, be future ready and able to move to other types of crops and monitor agriculture at a national level, thanks to the combination of AI and dashboard analytics.

# 2. Background

## 2.1. About the client / About the end user

The main consumers of the proposed AI-based crop disease detection system will be farmers and specifically small-scale farmers residing in rural and semi-urban areas. Such farmers are usually highly dependent on agriculture as their primary income and directly influenced by the crop diseases, climatic variations, and the lack of access to the expert assistance in a timely manner. Smallholder farmers do not have high-level technical expertise or access to agricultural extension services on a regular basis, and therefore, early detection and correct treatment advice is a serious problem. The system is thus expected to be straightforward, cheap yet accessible so that farmers can post crop leaves images via a mobile phone or web interface and get instant disease forecasting and treatment prescriptions.

Other secondary users such as agricultural experts and system administrators are also needed in the system as they are instrumental in ensuring that the platform remains accurate and reliable. Agricultural specialists will ensure that the classification of the diseases, the recommendations of treatment options, and the knowledge base are updated according to the emergence of crop diseases, seasonality and agricultural practices in different regions. Their participation will see to it that the system offers scientifically valid and context-specificate advice to farmers.

The administrators of the system will take care of overall management and monitoring of the platform. They have the responsibility of maintaining, managing user accounts, monitoring system performance, providing data security, as well as updating systems. The administrators are also able to look into data and disease trends generated by the system in aggregation and can be used to facilitate decision-making, policy formulation, and even future upgrades to the platform. All these groups of users make the system accurate, scalable, and sustainable agricultural development beneficial.

## 2.2. Understanding the solution

The solution offered by the author is implemented in the form of Artificial Intelligence (AI), the newest web technologies, and cloud computing to provide a smart and conveniently designed crop disease detection system. The system will assist farmers in early detection of crop diseases and timely treatment advice on them, hence minimize loss of crops and enhance agricultural yield.

The workflow of the system starts with the acquisition of images where the users will be allowed to upload pictures of crop leaves via web or mobile application. These images are subjected to an image preprocessing step, which involves image resizing, image normalization and image noise reduction to maintain uniformity and enhance model accuracy. The preprocessed images are sent to the AI inference part where a trained Convolutional Neural Network (CNN) identifies visual patterns and labels the disease or verifies healthy crops.

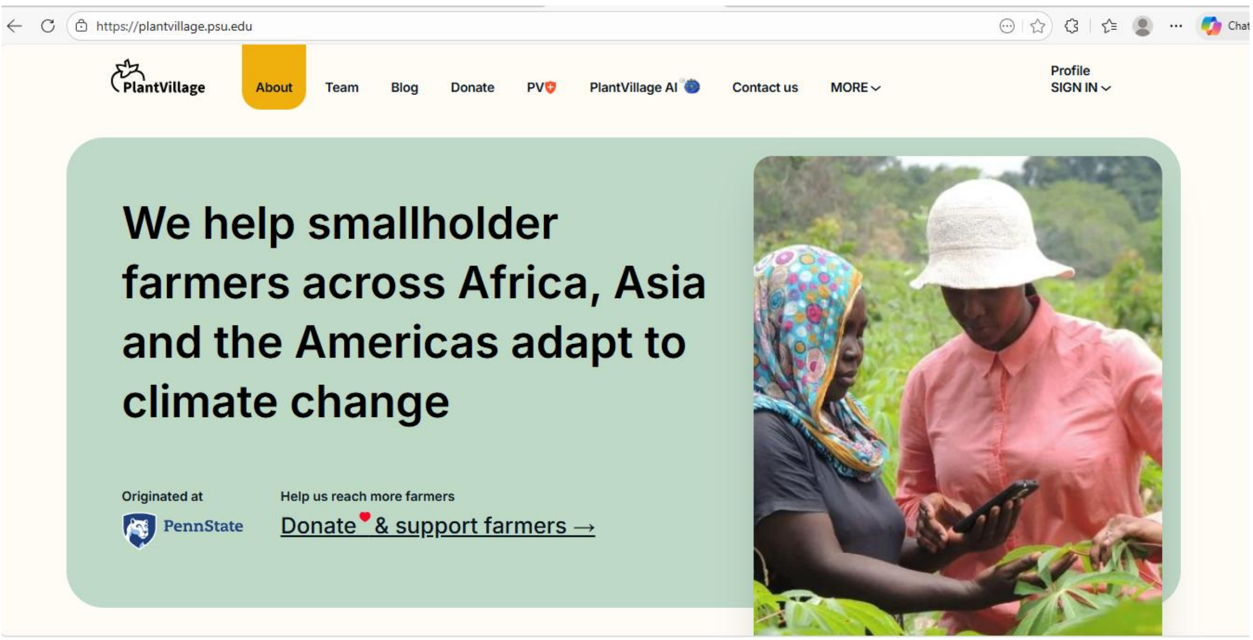
After generating the prediction, the system goes to the result generation stage where the identified disease name, the level of confidence, information on the severity is assembled. The system provides custom-made recommendations based on the diagnosed illness, including organic treatment recommendations, chemical recommendations, fertilizer recommendations, and preventive recommendations. Agricultural experts curate and validate these recommendations in order to guarantee the reliability and the relevance of the recommendations to the region.

The solution also gives great consideration to usability, which makes the interface easy and not hard to use by people with minimal technical knowledge. It is accurate with the help of ongoing training of the models and changing of the data and is scalable with the well-known cloud-based infrastructure that enables the system to operate with the increasing number of users and data satisfactorily. Moreover, the system will also contain Progressive Web App (PWA), whereby it can store the results and recommendations that have already been seen by the user, and they can use them even without internet access, which is especially useful in those locations where the net connection is weak or unreliable.

In general, the concerted solution guarantees the creation of a solid, scalable, and user-friendly solution that will close the distance between the high level of AI technologies and the actual demand of agriculture.

## 2.3. Similar Projects

### 2.3.1. Project 1 – PlantVillage



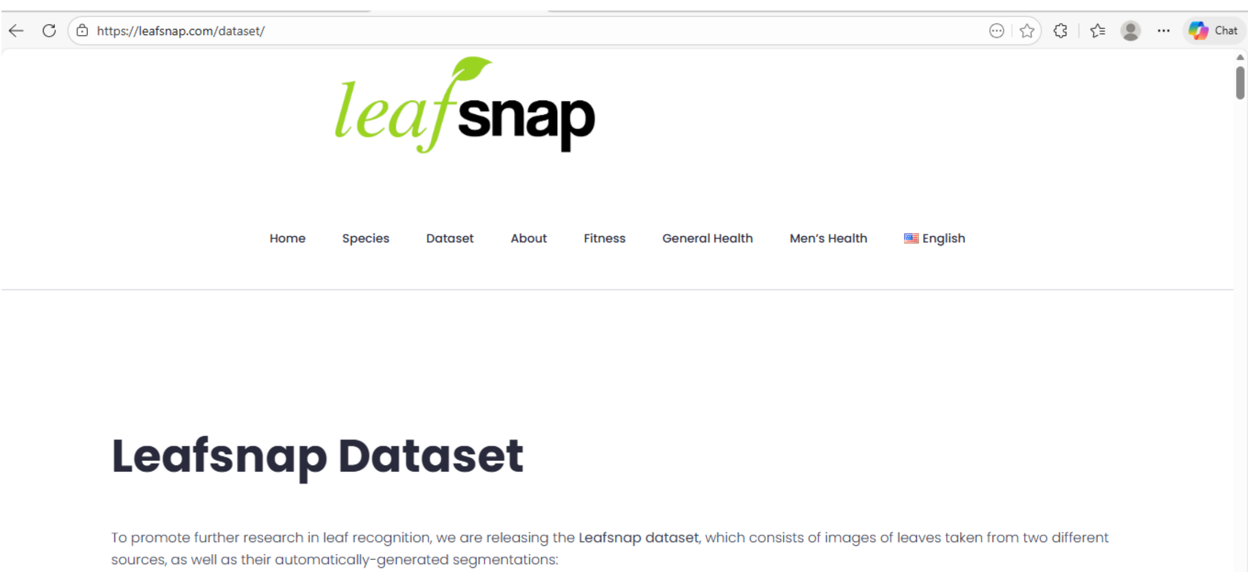

Link: [plantvillage.psu.edu](plantvillage.psu.edu)

PlantVillage PlantVillage is a globally known open-source dataset and deep learning-powered platform created to assist in identifying plant diseases with the help of images. It offers a big and well organised set of labelled pictures of plant leaves that have been widely used to educate and assess Convolutional Neural Network (CNN) models. PlantVillage is highly successful in detecting a variety of crop diseases because of the quality and volume of its data, and it has been an important part of agricultural AI research.

Although the performance of the PlantVillage is good in disease classification, it is rather oriented on disease identification and not end-user support. The site provides a few localized treatment guidelines, and it fails to sufficiently take into account area-specific agriculture, weather, or resource supply. Also, the system is less farmer-focused and more research-driven, not having such features as simpler user interfaces, offline capabilities, support in multiple languages, and monitoring disease history.

Consequently, although the idea of PlantVillage is an ideal starting point to train models and benchmark, it does not take the practical requirements of smallholder farmers into consideration to the full extent. This drawback shows the need of a more holistic solution that would include proper detection of the disease along with localized guidance and actionable and usability effects, which this proposed system would offer.

### 2.3.2. Project 2 – LeafSnap



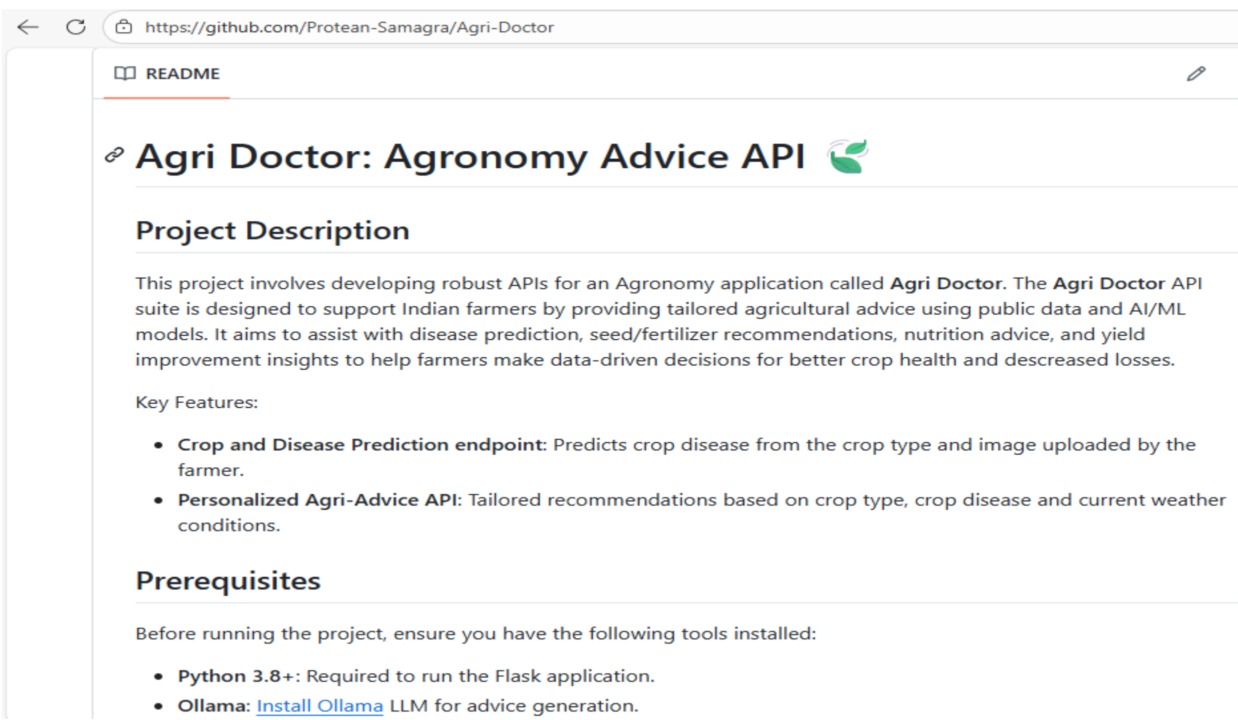

Link: [Leafsnap Dataset - LeafSnap]

LeafSnap is an application created by the company depending on image recognition and machine learning algorithms to identify plants by their leaves with the use of an image. The system matches visual data of plant leaves like shape, texture, and patterns of veins in a pre-existing database and uses the information to classify the plant species. LeafSnap has become popular in educational, botanical and environmental applications.

Nonetheless, the main task of LeafSnap is not the diagnosis of crop diseases, but the classification of plants. The application does not interpret symptoms associated with the health of plants or offer information on the causes, severity, and treatments of the diseases. It also does not have agricultural decision support capabilities as far as preventive advice, fertilizer advice or currently validated farmer solutions.

Therefore, though LeafSnap shows that image-based plant recognition works, there are no issues regarding how farmers can practice the idea of managing crop diseases in practice. This difference also serves as the reminder of the necessity of a special system aimed at detecting the disease, suggesting treatment, and providing support to the farmers, which is the idea of the given project.

### 2.3.3. Project 3 – AgriDoctor


https://github.com/Protean-Samagra/Agri-Doctor

README

Agri Doctor: Agronomy Advice API

Project Description

This project involves developing robust APIs for an Agronomy application called **Agri Doctor**. The **Agri Doctor** API suite is designed to support Indian farmers by providing tailored agricultural advice using public data and AI/ML models. It aims to assist with disease prediction, seed/fertilizer recommendations, nutrition advice, and yield improvement insights to help farmers make data-driven decisions for better crop health and descreased losses.

Key Features:

- **Crop and Disease Prediction endpoint:** Predicts crop disease from the crop type and image uploaded by the farmer.
- **Personalized Agri-Advice API:** Tailored recommendations based on crop type, crop disease and current weather conditions.

Prerequisites

Before running the project, ensure you have the following tools installed:

- **Python 3.8+:** Required to run the Flask application.
- **Ollama:** Install Ollama LLM for advice generation.

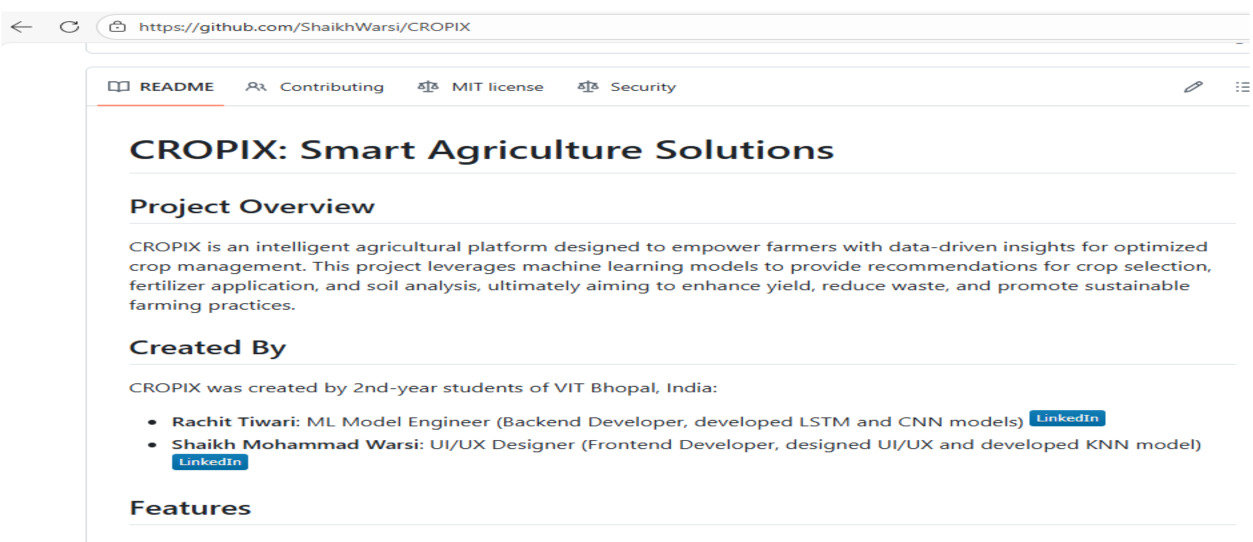
Link: Protean-Samagra/Agri-Doctor

AgriDoctor is an agricultural decision support system which gives information about crop disease via expert system technique. The interface depends on logic-based rules and human expertise, where the user manually enters the symptoms he/she sees (e.g., the color of the leaves, spots, patterns of growth). According to these inputs, the system compares the symptoms with a set of rules stored to detect potential diseases and propose general management principles.

Although AgriDoctor can be handy in the delivery of expert-validated knowledge, it remains very susceptible to error in case of manual input that is given by people. The farmers will not be able to identify or describe symptoms appropriately thus making incorrect or incomplete diagnoses. Also, the system is not very adaptable to new or changing diseases and it is hard to scale unless expert intervention continues, as it is rule-based.

Moreover, AgriDoctor does not have an AI-based image analysis, automatic learning, and performance optimization in real-time. Consequently, even though it can be a useful source of expertise, it is not as automated, accurate, and user-friendly as it needs to be to be adopted by smallholder farmers in large scale. These shortcomings indicate the benefits of the suggested system, combining AI-powered image identification and professional verification of the recommendations to provide a more convenient and efficient solution.

### 2.3.4. Project 4 – Cropix



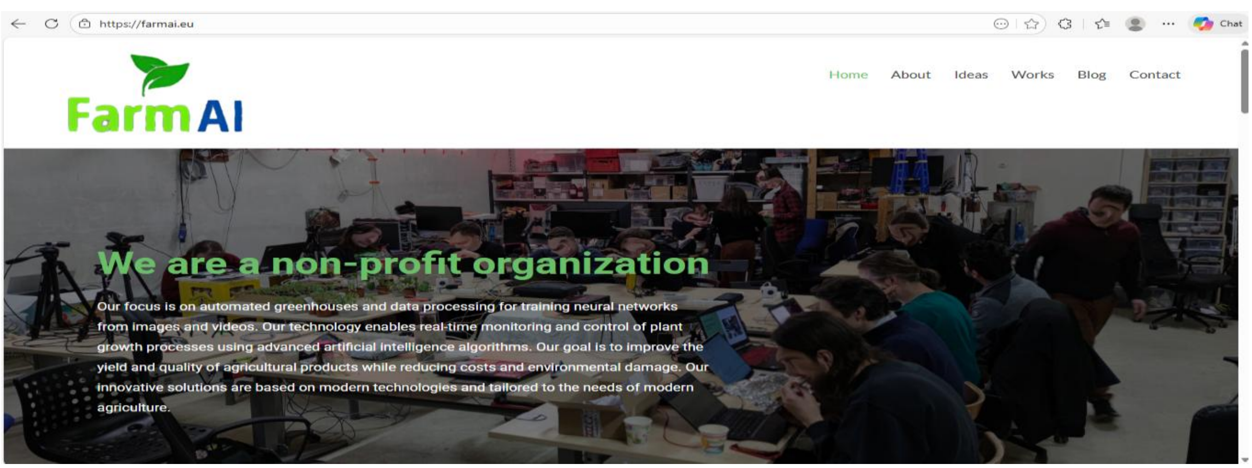
Link: [ShaikhWarsi/Cropix: An AI-driven solution that uses data insights to help farmers increase their productivity and efficiency.]

Cropix is a commercial agricultural technology platform offering AI-powered crop disease detection and advanced analytics and farm monitoring technology. The system has the ability to detect plant diseases based on the analysis of photographs and also provides other services like crop health, yield information and data-informed decision support. Cropix is very acceptable in large scale farms and agribusinesses because of its advanced analytical functions.

Cropix is also expensive, though it has technical advantages, and may be necessary to include subscription or special hardware or integration on an enterprise level. This renders the platform not accessible to the smallholder farmers, especially in the rural and developing areas where financial and technological capabilities are scarce. Moreover, the system can end up being too complex to effectively use by farmers who are not very digital literate.

Consequently, although Cropix can show the promise of high-tech AI and analytics in the agricultural sector, it fails to meet small-scale farmers in the context of their affordability and accessibility requirements. This constraint supports the necessity of such a solution that is cheap, user-friendly, and publicly available, which the offered solution should offer by integrating AI-driven disease detection with features such as farmer-focused design and offline support.

### 2.3.5. Project 5 – FarmAI Research Systems



Link: FarmAI – Neural network training, automated greenhouses

FarmAI research systems are a body of scholarly and practical AI-based applications that have been proposed to identify crop diseases with state-of-the-art machine learning and deep learning algorithms. Such systems are frequently highly classified on controlled environments and are a source of innovation and research on agricultural technology. They are typically tested in comparison to benchmark datasets and are used to investigate new algorithms, architectures and optimization methods.

Nevertheless, the majority of the FarmAI research systems are not ready to be deployed and do not consider practical issues to apply in practice. They are usually high-computational processes, regulated image input, and laboratory-based processes and thus impracticable to direct farmers. Moreover, these systems are typically not user friendly, not local in their recommendations, offline, and long term strategy of maintaining the system.

Therefore, although FarmAI research systems show the technical possibility and potential of AI in crop disease identification, they do not bridge the research and the application in the real-life agricultural contexts. This constraint makes the need to come up with a solution that is both highly accurate and at the same time focuses on usability, accessibility, scalability and real-world implementation as the primary goals of the proposed system.

## 2.4. Comparative Review on Similar Projects

A thorough analysis of the current crop disease detection and agricultural decision-support systems reveals that even though there has been considerable advancement in employing artificial intelligence and expert knowledge in agriculture, most solutions are applied to a single component of the issue. Consequently, it is evident that there is still a significant disparity between the technically developed systems and solutions that are actually applicable and available to the farmers in the actual real world situations.

PlantVillage and other research systems of FarmAI have shown that, with properly trained deep learning models, there is a high potential of predicting diseases in crops with high accuracy using leaf images. These systems are also useful tools in the academic research and have served to set benchmark datasets and evaluation standards. They are however, mostly geared into controlled conditions and testing. They are usually not deployment-ready, including user-friendly interfaces, localized treatment advice, multilingual supports, and offline accessibility. These restrictions are very limiting to the practical usefulness of smallholder farmers, particularly those who reside in rural regions with little connectivity and technical literacy.

Applications such as LeafSnap also portray the issue of scope limitation. LeafSnap is a successful project with its main aim being classification of plant species though it takes advantage of image recognition to identify the species of plant instead of crop health management. It does not examine the signs of diseases or offer treatment information, which is not appropriate to farmers who require actionable information to save their crops. That is why it is necessary to shift toward identification to the systems that are oriented on decisions and that directly assist the farming activities.

Other solutions based on expert systems like agri doctor do not follow this method but use pre-set rules and the knowledge of the expert. Although these systems have the potential to provide reliable information in case the inputs are precise, it relies on farmers who manually describe the symptoms. In reality, this is sometimes challenging, since farmers might fail to recognize the signs of a disease, or they might lack the technical terminology to do so. Also, systems that are based on rules are weak in adapting to new diseases and climate changes and to receive new agricultural conditions, which limits their scalability and long-term performance.

The most sophisticated side of the spectrum is commercial platforms such as Cropix which are a combination of AI-based disease detection, analytics and monitoring. Nevertheless, small-scale farmers are often unable to have them because they are very expensive, they are subscribed to and are also complicated to operate. These types of solutions are normally laid down in large farms or agribusinesses, with a large proportion of farming community poorly served.

Comparatively, the suggested system is to be set to balance between accuracy, affordability, usability and applicability to real-life. It combines AI-driven image detection with expertapproved, localized treatment suggestions based on local farming methods. The simplicity of the interface along with intuition is to guarantee easy use by the farmers with little technical skills. Moreover, the scalability is facilitated by the use of cloud services, and Progressive Web App (PWA) characteristics allow to access the results already seen offline and solve the issue of connectivity that is typical of rural environments.

The proposed solution is focused on the adoption of farmers and practical application, unlike research-only or high-cost commercial systems. It will fill the gap between technological advancement and the practical necessity of agriculture by focusing on the weaknesses of current projects and providing a cost-effective, inclusive, and impactful solution to the modern problems of crop diseases.

## 2.5. Methodology

One of the most important decisions in the life cycle of a software project is the choice of a development methodology because it directly determines the effectiveness, quality, and success of the system under development as well. In our situation, the project is the creation of an AI-based system of crop disease detection, which includes such sophisticated elements as the models of artificial intelligence, image-processing algorithms, and the web or mobile interface. The combination of these elements contributes to further complications of the project and, thus, the selection of the methodology becomes even more significant. There are seldom defined or set requirements in AI-based projects unlike in a traditional system. Machine learning models change in accuracy, data quality and variety, and end-user expectations can vary considerably during development. As an example, pre-testing of the disease detection model can indicate the gaps in prediction accuracy or extra image preprocessing stages. Likewise, user feedback on the interface would necessitate modifications in the display of the system in terms of showing predictions and recommendations.

Due to such inherent variability, the methodology selected should be able to sustain flexibility, ongoing development, and improvement. It must enable the development team to try out alternatives of AI models, test algorithms, and take into account users without causing serious delays or rework. Meanwhile, the methodology must provide that the development is structured enough to be properly documented, tested and reliable of the system. Such two requirements of flexibility and organization are paramount to research and deployment-intensive projects such as ours.

We critically analysed various proven software development methodologies each of which are recognised such as the Waterfall model, Agile methodology, Scrum framework, Rapid Application Development (RAD) and Prototyping model to come up with the most appropriate methodology. All methodologies were evaluated on how they could cope with changing requirements, respond to creating iterative developments, encourage collaboration among stakeholders, and deliver functional increments in time. In doing this comparative analysis, we hoped to choose a methodology that would not only suit the experimental feeling of AI development but also give a clear outline of project planning, monitoring, and evaluation.

Finally, we concentrated on seeking a solution that would be flexible and controllable so that the system can be developed effectively and have high quality, usability, and reliability to the end-users.

### 2.5.1. Considered Methodologies

#### 2.5.1.1. Waterfall Model

The Waterfall model is an old fashioned model of software development that is linear and sequential in nature. The phased approach of requirements analysis, system design, implementation, testing, deployment, and maintenance should not be neglected and should be completed before moving to the next step. This model is suitable in cases of projects whose requirements are stable and well known (Pressman, 2014).

With AI-based systems, however, the requirements tend to evolve during the training and evaluation of the models. As an illustration, the accuracy of disease classification can be increased, the datasets can be enlarged, or other features can be ordered once preliminary testing is done.

Waterfall model is rigid and thus, it is hard to accommodate such changes without major rework. Consequently, the Waterfall model was not deemed appropriate to this project because it lacked flexibility and did not support the aspect of iterative improvement.

#### 2.5.1.2. Agile Methodology

Agile methodology is grounded on the principle of iterative and incremental development, according to which the system is developed in small and manageable steps with a continuous user feedback. Agile focuses more on flexibility, teamwork, and regular release of working software instead of large-scale advance documentation (Sommerville, 2016).

In the case of AI-based systems, Agile is especially useful, as it enables the training, testing, and refining of models in several cycles. User feedback and system reviews can be integrated into the next round of development within a short period of time. Agile also minimizes the chances of project failure since the problems are identified early and the system is also developed based on actual user requirements. Agile emerged as one of the good candidate methodologies due to these benefits.

#### 2.5.1.3. Scrum Framework

Scrum is an orderly structure of Agile that breaks the development into short cycles, called sprints, which can take two or four weeks. Every sprint is concerned with delivering a workable and possibly deliverable increment of the system. Scrum comes with specific roles like Product Owner, Scrum Master and Development Team, which enhance accountability and communication (Schwaber and Sutherland, 2020).

Scrum will particularly be appropriate in this project since it facilitates the development of the AI in bits, including uploading images and disease prediction algorithms and recommendation systems. Frequent sprint reviews and retrospectives allow checking the progress and risks continuously, and Scrum is one of the methods that can be used to address uncertainty that can arise in AI experimentation.

Development of software and systems at a high pace (RAD)

#### 2.5.1.4. Rapid Application Development (RAD)

RAD is concentrated on the quick development by the means of the rapid prototyping and minimum planning. Although the method is fast at the initial development stages, it usually compromises system architecture, documentation, and scalability (Pressman, 2014). Rad alone is not enough in case of an academic and research-oriented project which needs to be structured, analyzed, and maintained. So, it was not chosen as the main methodology.

#### 2.5.1.5. Prototyping Model

The Prototyping model entails developing prototypes of the systems to ensure that the requirements of the users are clearly known and to obtain feedback. The method can be useful in cases where the requirements are not defined initially. Nevertheless, overdependence on prototypes may slow down the final development and lead to inadequately organized systems in case prototypes are not redesigned but constantly modified (Sommerville, 2016).

### 2.5.2. Selected Methodology – Scrum

After carefully analyzing all the considered methodologies, the Scrum Framework under the Agile methodology was selected as the most suitable approach for the AI-based Crop Disease Detection System. This choice was driven by the project's iterative nature, the need for adaptability, and the structured management approach that Scrum provides. The development of this system involves multiple complex components, including AI model experimentation, software integration, and user interface development. Each of these components requires continuous testing, validation, and feedback, which Scrum accommodates exceptionally well through its sprint-based iterative cycles. Unlike traditional linear methodologies, Scrum allows flexibility to adjust requirements and refine deliverables based on ongoing evaluation and stakeholder input, which is essential when working with AI models that may need repeated tuning for optimal accuracy.

Scrum organizes the project into short, time-boxed iterations called sprints, each lasting a few weeks and focusing on delivering specific functionalities. For this project, sprints are planned around critical milestones such as dataset preparation, AI model training, frontend and backend development, system integration, and testing. Breaking the work into manageable increments ensures that progress is measurable and any potential issues are identified early, preventing delays or major setbacks. Daily standup meetings provide a platform for immediate communication among team members, facilitating quick resolution of obstacles and enhancing collaboration. At the end of each sprint, sprint reviews and retrospectives are conducted to evaluate completed work, gather feedback, and plan improvements for the next sprint. This continuous review process allows for refinement of both the AI models and the application interface based on performance results and user feedback, ensuring that the system evolves toward optimal functionality.

The adoption of Scrum provides several key benefits for this project. Firstly, flexibility allows the team to adapt to evolving requirements, unforeseen challenges, and iterative improvements in AI model accuracy. Secondly, incremental delivery ensures that functional components are available for testing and evaluation at the end of each sprint, providing tangible results and early validation. Thirdly, Scrum encourages collaboration and communication among all stakeholders, which is critical for aligning AI performance, software development, and user interface design. Additionally, Scrum supports risk management by identifying and addressing issues early, thereby reducing the likelihood of major failures during the project. Finally, the methodology facilitates user feedback integration, allowing potential users, advisors, or stakeholders to test early versions of the system, ensuring that the final product meets real-world needs effectively.

By applying the Scrum Framework, the development of the AI-based Crop Disease Detection System is structured, efficient, and adaptable. It ensures that the project progresses in a coordinated and organized manner while maintaining high quality and timely delivery. The iterative approach enables continuous improvement, better integration of AI and software components, and a user-centered design, making Scrum the most appropriate methodology for this innovative and technically challenging project.

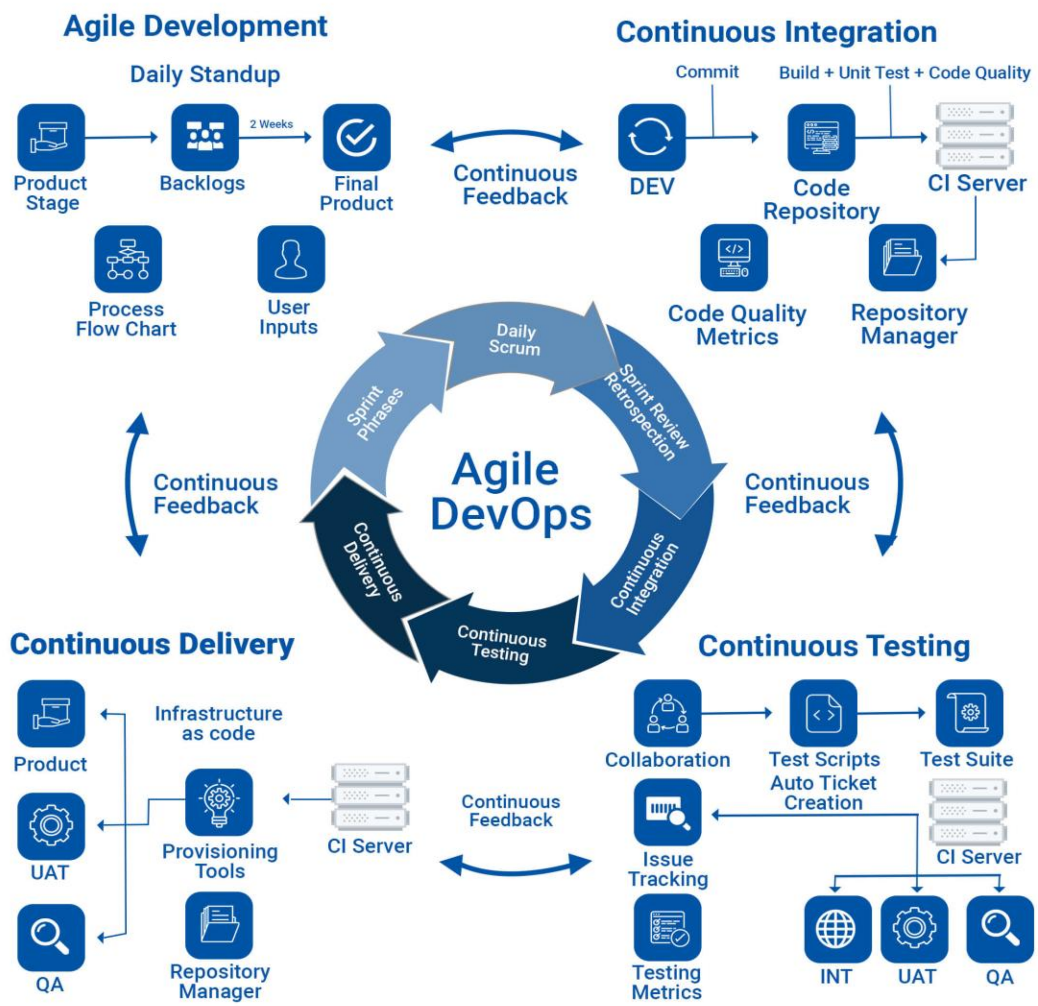



*Figure 1: Selected Methodology*

At last what we can say is that we chose the Scrum framework in this project after we have carefully considered the advantages and disadvantages of all the methodologies that were taken into consideration. Scrum offers flexibility necessary in the training of AI models, performance optimisation, and system requirement development. The sprint based framework will enable us to build upon and test the critical parts of the system, including precisely detecting diseases, prescribing treatment, and visualisation dashboards, in a step-by-step fashion.

The frequent review of sprints provides the focus on the project goals, whereas retrospectives promote the constant improvement of the process. Moreover, Scrum fosters the reduction of risks as it allows detecting technical problems and user concerns at the beginning of the development. In general, Scrum is a structured, flexible, and feedback-driven approach that is the most suitable methodology in the successful development of this AI-driven system.

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## 2.6. Review of the Technical Aspects

This project will be technically implemented by applying the latest machine learning algorithms, along with modern web development technologies, to develop a high-performance cost-effective scalable and user-friendly AI-based crop disease detection system. The central element of the system is the Convolutional Neural Networks (CNNs), a form of a deep learning model that is specifically developed to perform the tasks of image recognition and classification (LeCun et al., 2015). CNNs are also great in the extraction of hierarchical features automatically in images e.g. the pattern and color variations as well as anomalies in the texture of leaves, which are crucial in the accurate identification of crop diseases. Training the CNN model on a large, labeled set of leaf images allows the system to attain high accuracy in classification, and is computationally efficient to be deployed in the real world.

To develop the application, we have used the MERN stack that includes MongoDB to be flexible and scalable in data storage, Express.js and Node.js to be able to use server-side logic and API-related tasks, and React.js to create a dynamic and responsive user interface (Flanagan, 2020). This stack gives us the ability to build a full-stack web or mobile application in which the user can simply post images of crops, see the diagnostic results and obtain treatment recommendations. RESTful API architecture allows effective exchange of information between the front and the backend through the stateless, secure, and efficient communication (Fielding, 2000). The role of REST APIs is to transmit images to the server to process them and retrieve the results of the prediction through the AI model and store the information related to the past, which is important in monitoring disease trends and decision-making.

Cloud infrastructure addresses scalability, reliability of the system by offering on-demand computational resources and high availability (Armbrust et al., 2010). This is because cloud deployment results in the system being able to support an extensive number of users at a given time, large-scale inference of models without latency factors and scaling to additional expansions such as the inclusion of more crop types or diseases or additional analysis functionalities. Using the services of the clouds, we will also be able to deploy automated backups, system monitoring, and automatic updates that result in the increased maintainability and resilience of the system.

On the whole, the combination of CNN-based AI, the MERN technology stack, RESTful APIs, and cloud infrastructure is a guarantee of the technical soundness of the system, its flexibility, and the ability to provide end-users with real-time, accurate, and actionable results. Besides serving the short-term goals of crop disease detection, such a combination provides a solid base of future improvement and scalability.

# 3. Work Till Date

## 3.1. Requirements Gathering

The need to capture requirements is an extremely important first stage in the software creation since it defines the purposes that the system will be required to perform, its limitations, and the demands of the users (Sommerville, 2016). In this case of AI-driven crop disease detector system, the requirement must be collected properly to make the system functional, user-friendly and technically viable.

Functional requirements are the statements of what the system has to do. In this project, such essential functional requirements include the possibility to upload crop images, detect and classify them with trained CNN model, and recommend useful treatment options, store past prediction results, and make visualizations to demonstrate trends in crop diseases (LeCun et al., 2015). Other requirements involve user account management, secure authentication and the ability to support multiple users at the same time using cloud infrastructure so as to make it accessible and reliable.

Non-functional requirements are concerned with the performance of the system, its usability, scalability and security. Since the target audience is the farmers who will certainly exhibit different degrees of technological skills, the system should be user-friendly, responsive, and cross-platform and cross-network (Pressman, 2014). The performance requirements are rapid and precise forecasting and reliability and scalability is achieved through the use of the cloud application so that the system can support the uploading of more images at a given time. Sensitive user and crop data is secured by security mechanisms such as encryption of data and authentication of users.

In order to get these requirements, we employed several methods. The interviews with stakeholders (farmers, agricultural specialists, system administrators) assisted in finding out the key needs, workflow and challenges. The quantitative data of the user expectations in terms of accuracy and response time and desired features were provided through survey and questionnaires (Wiegers and Beatty, 2013). Besides, literature analysis and comparisons with current AI-based systems in detecting crop diseases guided the technical requirements, revealing the best practices in the implementation of CNNs models and interface design.

The results of this process were documented as a requirements specification document that acted as a guide during the system design and development process. Through the systematic collection and verification of the requirements, we were able to be confident that the end system will be in line with the needs of the users, lessen chances of miscommunication, and provide feasible, quality solutions to crop disease management.

### 3.1.1. Software Requirement Specification (SRS)

The Software Requirement Specification (SRS) serves as a formal document that clearly defines the functional and non-functional requirements of the system by providing a blueprint for design, development, testing and deployment (Sommerville, 2016). For this AI-driven crop disease detection system, the SRS ensures that all stakeholder, developers, users and project supervisors have a shared understanding of the system's objectives and capabilities. By documenting requirements in a structured manner, the SRS reduces ambiguity, facilitates planning and improves the likelihood of project success.

#### 3.1.1.1. Functional Requirements

Functional requirements are those requirements that outline the particular tasks or operations that the system has to execute to produce the functionality intended by it. These are the primary requirements that are at the heart of the value of the system to the users.

- User Authentication: The system should allow users to create their accounts, log in and manage their personal profiles. They are made to ensure that the system and sensitive data stored therein are only accessible by people who are allowed access. Firewalls and encryption keys are necessary to ensure the integrity and trust of the users (Pressman, 2014).

- Image Upload and Disease Detection: The users should be allowed to add images of crop leaves either through the mobile or web interface. This system works with these images using a trained Convolutional Neural Network (CNN) model that can automatically detect the patterns of various diseases and the crop condition identified on the image. This will enable farmers to get valid and timely diagnoses, which will enhance crop management and cut down on losses (LeCun et al., 2015).

- Recommendation Delivery: Once the disease is detected, the system must offer recommendations that are to be implemented such as treatment, fertilisers or organic intervention. This will make sure that users make informed decisions through AI predictions that will translate data into feasible instructions on how to take care of crops.

- Data Management and History Tracking: The system should be able to store user interactions, uploaded images and prediction results in a well organized format. This historical information enables one to analyze trends in the diseases with time, track recurrent problems, and assess the system operation hence facilitating improved planning and decision-making.

- API Integration: RESTful API is needed to enable safe and effective communication between the frontend and the backend system. Such APIs are used to carry out actions like transmitting image data to be processed, retrieving prediction, and update records in history which make the interactions between systems to be seamless and responsive (Fielding, 2000).

### 3.1.1.2. Non-Functional Requirements.

Non-functional requirements define quality attributes, constraints, and system characteristics required to ensure its functional operations. They define the effectiveness of the system and the experience of the users.

- Performance: The system should be able to yield fast and accurate predictions, even in the case of numerous users using it at the same time. The cloud-based infrastructure means that the system is able to scale dynamically to meet different workloads to provide the same response time (Armbrust et al., 2010).

- Usability: The system is meant to be used by farmers, including less technical, thus the interface needs to be user-friendly, easy to use, and convenient. Very simple steps to upload the image, clear instructions, and simple interpretation of results lead to the maximization of the adoption.

- Reliability and Availability: The system must have high uptime and must be operating continuously and must recover gracefully in case of errors or some unexpected failures. Dependable performance is critical to applications in the real world agricultural setup whereby timely information may avert losses of crops.

- Security: The user data (images and past predictions) should be secured with the help of encryption, safe authentication, as well as adherence to privacy standards. The security systems are used to protect sensitive data and avoid unauthorized access.

- Maintainability and Extensibility: The system design must accommodate future modification, inclusion of new types of crops, disease models, and inclusion of new advanced analytics. This guarantees sustainability in the long run and flexibility regarding the changing agricultural requirements.

The comprehensive blueprint of the SRS is offered through a rigorous definition of functional requirements, non-functional requirements; hence, it guides the development, testing, deployment, and maintenance. It makes sure that the system is not only in the expectations of the users but works effectively, is in a secure position, and can be modified to future improvements.

### 3.1.2. Product Backlog

Product Backlog Product Backlog is the list of those features, requirements, additions, or fixes needed in a software system, in order and constantly changing. It is the sole provider of the requirements upon any alterations to be done to the product and is fundamental in the Scrum model. Product Backlog items are usually defined as user stories, a system functionality that is described through the eyes of end users or stakeholders (Schwaber and Sutherland, 2020).

Product Backlog is also prioritized in terms of business value, risk and dependency where the most important features are built at the beginning. Agile development does not have a fixed backlog, but the backlog is enhanced on a regular basis based on the changing user requirements, feedback, and project constraints. When the Sprint Backlog is being planned, items are taken out of the Product Backlog and into the Sprint Backlog, which allows it to be developed incrementally and iteratively.

The functional requirements that are reflected in the Product Backlog in this project include user authentication, AI-based crop disease detection, treatment recommendation, analytics, and administrative controls. The project is also well structured through its Product Backlog, which makes it have transparency, good planning, and adherence to the Agile and Scrum concepts during the development lifecycle.

*Table 2: Table of Product Backlog*

| ID | User Story | Priority | Sprint | Story Point |
|---|---|---|---|---|
| US001 | As a farmer, I want to register an account so that I can use the system securely. | High | 1 | 3 |
| US002 | As a farmer, I want to log in using my credentials so that I can access my dashboard. | High | 1 | 2 |
| US0003 | As a farmer, I want to upload an image of a crop leaf so that the system can detect diseases. | High | 1 | 5 |
| US004 | As a system, I want to preprocess uploaded images so that AI predictions are accurate. | High | 1 | 5 |

| | | | | |
|---|---|---|---|---|
| US005 | As a system, I want to classify crop diseases using a CNN model. | High | 2 | 8 |
| US006 | As a farmer, I want to view disease prediction results so that I know the crop condition. | High | 2 | 3 |
| US007 | As a farmer, I want to see a confidence score with predictions so that I can trust results. | Medium | 2 | 2 |
| US008 | As a farmer, I want to receive treatment recommendations so that I can cure the disease. | High | 2 | 5 |
| US009 | As a farmer, I want organic treatment suggestions so that I can avoid chemical damage. | Medium | 3 | 3 |
| US010 | As a farmer, I want pesticide recommendations with dosage so that treatment is safe. | High | 3 | 5 |
| US011 | As a system, I want to store prediction history so that farmers can track past results. | High | 3 | 5 |
| US012 | As a farmer, I want to view my prediction history so that I can monitor crop health. | Medium | 3 | 3 |
| US013 | As a farmer, I want disease trend charts so that I can identify frequent issues. | Medium | 4 | 3 |
| US014 | As a system, I want to generate analytics reports so that trends are visualized clearly. | Medium | 4 | 5 |

| | | | | |
|---|---|---|---|---|
| US015 | As an admin, I want to manage crop and disease datasets so that AI accuracy improves. | High | 4 | 8 |
| US016 | As an admin, I want to retrain AI models so that predictions stay up-to-date. | High | 5 | 8 |
| US017 | As an admin, I want to manage users so that system access is controlled. | Medium | 5 | 3 |
| US018 | As a farmer, I want notifications for disease alerts so that I act on time. | Medium | 5 | 3 |
| US019 | As a system, I want to support mobile devices so that farmers can use phones easily. | Medium | 6 | 5 |
| US020 | As a system, I want offline support (PWA) so that farmers can use it in low network areas. | Medium | 6 | 5 |
| US021 | As a farmer, I want crop calendar suggestions so that I plan farming activities. | Low | 6 | 3 |
| US022 | As a farmer, I want to participate in a discussion forum so that I can share knowledge. | Low | 7 | 3 |
| US023 | As an expert, I want to reply to farmer queries so that proper guidance is given. | Medium | 7 | 3 |
| US024 | As a system, I want secure authentication so that user data is protected. | High | 7 | 5 |

| | | | | |
|---|---|---|---|---|
| US025 | As a system, I want cloud deployment so that the application is scalable and available. | High | 8 | 8 |

### 3.1.2.1. Sprint Backlog

The Sprint Backlog is an extension of the Product Backlog that includes the chosen user stories, tasks and technical activities that the development team promises to accomplish within a particular sprint. It is the work to be implemented in a specific duration of time, normally one to four weeks, and developed at the sprint planning meeting.

Sprint Backlog is a short term implementation plan in Scrum methodology. It contains the high-priority user stories selected among the Product Backlog and their approximate effort (story points) and implementation information. Sprint backlog belongs to the development team and is regularly updated as the work is developed during the sprint.

In this project, Sprint Backlog is used to group user stories, including image upload, AI-based disease detection, treatment recommendation, analytics, and system deployment into several sprints. This method guarantees the gradual improvement, constant testing, and initial delivery of fundamental capabilities. With the help of a Sprint Backlog, the project is transparent and better time-managed, and it can constantly adjust itself on the feedback and performance assessment (Schwaber and Sutherland, 2020).

The Major Characteristics of a Sprint Backlog.

- Includes some user stories chosen in the Product Backlog.
- Time based on a definite sprint period.
- Effort estimation uses story points.
- Concentrates on performance and action.
- Change on a daily basis during the sprint.

**Sprint 1 – User Access & Core Upload (Foundation Sprint)**

**Sprint Goal:** Enable user registration, login, and image upload.

| ID | User Story | Story Point |
|---|---|---|
| US001 | Farmer registration | 3 |
| US002 | Farmer login | 2 |
| US003 | Upload crop leaf image | 5 |
| US004 | Image preprocessing | 5 |
| US024 | Secure authentication | 5 |

In Sprint 1 we were working on laying down the main backbone of the system. The primary goal of this sprint was to allow access and basic system interaction without conflicts. In this sprint, we introduced user registration and login system and security authentication systems. We also designed crop image upload option where farmers can post the image of the crop leaves to be analyzed. Further, the process of image preprocessing was also introduced to achieve the desired quality of uploaded images to guarantee the correct disease detection through AI. Every next system functionality was preconditioned by this sprint.

**Sprint 2 – AI Disease Detection**

**Sprint Goal:** Implement core AI prediction functionality.

| ID | User Story | Story Point |
|---|---|---|
| US005 | CNN-based disease classification. | 8 |
| US006 | View prediction results | 3 |
| US007 | View confidence score | 2 |
| US008 | Treatment recommendations | 5 |

Sprint 2 was supposed to bring in the essence of intelligence of the system. This sprint involved the incorporation of a Convolutional Neural Network (CNN) model to detect crop diseases using posted images. Other features we created are the ability of the farmers to see disease prediction outcomes and the confidence scores. Moreover, the treatment recommendations were worked out on the detected diseases. This sprint turned the system into a project that is an intelligent decision-support tool instead of being a simple application.

**Sprint 3 – History & Recommendation Enhancement**

**Sprint Goal:** Improve usability and monitoring.

| ID | User Story | Story Point |
|---|---|---|
| US009 | Organic treatment suggestions | 3 |
| US010 | Pesticide & dosage guidance | 5 |
| US011 | Store prediction history | 5 |
| US012 | View prediction history | 3 |

We improved the quality and utility of recommendations given to farmers in Sprint 3. The system was spread to provide organic and chemical treatment recommendations including the appropriate dosage. We also added an ability to save the history of predictions and provide the user with an opportunity to see their previous disease detection outcomes. This sprint enhanced the traceability and aided farmers in tracking crop health with time.

**Sprint 4 – Analytics & Admin Control**

**Sprint Goal:** Enable analytics and administrative control.

| ID | User Story | Story Point |
|---|---|---|
| US013 | Disease trend charts | 3 |
| US014 | Analytics report generation | 5 |
| US015 | Dataset management | 8 |

Sprint 4 was concerned with data analysis and system management. In this sprint, we installed analytics dashboards, which show disease trends and visual reports. Functionalities like the administration of datasets were also built and enabled administrators to maintain and enhance quality training data. This sprint helped monitor, control, and make data-driven decisions in the system.

**Sprint 5 – Model Optimization & Notifications**

**Sprint Goal:** Improve AI accuracy and user engagement.

| ID | User Story | Story Point |
|---|---|---|

| | | |
|---|---|---|
| US016 | AI model retraining | 8 |
| US017 | Admin user management | 3 |
| US018 | Disease alert notifications | 3 |

Sprint 5 was devoted to enhancing the accuracy of the system and the interaction with the user. The retraining functionality of AI models was introduced to make sure that the disease detection model will be accurate with the arrival of new data. The system access controls were also added by administering the users of the administration. Also, notification of the farmers concerning the important cases of diseases was incorporated to ensure prompt intervention.

**Sprint 6 – Mobile & Offline Support**

**Sprint Goal:** Accessibility in rural environments.

| ID | User Story | Story Point |
|---|---|---|
| US019 | Mobile responsiveness | 5 |
| US020 | PWA offline support | 5 |
| US021 | Crop calendar suggestions | 3 |

Sprint 6 was aimed at ensuring the system was available in rural and under-connectivity areas. In this sprint, we made the user interface friendly to the mobile device and added Progressive Web App (PWA) features that would allow the user to use it without an internet connection. Crop calendar recommendations were also brought to assist farmers take care of farming activities. This sprint this enhanced the ease of use and access.

**Sprint 7 – Community & Expert Interaction**

**Sprint Goal:** Knowledge sharing and expert advice.

| ID | User Story | Story Point |
|---|---|---|
| US022 | Farmer discussion forum | 3 |
| US023 | Expert replies | 3 |

In the 7 th sprint, we were working on knowledge sharing and expert support. The discussion forum was introduced where farmers should post questions and experiences. Farmers were empowered to answer such questions and offer professional opinions. This sprint improved teamwork and created an encouraging agricultural community of the system.

**Sprint 8 – Deployment**

**Sprint Goal:** Make system production-ready.

| ID | User Story | Story Point |
| --- | --- | --- |
| US025 | Cloud deployment | 8 |

The last sprint which is Sprint 8 was devoted to the system deployment and readiness to the real-life application. This sprint saw the application moved to a cloud environment in order to guarantee scalability, availability and performance. The system was also tested, configured and optimized to finalize its functionality and non-functionality requirements. This sprint was the final stage of the development lifecycle.

### 3.1.3. Pre-Survey

In order to make sure that the AI-based crop disease detection system can meet the needs of its target users, the pre-survey was performed on 30 respondents. The main objectives of this survey were to receive information about the difficulties that farmers faced to deal with crop diseases, their access to technology, their knowledge of mobile applications, and their expectations of an AI-powered solution. A pre-survey like this is an essential part of a user-centered design since it helps the development team to prioritize system functionality with the real-world requirements instead of making assumptions and relying on secondary research only (Sommerville, 2016).

#### 3.1.3.1. The demographics and occupation of the respondents will be presented in it.

The respondents of the survey were of varying occupations and mostly were students (76.7%); only a few numbers were farmers, business owners, housewives, and those in other occupations. Students are the majority but even though this means that the survey will still be valuable as a number of them have experience in agricultural practices, are related to farming household or even have interest in agricultural technology. This is due to the diversity of the respondents that assists in the interpretation of the system with regard to its usability and accessibility among a variety of potential users.

### 3.1.3.2. Prevalence of Frequency of Crop Disease and Current Practices.

The poll established that most respondents are concerned about crop disease:

- Occurrence frequency: 36.7% occasionally experience crop diseases, 46.7% infrequently and 10% often. Very few people (0.10 percent) stated that they had never encountered the problem of crop diseases.

- The methods of identification: The methods that farmers use to identify agro-pests are mostly asking specialists (40%), visual observation (40%), and online tools like YouTube or websites (56.7%). Only a few respondents (6.7) do not have a way of identifying crop disease.

- These findings suggest that although traditional knowledge and expert consultation are still significant, there is a strong need to have quicker, more steady and convenient methods of detecting diseases especially where the advice of experts may not be easily accessible.

### 3.1.3.3. Technology Access and Familiarity.

Mobile apps familiarity: The respondents familiar with agricultural apps were only very familiar with them (16.7) and 26.7 were somewhat familiar with the apps and 56.7 were not familiar.

- Favorite devices: 63.3% were using smartphones, followed by laptops (16.7%), and tablets (13.3%).

- The above observations demonstrate that it is necessary to create a mobile-first system that is lightweight, user-friendly, and does not demand high-level technical skills.

### 3.1.3.4. Expectations and Preferences of the users.

Respondents gave their perceptions of what they would want an AI-based crop disease detection application to offer:

- Early detection: 36.7% thought that early detection of the disease would help avoid losses in crop and 53.3% thought that it could help, an indicator that most of the users understand the importance of timely information.

- Interest in using the app: 53.3% will certainly use such an app and 46.7% may use it which shows that the rate of adoption would be high.

- Significance of accuracy: 55.2 percent found the importance of accuracy to be extremely vital, the importance of predictive reliability is critical as thought by 34.5 percent.

- Suggestions on treatment: 33.3% thought treatment suggestions to be necessary and 60% believed they were optional.

- Offline ability: 75.9% wanted the use of the app offline, which indicates the issue of connectivity in rural locations.

- Language preferences: 63.3% were more inclined towards local and language support, 23.3% were inclined towards Nepali, and 13.3% were inclined towards English, which demonstrates the value of multi-language support.

#### 3.1.3.5. Crop Types and Features

Vegetables (66.7%), fruits (33.3%), rice (13.3%), and wheat (13.3%) are mostly grown by the respondents. In terms of the features of the apps, 46.7 percent desired to have all-inclusive features (disease identification, treatment recommendations, prevention tips, and weather warnings), whereas the remaining preferred particular features such as disease identification (23.3 percent).

#### 3.1.3.6. Willingness and Concerns over Payments.

Payment of service: 63.3% were ready to pay a small amount of money to be accurately detected with the disease, which showed the perception of the value and the possibility to be monetized.

- Fears related to the adoption of AI: The biggest fears were related to the trust (63.3%), accuracy (56.7%), ease of use (43.3%), and costs (36.7%).

#### 3.1.3.7. System Design Implications.

The outcomes of the pre-survey have a direct effect on the design of the system and the functionality:

- Mobile first and user-friendly interface, particularly to the non-technical users.
- Detection of diseases with high accuracy using CNN to cover the issues of trust and reliability.
- Prescription advice and preventive measures to facilitate action planning.
- Offline feature to deal with low connectivity.
- Multi-language assistance to increase accessibility to a variety of users.
- Guarantee the use of secure authentication and storage of data to create confidence and safeguard confidential information to the user.

To sum up, the pre-survey confirms the need and the possibility of an AI-based system of crop disease detection. It identifies the preferences of the users, their possible adoption rates, and the main features that will make it more effective, trusted, and usable (Pressman, 2014; Sommerville, 2016). It has these implications making the system practical, user-friendly, and something that fits the real-life problems in agriculture.

## 3.2. System Design

The system design phase is a foundational stage in software development where the project's requirements are transformed into a structured framework that directs how the system will be built. This phase goes beyond simply listing features; it defines *how* the system's components will work together, how data will flow among them, and how different users will interact with the system overall. In effect, system design creates a blueprint that guides the development team from concept to implementation, reducing ambiguity and helping prevent costly mistakes later in the project lifecycle (Prepbytes, 2023; Shocodevin, 2025).

A well-designed system does not emerge by accident it is the result of careful planning that balances clarity, modularity, and scalability, which are crucial quality attributes:

- **Clarity:** At its core, clarity in system design ensures that every component and interaction within the system is well defined. A clear design helps developers and stakeholders understand exactly how different parts of the system relate to one another and what role each component plays. This shared understanding eliminates confusion during implementation and makes it easier to communicate design decisions among team members. Architectural design serves as this clear roadmap by specifying the overall structure, components, and communication paths in a system (Tpoint Tech, 2025).

- **Modularity:** Modularity means dividing the system into smaller, independent units or modules such as frontend, backend, AI services, and databases — where each module handles specific functionality. This separation makes complex systems manageable by isolating responsibilities, lowering interdependencies, and allowing developers to work on individual modules without risking unintended impacts on the entire system. Modularity also improves maintainability because changes in one module do not necessitate widespread changes elsewhere (Wikipedia, 2025).

- **Scalability:** Scalability refers to the system’s capacity to handle growth in terms of users, data, and operations without significant performance degradation. Designing for scalability means anticipating future increases in workload and choosing architectural patterns that support expansion without requiring a complete redesign. This is crucial for systems expected to serve many users or process large volumes of data, such as agricultural platforms used by thousands of farmers. Scalability ensures the system can adapt and evolve over time while maintaining performance and reliability (Shocodevin, 2025; Wikipedia, 2025).

By focusing on these principles during the system design phase, the project gains a solid technical foundation that aligns with both immediate requirements and future goals. A detailed design helps the development team plan effectively, improves overall communication, and ensures the system can be extended, maintained, and scaled with minimal friction as new features are added or usage increases (Prepbytes, 2023; Tpoint Tech, 2025)

### 3.2.1. System Architecture

The system architecture is an essential roadmap, which determines the structure of the system and how various parts of the system will relate to each other to realize the desired functionality. In the present project, we have implemented the layered architecture, which splits the system into distinct interdependent layers. This will guarantee modularity, scalability, maintainability, and clarity as well as provide an avenue to enhance or upgrade in future without interfering with any existing functionality (Shocodevin, 2025).

Functionality and Layer Breakdown.

- Frontend Layer
  The interface layer is the interface point between the system and the users who are farmers, the agricultural experts and the administrators. This interface is meant to be simple and easy to use, enabling farmers to post photos of crops, get predictions of diseases, and have treatment plans. AI predictions can be validated by experts and guided, and knowledge bases can be updated, and admins can be able to monitor the overall system activity. This way, we can guarantee that all modifications to how the system is being displayed, including redesigning forms or dashboards, does not affect the functionality of the backend code or AI processing (Prepbytes, 2023).

- Backend Layer
  The backend layer is a system layer that handles the business logic of the system. It receives the user requests in the front end and interprets them based on system rules and sends messages to the AI service and the database. An example would be when a farmer sends a crop picture, the backend will validate this input, send it to the AI layer to analyze it and the prediction results will be stored back to the rear of the frontend and displayed. The location of business logic in the backend allows the processing of all data to be consistent, secure and maintainable (Tpoint Tech, 2025).

- AI Service Layer
  The layer of AI service will involve making intelligent decisions in the system. It uses machine learning algorithms, including Convolutional Neural Networks (CNNs), to identify crop diseases and provide the correct treatment options. When an image has been passed down the backend to this layer, the AI model works on it, identifies possible diseases, and returns actionable information. It is possible to enhance the models, add more training data, or add predictive analytics, and leave other system layers unchanged by isolating the AI services into a separate layer (Shocodevin, 2025).

- Layer

The database layer is a layer that is focused on the storage and management of structured data. These are user profiles, crop data, disease history, prediction history, and analytics outcomes. This system will allow the storage and efficient querying of diverse information, as it will manage data in a distinct layer, and allow scaling that can be used as the number of farmers and crop data grows. This division also permits the AI as well as the backend layers to run without having to engage directly with the raw data storage, enhancing performance and reliability (Wikipedia, 2025).

The benefits of the Layered Architecture.

i. The layered approach has a number of advantages:

- Maintenance: Developers are able to work on a single layer without impacting the other thus minimizing errors and making the process of testing less complicated.

- Scalability: It is possible to scale the use of a specific layer to increase the number of users, crops, or AI models as required. As an illustration, when an increasing number of farmers are on the platform at the same time, the back-end and database may be scaled without altering the frontend or AI models.

- Upgrades in the Future: It can be upgraded to accommodate new AI algorithms or new features or redesign user interfaces without involving the system itself, which makes the system flexible and adaptable to future demands.

Attainable Workflow.

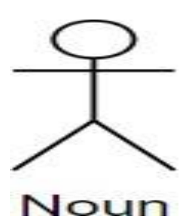
As an example, in case a farmer uploads a leaf image, the request is received by the frontend layer which gives a confirmation interface first. The picture is then forwarded to the backend where it is validated and preprocessed. Then, the AI service layer interprets the image to identify diseases and provide recommendations. Lastly, the predicted value and the history of the interactions are stored in the database layer and the results are returned back on the backend to the frontend so that the farmer may see. This is an illuminating flow of the importance of each layer in the process of the smooth, precise, and efficient functioning.

### 3.2.2. Use Case Diagram

A Use Case Diagram is a visual representation of the functional requirements of a system, highlighting the interactions between **actors** (users or external systems) and the system itself. It captures what tasks the system can perform, who can perform them, and how the system responds to different user actions (Sommerville, 2016). In the AI-Powered Crop Disease Detection and Recommendation System, the use case diagram demonstrates the dynamic interactions between farmers, administrators, agricultural experts, and the AI system. Farmers can submit images of crops to detect diseases, receive automated treatment recommendations, and track the health of their crops over time. Experts can validate disease diagnoses and update treatment suggestions. Administrators manage user accounts, monitor system performance, and ensure data integrity. The AI system acts as a central processing entity, analyzing submitted crop images, detecting diseases using machine learning models, and generating relevant recommendations for treatment. This use case representation helps stakeholders understand the system's functional capabilities, clarifies user roles, and serves as a foundation for system design and development (Jacobson et al., 1992).

#### 3.2.2.1. Notation used in use case

**i. Actor**




Actors give the various kinds of users/external systems expected to communicate with the system. They may be individuals, other applications or computer hardware that interact with the system to get things done. The system presents certain roles to different actors and each communicates with the other to reach a goal. Actors in a use case diagram are usually depicted as a stick figure outside the system boundary. As an example, a customer, administrators, and sellers are the actors likely to appear in an e-commerce system. The naming convention is noun.

### ii. Use Case

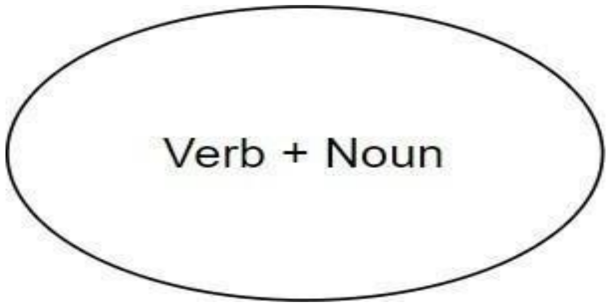



*Figure 4: Notation for Use Case*

Describing how the system responds to actor requests, use cases detailed the actual actions/tasks performed by the system in response to such requests. Every use case symbolizes a functional requirement, or a specific service that the system provides to the persons. Within the system bound, use cases are shown as horizontal ovals (or ellipses) with the name of the functionality written in them. As an example, use cases could consist of the following in the same context of the e-commerce example: such as Register Account, Place Order, Manage Products, and Generate Sales Report. These assist in explaining what the system should perform on the part of the users.

### iii. System boundary

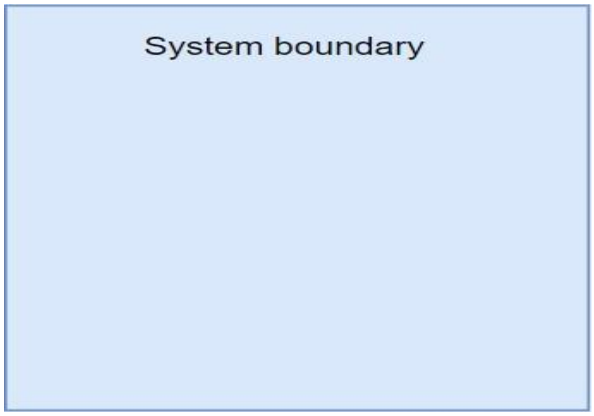



*Figure 5: Notation for System boundary*

The system boundary controls what is within a system and excludes actors outside a system. It is signified by a great rectangle (box) that contains all the use cases and depicting what belongs to the system or what is outside of it. This is because usually the name of the system is written in the top part of the rectangle. The presented visual representation assists the stakeholders to make a clear distinction between internally defined behaviors and externally defined behaviors and thus have a well-scoped system design.

### iv. Association

*Figure 6: Notation for Association*

An association is shown by a solid straight line between an actor and a use case. It means that the actor is involved in the use case, either as a producer of the call that leads to use case or as a consumer of some results of the use case. It is in most cases that they do not have an arrowhead as their lines unless the direction is required to be indicated. The use case diagram depends on associations to resemble its foundation based on the direct interaction of the system and the user (or external element).

### v. Include

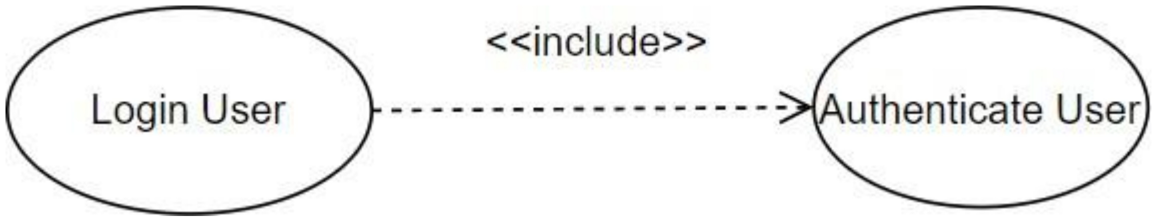



*Figure 7: Notation for Include*

The include relationship is represented by a dashed arrow labelled «include» and whose direction is drawn as showing the base use case to the included use case. It refers to the fact that the behavior of the included use case will be always executed when executing the base use case. This prevents duplication as it is possible to share functions. In diagrams, the arrow is extended towards the inclusion use case and indicates that such an inclusion is mandatory.

**vi. Extends**

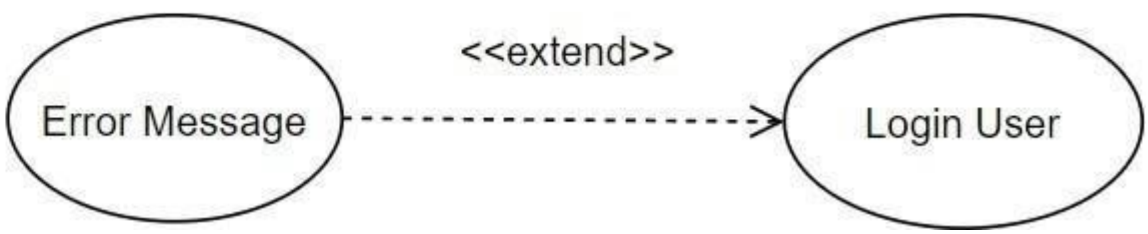



*Figure 8: Notation for Extends*

The extend association is also illustrated using a dashed arrow but the direction is in opposite as labelling whereby the label is extended and the extended use case points at the base use case. It describes optional or contingent behavior, which can be activated, only under some conditions. It means that the base use case can remain short and extensions can be addressed individually. The direction of the arrow, against which optional functionality, directs to the major covering use case.

### vii. Generalization

*Figure 9: Notation for Generalization*

Generalization is shown as a solid line having a hollow triangle arrow at the end with the more specific actor or use case on the tip. To actors, this implies that all the roles and responsibilities of the parent actor are transferred to the child actor. To be used, it indicates that the specialized use case is the special form of the generalized one. The visual provides reuse of actors/behaviors and hierarchy of actors/behaviors

## 3.2.2.2. Actor and Use Case

1**. Actors:**

- Farmer
- Admin
- Expert
- AI System

**2. Use Cases:**

**i. Farmer:**

- Register
- Login
- Upload Crop Image
- View Disease Detection Result
- View Confidence Score
- View Treatment Recommendations
- View Prediction History
- Receive Disease Notifications
- Participate in Discussion Forum

### ii. Admin

- Manage Users
- Manage Crop Disease Dataset
- Retrain AI Model
- View Analytics Reports

### iii. Expert

- View Farmer Queries
- Reply to Farmer Queries

### iv. AI System

- Preprocess Image
- Detect Crop Disease
- Store Prediction History
- Generate Analytics
- Send Notifications

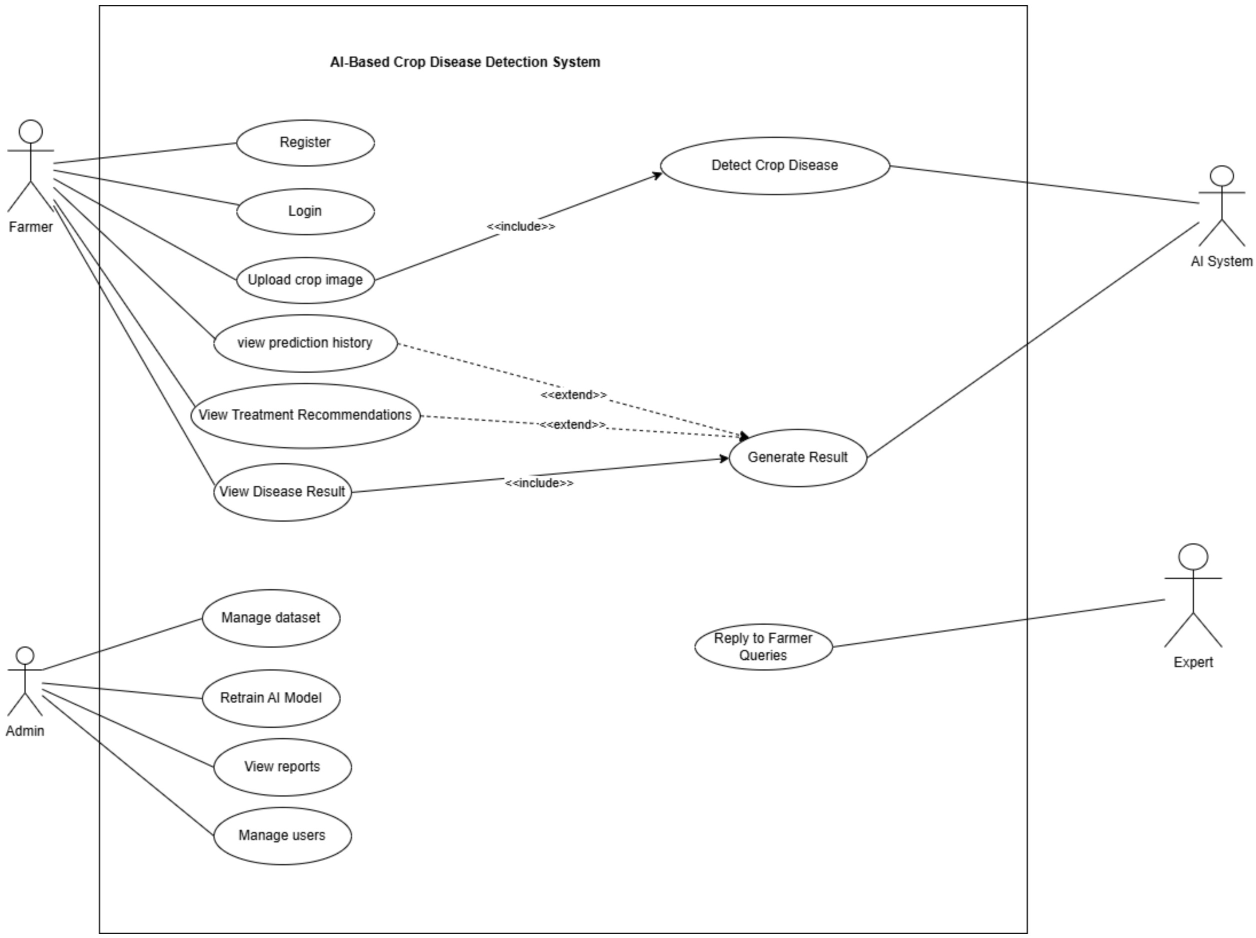



*Figure 2: Usecase diagram*

Aligned User Stories:
US001, US002, US003, US006, US007, US008, US011, US012, US015, US016, US017, US022, US023, US024

This use case diagram illustrates the interaction between farmers, administrators, experts, and the AI system. It highlights the functional requirements of the system, including disease detection, treatment recommendation, dataset management, and expert interaction.

### 3.2.3. Entity-Relationship Diagram (ERD)

The ERD for the AI-Powered Crop Disease Detection and Recommendation System presents the structural layout of the database layer supporting all the AI-driven prediction operations, user roles, farming knowledge-base storage, community interaction, and system analytics. Each entity defines a data object in the system along with its properties, and relationship mapping ensures seamless workflow between prediction, treatment suggestion, calendar management, and expert advisory support.

The architecture is scalable and supports various user categories, model upgrades, dataset retraining, automated notifications, and role-based data access.

#### 3.2.3.1. Entities and Attributes

In a database system, entities represent the primary objects or components about which data is stored. These can be people, objects, events, places, or concepts that exist within the system and are of some significance to the application. Each entity will be considered as an independent data unit, and in a relational database, it is normally mapped to a table. For instance, in an AI-powered crop disease detection system, major data elements such as User, Crop, Disease, Farmer, and AI_Model are the entities that the system interacts with and maintains information regarding.

Attributes, on the other hand, describe characteristics or properties of each entity. They define what type of information is stored within an entity, and in a database, they would be represented as fields or columns within a table. Each entity contains multiple attributes in order to uniquely define its behavior and structure. Such an example could be the User entity, which might have full_name, email, contact_number, and role as its attributes. These define the particular identity and details of any user. Put simply, entities describe what the system is storing data about, while attributes describe what detail of that entity is being stored.

**1. User**

user_id (PK)

full_name

email

password_hash

contact_number

role

date_created

**2. Admin**

admin_id (PK)

user_id (FK)

permissions

last_login

created_at

**3. Expert**

expert_id (PK)

user_id (FK)

specialization

certification_id

verified_status

rating

**4. Farmer**

farmer_id (PK)

user_id (FK)

location

land_size

preferred_crop

**5. Crop**

crop_id (PK)

crop_name

type

growth_duration

suitable_climate

soil_type

**6. Disease**

disease_id (PK)

disease_name

crop_id (FK)

severity_level

description

common_symptoms

**7. AI_Model**

model_id (PK)

model_version

training_dataset

accuracy_score

deployment_status

updated_at

**8. Prediction**

prediction_id (PK)

user_id (FK)

disease_id (FK)

confidence_score

image_path

prediction_timestamp

**9. Treatment_Recommendation**

recommendation_id (PK)

disease_id (FK)

treatment_type

product_name

dosage_instruction

eco_friendly_option

**10. Fertilizer**

fertilizer_id (PK)

fertilizer_name

crop_id (FK)

soil_requirement

application_method

recommended_season

**11. Pesticide**

pesticide_id (PK)

product_name

chemical_composition

application_frequency

safety_precautions

disease_id (FK)

**12. History_Record**

record_id (PK)

user_id (FK)

prediction_id (FK)

status

notes

record_created

**13. Crop_Calendar**

calendar_id (PK)

crop_id (FK)

planting_month

harvesting_month

season_alerts

**14. Notification**

notification_id (PK)

user_id (FK)

message

notification_type

status

created_at

**15. Forum_Post**

post_id (PK)

user_id (FK)

title

content

image_path

created_date

**16. Forum_Reply**

reply_id (PK)

post_id (FK)

user_id (FK)

reply_content

created_at

upvotes

**17. Dataset**

dataset_id (PK)

source_name

total_images

labels_count

last_updated

**18. Analytics_Report**

report_id (PK)

admin_id (FK)

report_type

generated_date

affected_region

key_metrics

**19. Region**

region_id (PK)

region_name

climate

humidity_level

soil_ph

risk_index

**20. Subscription**

subscription_id (PK)

farmer_id (FK)

plan_type

payment_status

valid_until

### 3.2.3.2. Relationship Mapping (Well-Formatted + Clear Representation)

#### i. User Role Relationship

A User account can exist as a Farmer, an Expert, or an Admin

**Type:** Inheritance

**Cardinality:** 1 User = 1 specific role

#### ii. Crop & Disease

One Crop may have multiple associated Diseases

**Type:** One-to-Many

**Cardinality:** Crop (1) → Disease (Many)

iii. **Disease & Recommendations**

One Disease can have several Treatment Recommendations

**Type:** One-to-Many

**Cardinality:** Disease (1) → Recommendation (Many)

iv. **Disease & Pesticides**

A Disease may require more than one Pesticide product

**Type:** One-to-Many

**Cardinality:** Disease (1) → Pesticide (Many)

v. **User/Farmer & Prediction**

A single user (or specifically a farmer) can upload multiple Predictions

**Type:** One-to-Many

**Cardinality:** User (1) → Prediction (Many)

vi. **Prediction & Disease**

Each Prediction corresponds to one Disease result

**Type:** Many-to-One

**Cardinality:** Prediction (Many) → Disease (1)

vii. **Prediction & History Record**

Every Prediction has one and only one History Record

**Type:** One-to-On

**Cardinality:** Prediction (1) → History Record (1)

viii. **Crop & Crop Calendar**

Each Crop maintains one Crop Calendar schedule

**Type:** One-to-One

**Cardinality:** Crop (1) → Crop Calendar (1)

### ix. User & Notifications

One User may receive many Notifications over time

**Type:** One-to-Many

**Cardinality:** User (1) → Notification (Many)

### x. User & Forum Posts

A User is allowed to write multiple Forum Posts

**Type:** One-to-Many

**Cardinality:** User (1) → Post (Many)

### xi. Post & Replies

A single Post can have several Replies from users

**Type:** One-to-Many

**Cardinality:** Post (1) → Reply (Many)

### xii. Dataset & AI Models

One Dataset can be used to train multiple AI_Model versions

**Type:** One-to-Many

**Cardinality:** Dataset (1) → Model (Many)

### xiii. Admin & Reports

Each Admin can generate multiple Analytics Reports

**Type:** One-to-Many

**Cardinality:** Admin (1) → Analytics Report (Many)

### xiv. Region & Farmers / Predictions

One Region can be associated with several Farmers and multiple Predictions

**Type:** One-to-Many

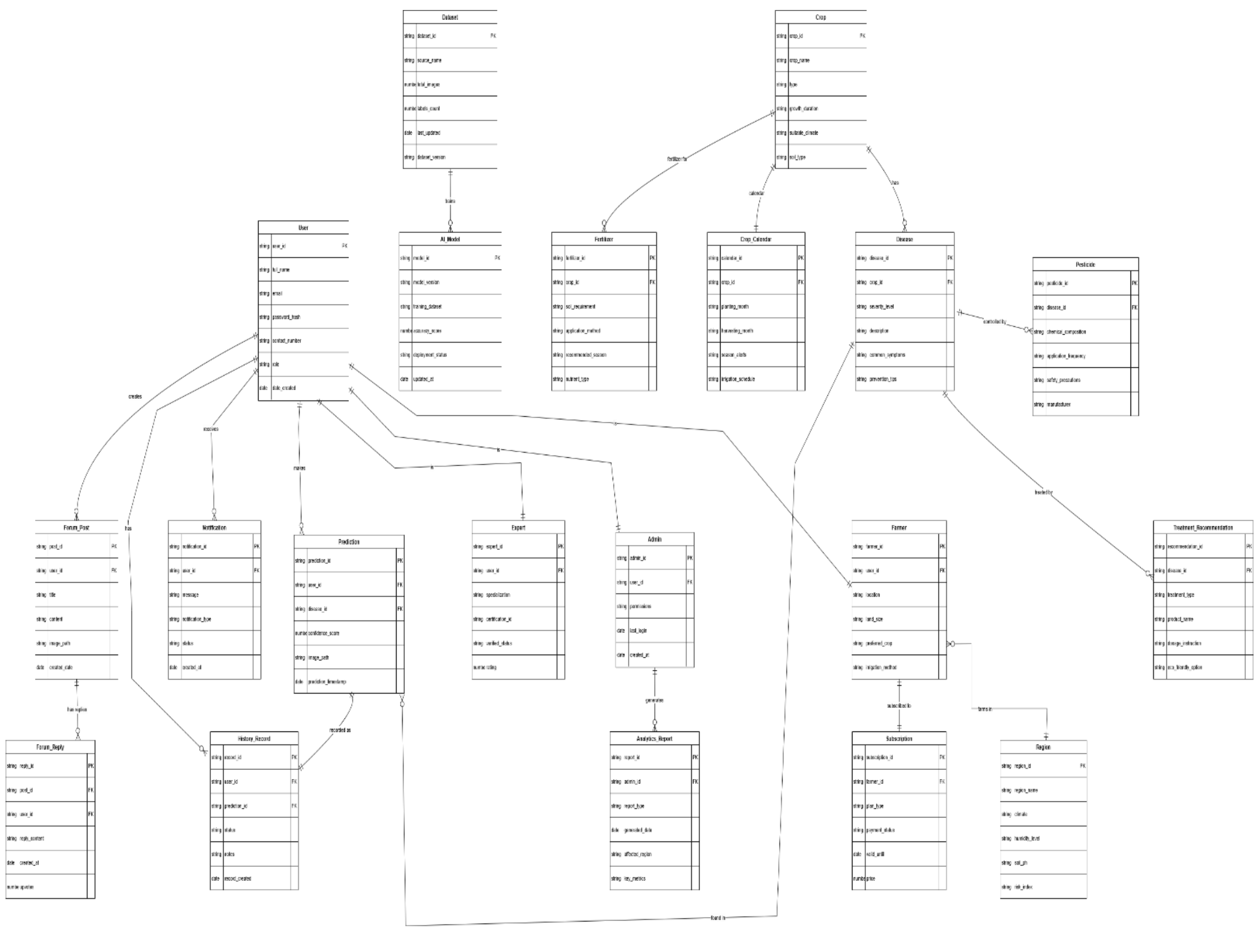
**Cardinality:** Region (1) → Farmer/Prediction (Many)

### xv. Farmer & Subscription

A Farmer may have one active Subscription plan

**Type:** One-to-One

**Cardinality:** Farmer (1) → Subscription (1)



## 3.2.4. Class Diagram

One of the most important objects in the design of objects is the class diagram. It gives a graphical depiction of the classes of the system, their attributes (data they store), methods (operations they perform) and their relationship with other classes. Basically it depicts the roadmap of the system, which is beneficial to the developers as it helps them comprehend how various parts of the system interact, the flow of data in the system, and the allocation of responsibilities. This way, it minimizes confusion in the development process and provides the system with a structure, maintenance and scalability (Shocodevin, 2025; Prepbytes, 2023).

i. Basic Classes and their duties.

1. Crop Class:

**Purpose:** Represents individual crops submitted by farmers.

**Key Attributes:** cropName, growthStage, imageData, diseaseStatus.

**Methods:** uploadImage(), requestDiseaseDetection(), viewRecommendations().

**Explanation:** Centralizing all crop-related data and operations in this class ensures consistency. For example, when a farmer uploads a leaf image, the Crop class handles storing the image and initiating disease detection.

2. Farmer Class:

**Purpose:** Represents farmers who use the system.

**Key Attributes:** farmerID, name, location, contactInfo.

**Methods:** submitCrop(), viewPredictionHistory().

**Explanation:** This class links farmers with their crops and predictions. One farmer can submit multiple crops, creating a **one-to-many relationship**. This setup also allows tracking each farmer's interactions and history within the system.

3. AIService Class:

**Purpose:** Encapsulates the machine learning logic for disease detection.

**Key Attributes:** modelType, trainingData, accuracy.

**Methods:** analyzeCropImage(), generateRecommendation().

**Explanation:** By isolating AI logic in its own class, the system can update models or add new predictive features without affecting other parts of the system. When a Crop object requests disease detection, this class processes the data and returns a Prediction object.

4. Prediction Class:

**Purpose:** Stores AI-generated results.

**Key Attributes:** predictionID, diseaseDetected, confidenceLevel, timestamp.

**Methods:** storePrediction(), retrievePrediction().

**Explanation:** This class maintains a history of predictions, enabling farmers and experts to track trends over time. It also allows analytics to identify frequently occurring diseases or high-risk crops.

5. Admin Class:

**Purpose:** Manages the system and oversees operations.

**Key Attributes:** adminID, role, permissions.

**Methods:** manageUsers(), viewAnalytics(), updateSystemSettings().

**Explanation:** Admins interact with other classes indirectly by monitoring user activity, managing data integrity, and controlling system configurations.

The interaction among classes within the system captures the interactions among various components to bring out the intended functionality. A Farmer may post several Crops which forms a one to many relationship and each Crop passes through the AIService that creates a Prediction object, so that all the crop information is processed in an orderly manner and is stored as a reference point whenever required. Admins monitor all the communication and may request reporting or management of any of the classes. Class diagrams do not only depict relationship diagrammatically, but also act as a guide to implementation in terms of how the code should be structured, how data travels between objects, and how the system can easily be extended, by adding new AI models or other types of users. Good relationships also enhance maintainability where modifications in a single class, including training of AI logic, does not affect other classes. As an example, when a farmer posts a new picture of the crop, the Crop object engages with the AIService to generate a Prediction object which is stored and connected to the farmer, and this illustrates how the object-oriented approach has grouped the operations together on logic, traceability and simplified the debugging process.

### 3.2.5. DFD Data Flow Diagram Level 0

Level 0 Data Flow Diagram (DFD) is an abstract level, high-level view of the system and it shows the flow of data between the key components and users. In contrast to detailed diagrams, Level 0 concentrates on the primary processes without internal logic and subprocesses demonstrating the general workflow of the system and is more comprehensible to stakeholders, developers, and users (Prepbytes, 2023; Shocodevin, 2025).

This system has farmers, experts, and administrators as its main users:

- The frontend side of the system allows farmers to enter crop data including images, type of crop, and a description of the crop growth.

- This input is fed to the AI service which then applies trained models to diagnose potential crop diseases and provide treatment suggestions.

- After the processing, the results are passed to the farmers through the frontend interface.

- The system allows experts to check the predictions of the AI to make additional treatment proposals and update the knowledge base, with the recommendations being accurate and reliable.

- Admins also track system activity, oversee users, and report and make decisions with analytics dashboards.

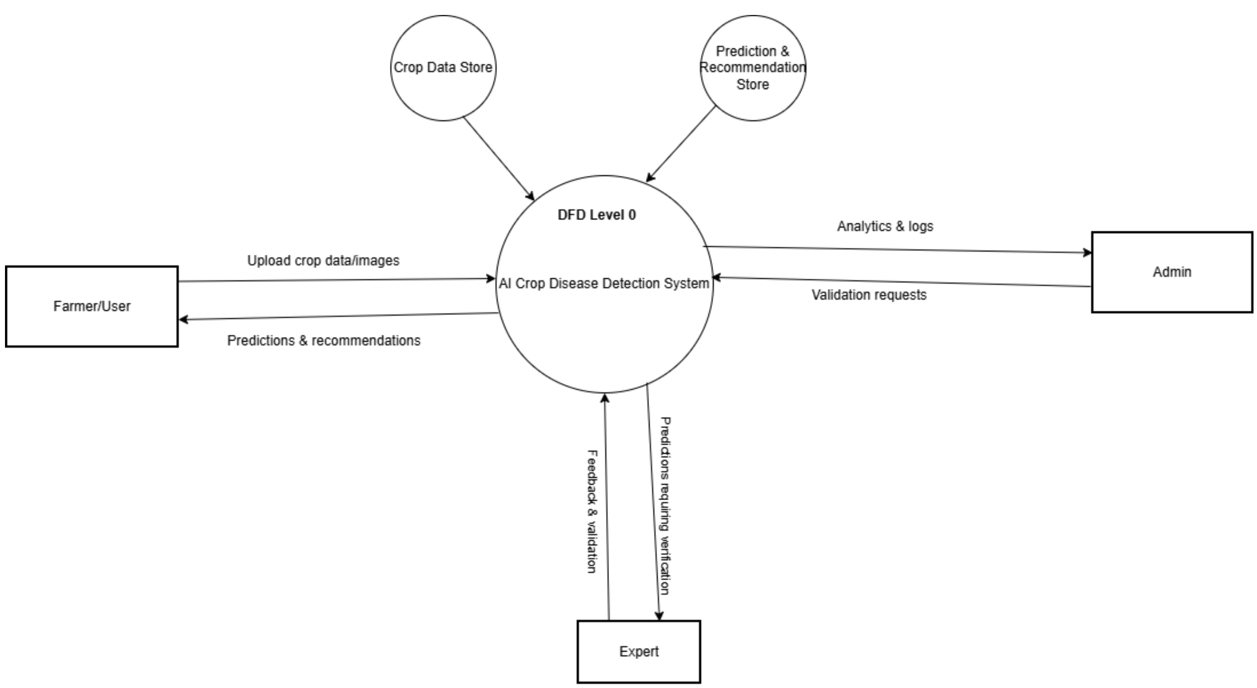



*Figure 3: DFD Level 0*

The Level 0 DFD will focus on the system boundaries which indicate what entities communicate with the system and where the data flows between them. It is essential in keeping track of the key interactions and does not over-determine the design with internal aspects. As an illustration, uploading a picture of a crop by a farmer will prompt the DFD to capture the following flow: Farmer → System → AI Service → Prediction → Farmer and also indicate that Experts and Admins can use the system to verify and supervise the system. This gives a clear guide to the developers on how to introduce the system effectively and also so that all the important processes can be viewed at a glance.

### 3.2.6. Any other UML for overall system can be added here

Additional UML diagrams such as activity diagrams and sequence diagrams were prepared to illustrate detailed workflows and interactions over time. Activity diagrams help in visualizing step-by-step processes like the flow from image upload to prediction delivery while sequence diagrams capture the chronological order of interactions between users, the backend and AI services. These diagrams collectively ensure all stakeholders and developers share a common understanding of system behavior.

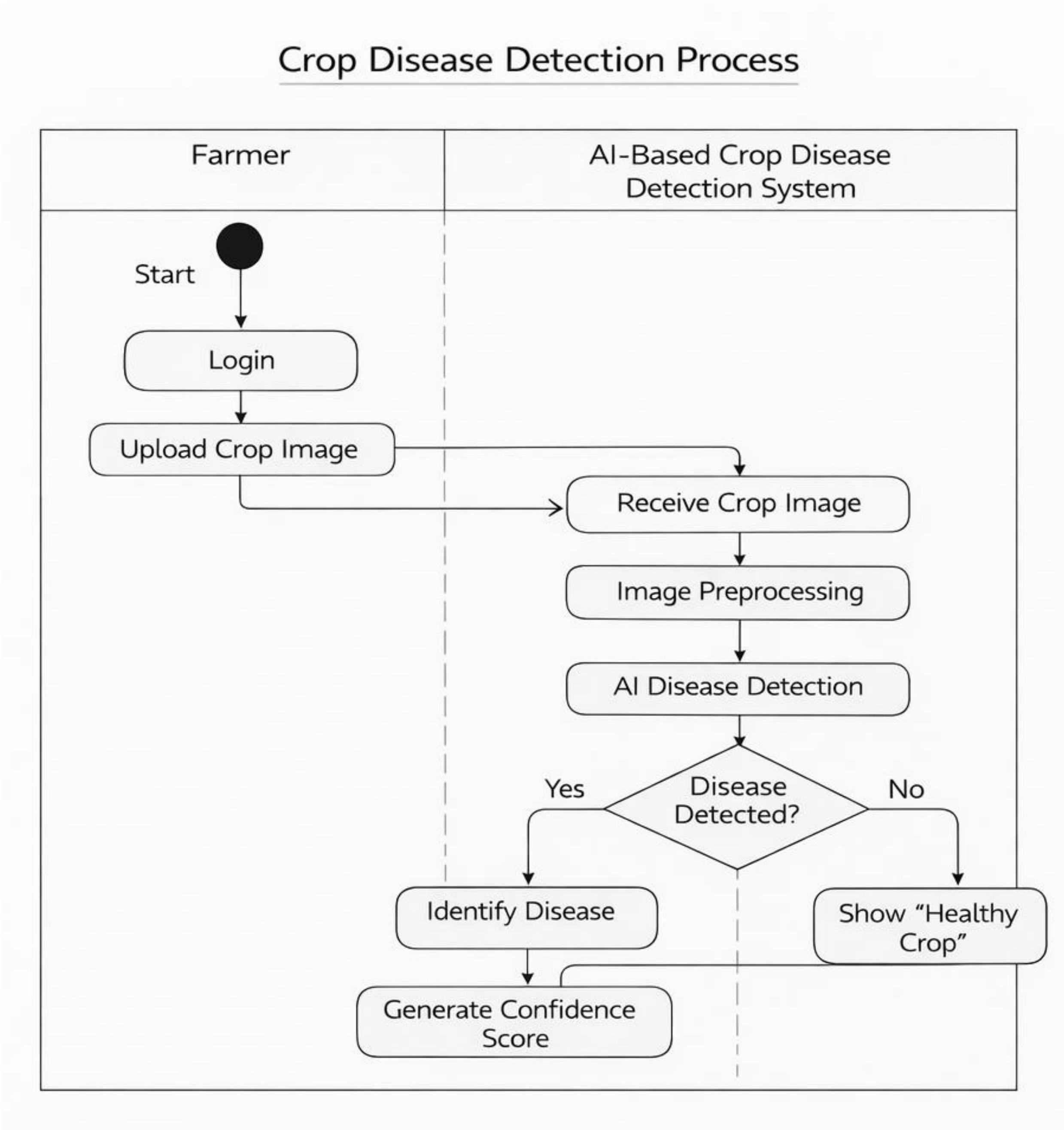
#### 3.2.6.1. Activity Diagram

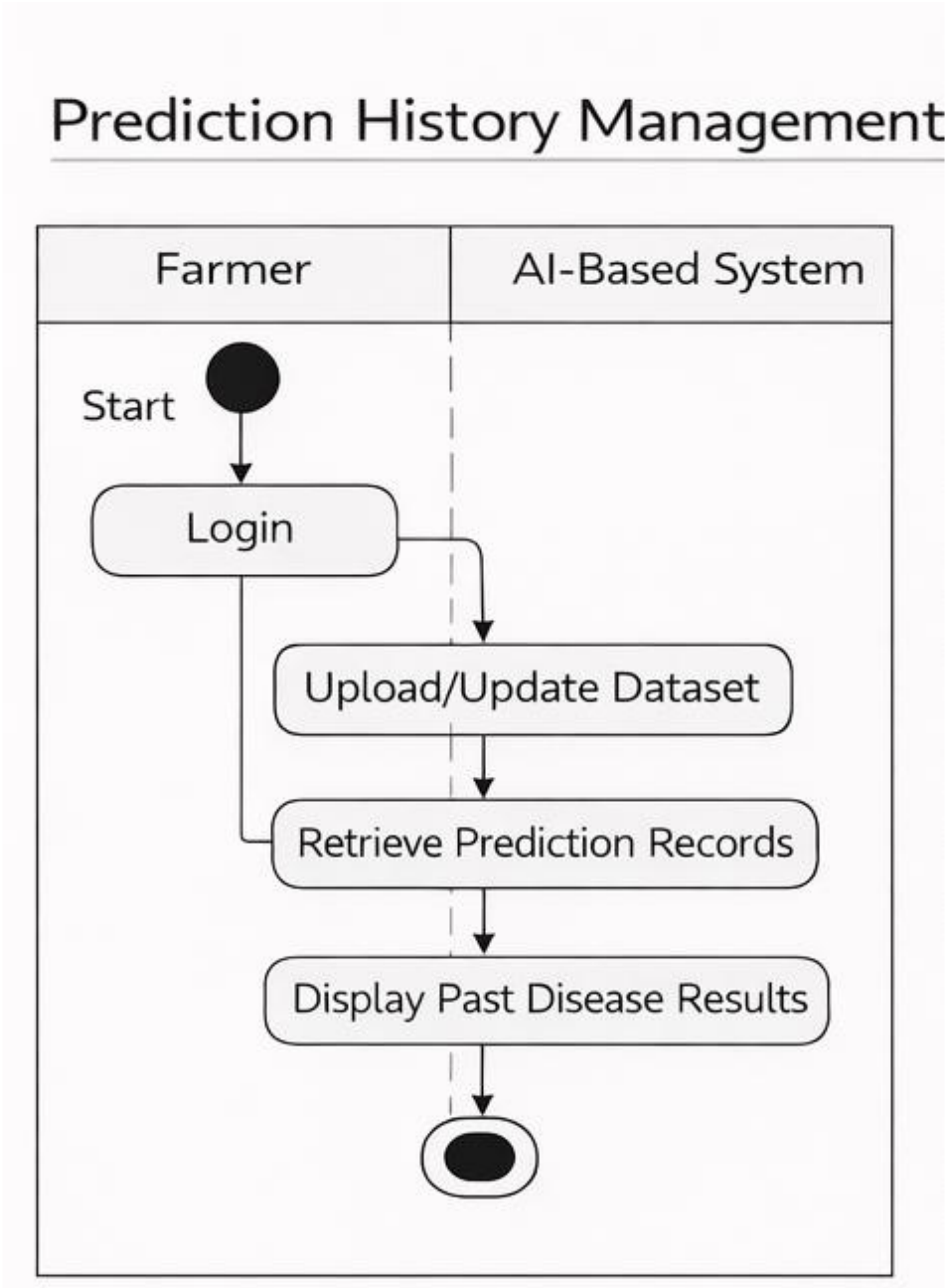
An Activity Diagram is a behavioral diagram in UML that represents the workflow of activities within a system. It shows the step-by-step progression of control from the initiation of a process to its completion, illustrating the sequence of actions, decision points, parallel activities, and synchronization between different tasks (Booch et al., 2005). Activity diagrams are particularly effective for modeling business processes, system workflows, and use case realizations, as they visually clarify how various activities are executed in response to specific events. They provide stakeholders with a clear understanding of the dynamic behavior of the system by illustrating how actions are coordinated and how control transitions between them.

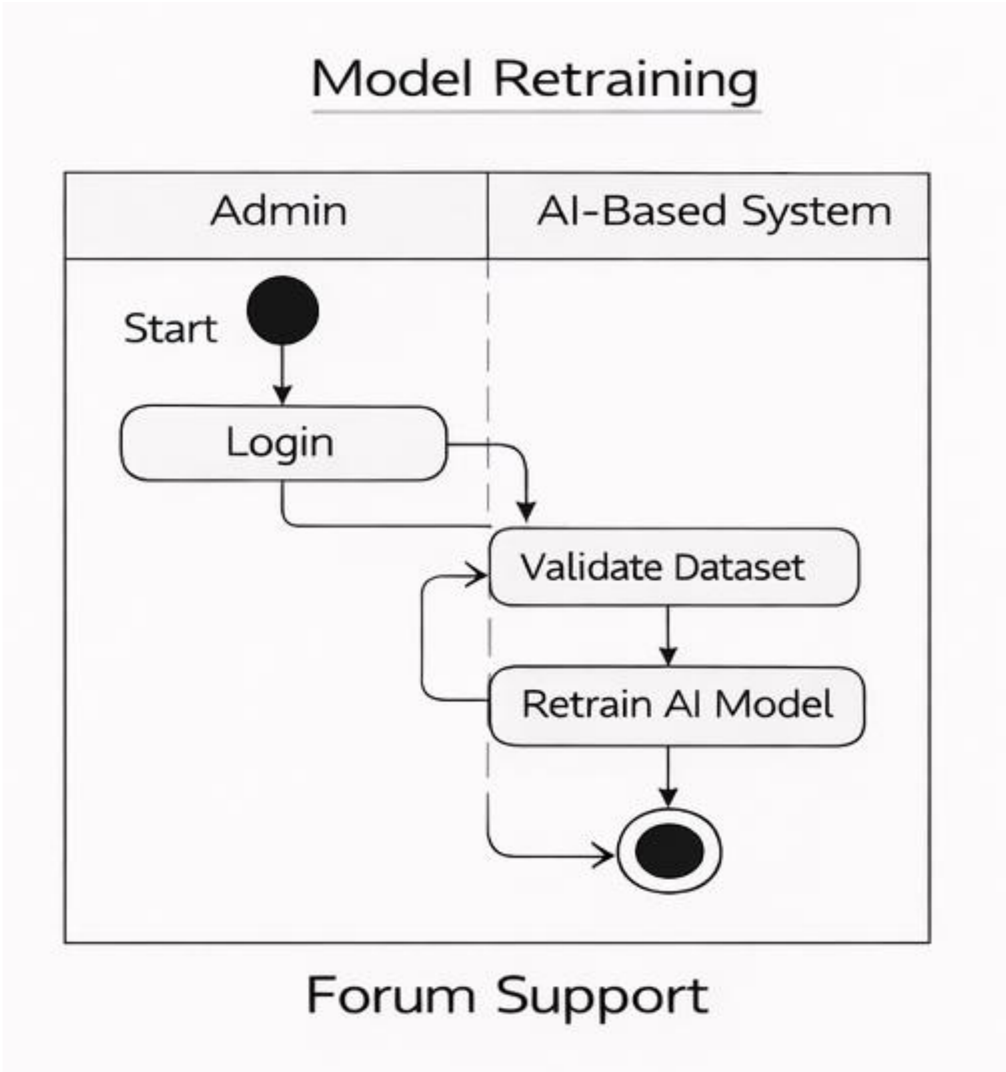
In the AI-Based Crop Disease Detection and Recommendation System, an activity diagram models the workflow beginning with image upload by the farmer, followed **by** image preprocessing**,** disease detection using the AI model**,** result generation**,** and finally the display of treatment recommendations. By representing these steps, the activity diagram helps ensure a clear understanding of system operations, supports effective system design, and enables validation of the processes, thereby enhancing both usability and reliability of the system (Ambler, 2004).

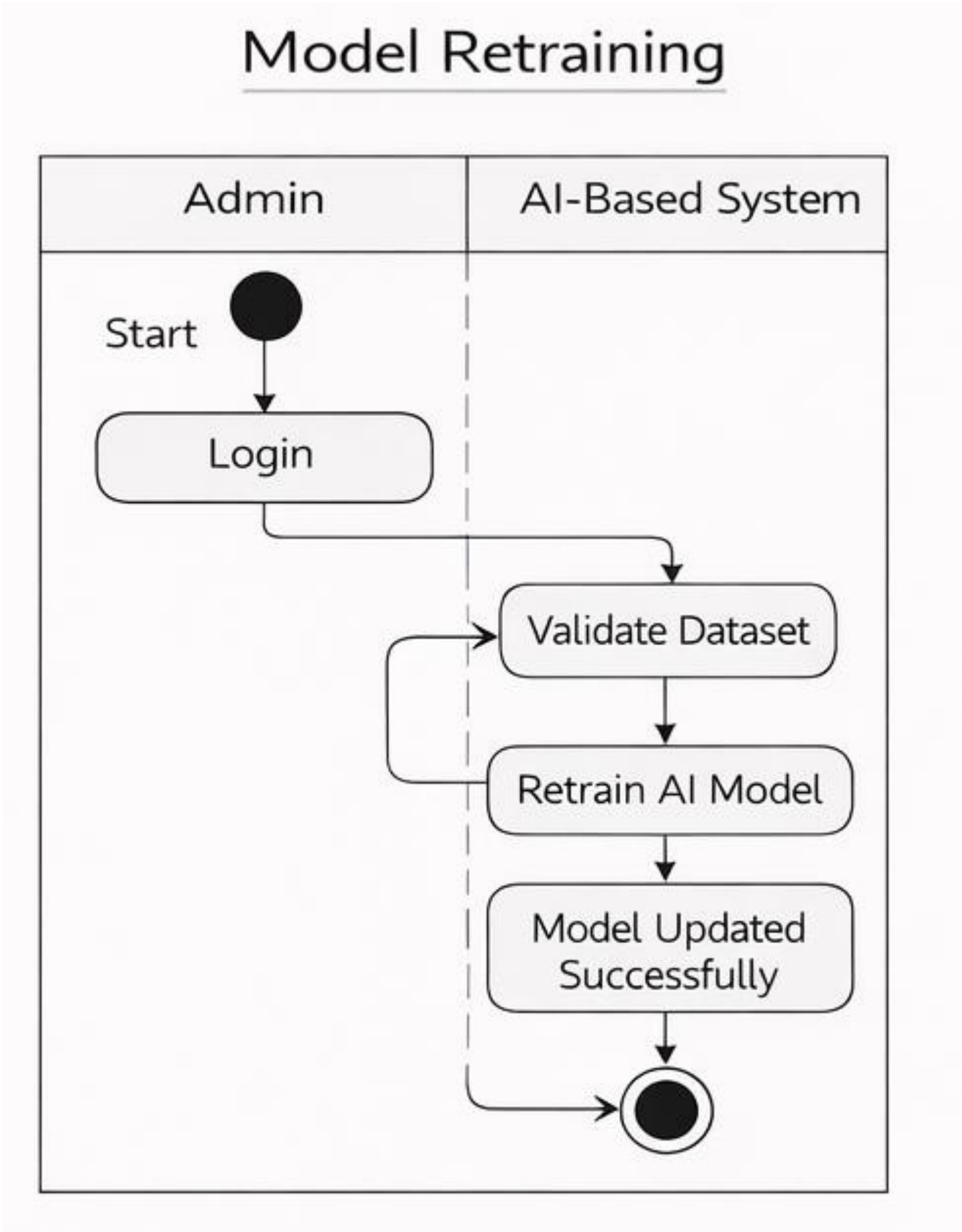
*Table 4: The Notation of Activity diagram*

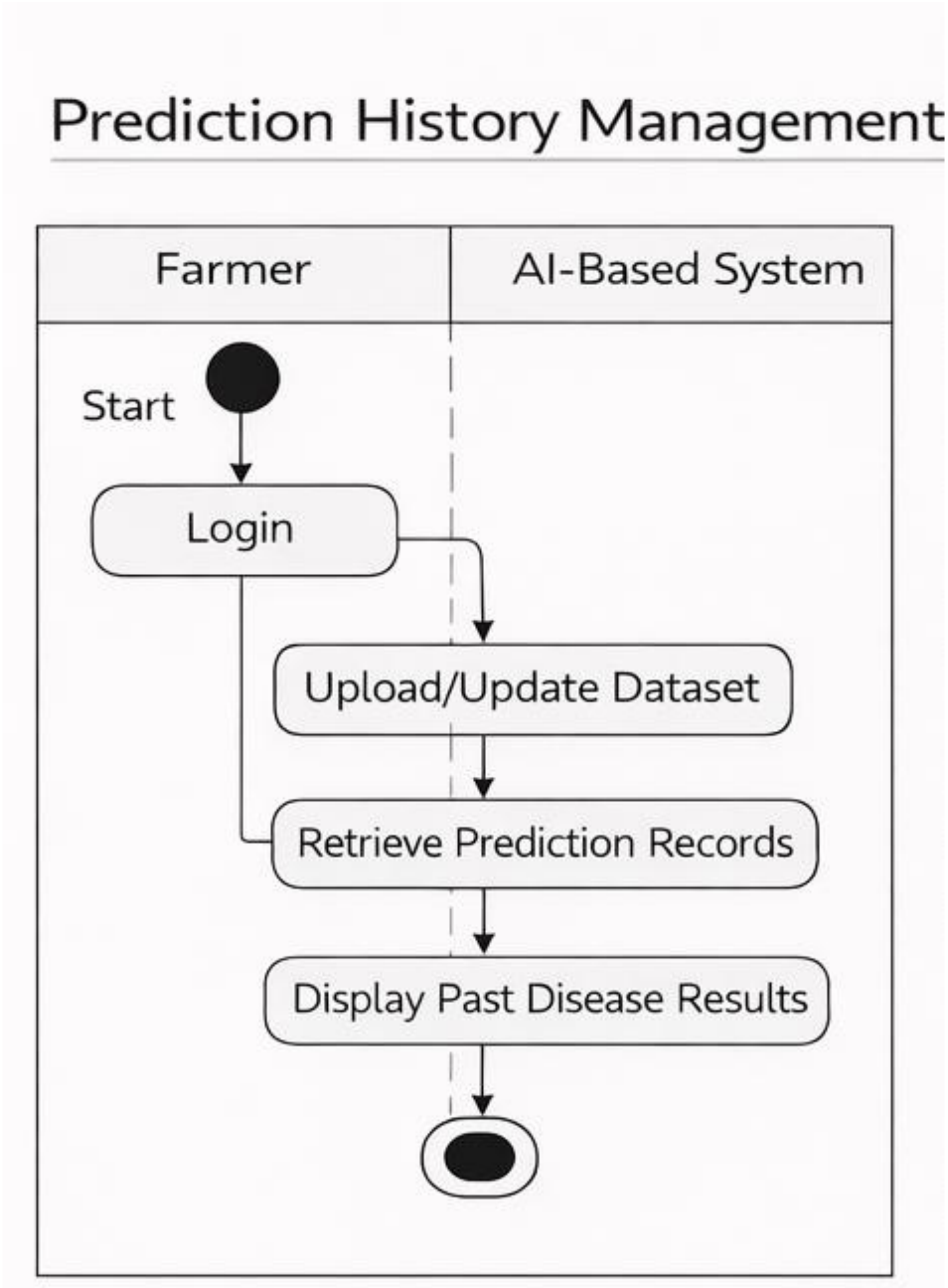
| **Sr. No** | **Name** | **Symbol Description** |
|---|---|---|
| 1. | Start Node | ● (Filled black circle) |
| 2. | Action State | ▭ (Rounded rectangle) |
| 3. | Control Flow | → (Arrow line) |
| 4. | Decision Node | ◇ (Diamond shape) |
| 5. | Fork | ┬ (One line splitting into multiple arrows) |
| 6. | Join | ┴ (Multiple lines joining into one line) |
| 7. | End State | ◎ (Black circle inside a white circle) |


Crop Disease Detection Process
Farmer
AI-Based Crop Disease Detection System
Start
Login
Upload Crop Image
Receive Crop Image
Image Preprocessing
AI Disease Detection
Disease Detected?
Yes
No
Identify Disease
Show "Healthy Crop"
Generate Confidence Score


Prediction History Management
Farmer
AI-Based System
Start
Login
Upload/Update Dataset
Retrieve Prediction Records
Display Past Disease Results


Model Retraining
Admin
AI-Based System
Start
Login
Validate Dataset
Retrain AI Model
Forum Support


Model Retraining
Admin
AI-Based System
Start
Login
Validate Dataset
Retrain AI Model
Model Updated Successfully

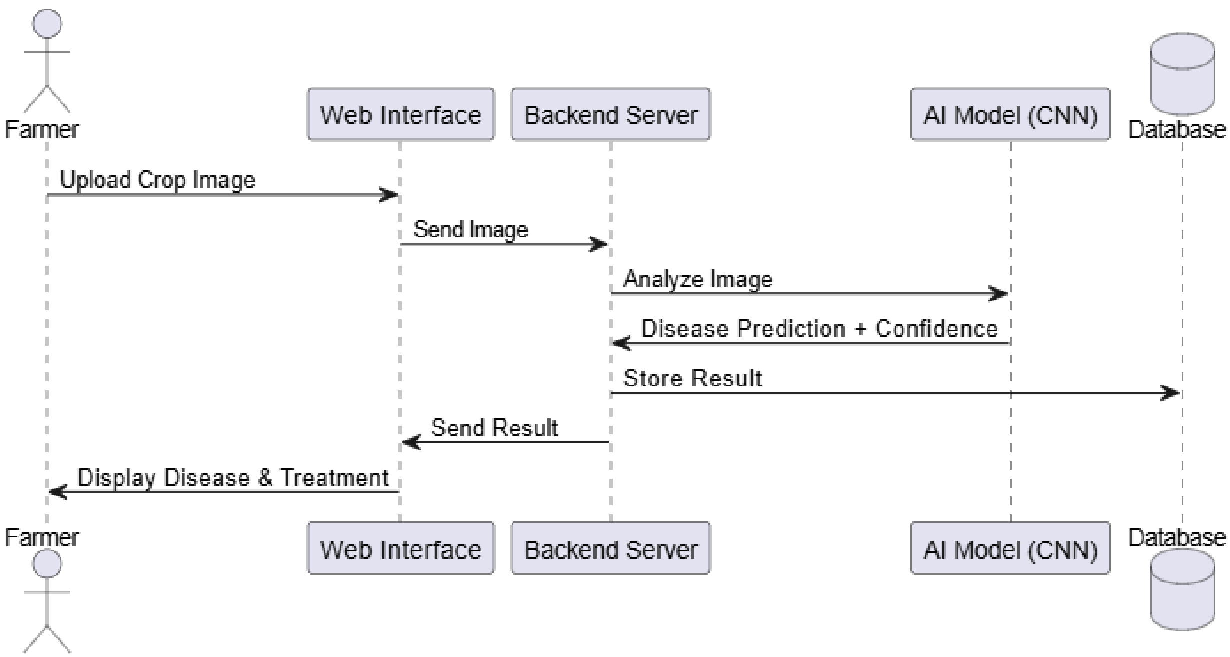



**Aligned User Stories:**
US003, US004, US005, US006, US007, US008, US011

#### 3.2.6.2. Sequence Diagram

A Sequence Diagram is a type of UML interaction diagram that illustrates how objects or system components interact with each other over time. It focuses on the order of messages exchanged between different entities to accomplish a specific functionality or use case. The diagram presents participants as lifelines arranged horizontally, while messages are shown as arrows that indicate the direction and sequence of communication from top to bottom, representing the flow of time (Booch et al., 2005).

Sequence diagrams are particularly useful for modeling dynamic behavior, validating system logic, and understanding how responsibilities are distributed among system components. They help developers and stakeholders visualize system processes, identify dependencies, and ensure that interactions occur in the correct order. By clearly depicting method calls, responses, and control flow, sequence diagrams support effective system design and improve communication among development teams (Sommerville, 2016).




## 3.3. Sprint 1 / Prototype 1 – Login and Register

### 3.3.1. Sprint Planning

The main objective of Sprint 1 was to have a safe and efficient user authentication procedure. This included allowing the users to create an account on the site and later log in using authenticated accounts. Tasks that were identified during the sprint planning session included designing user interface, setting up back end APIs, and integrating database where user credentials are stored safely and making sure that there were validation mechanisms in place. Each task was allocated story points to estimate effort and the team settled on realistic targets that were capable of being attained within the sprint.

## 3.3.2. Design

### 3.3.2.1. High-Level Use Case

The top-level use case of Sprint 1 focused on the user registration and user login. This use case recognizes actors (users) and the main interactions, which are necessary to access the system. At this point, what was required is to find out what the user wants of the authentication process: simple, intuitive and secure method of creating an account and signing in.

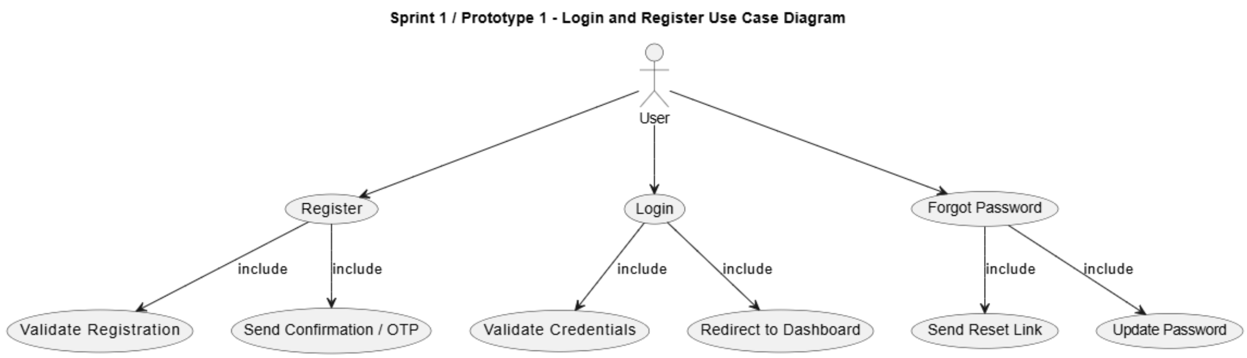



*Figure 4: Sketch of usecase diagram of sprint 1*

**Description:** This use case involves the user getting authenticated (logins and registration).

i. Actors:

- User/Farmer- This is the main user who desires to use the system.

ii. System -The AI Crop Disease Detection Platform in charge of authentication.

**Description:**

This use case is concerned with the operations of the creation of a new account (registration) and the entry into the system with the usage of a login. The system is secure and enables a user to access the system based on the user credentials and feeds them with proper feedback as successful or failed attempts. This is the baseline of the entry gate into the system, which provides the opportunity to further engage AI services and features.

**Preconditions:**

- The user is allowed to access the internet and system interface (web or mobile).
- Registration is done using a valid e-mail or phone number.

**Postconditions:**

- The creation of user account is successful and it is stored in the database.
- Authenticated user-users are allowed into functionalities that are authorized.

**High-Level Flow :**

- User enters into the system login or registration interface.

**For registration:**

- Personal information (name, email, password) is entered by the user.
- System verifies the input and verifies duplicate accounts.
- System develops the account and dispatches the confirmation (email/OTP).

**For login:**

- User provides credentials (email/username and password).
- System checks the credentials using stored data.
- A successful login will redirect a user to the dashboard, whereas a failed one will display an error message.

**Alternate Flows:**

- Forgot Password /Reset: The user sends an email/OTP request to reset his password.
- Invalid Credentials: This is used by the system to block a user and show an error message.

**Business Rules:**

- The passwords should be of appropriate security standards (length, complexity).
- Where there are duplicate registrations with the same email or phone number, they are not permitted.

### 3.3.2.2. Expanded Use Case

The widened use case divides the authentication process into specific steps. To register, it will involve entry of user details, checking email and password format, storing a hash of the credentials in the database, and giving the user a response. To ensure safe access, to do a login, it expounds the process of checking credentials, with invalid attempts, and session/token-based authentication.

**Name of Use Case:** User Authentication -Login and Registration.

**Actors:**

i. **Primary Actor:** User/Farmer - The user interacts with the system in order to access an account or create one.
ii. **Secondary Actor:** System- checks credentials, handles registration, issues confirmation/OTP and redirects users to authorized locations.

**Description:**
This use case outlines the steps to the process of registering a new user account in a secure way and logging in to the system. The system has user identity checks, secure credentials storage and authenticated users can only access the additional capabilities of AI Crop Disease Detection Platform. It is the basis of the access of the users and is essential to security and customized user experiences.

**Preconditions:**

- The system interface (web or mobile application) is available to the user.
- The system can be operational and linked to the database.
- To make a registration, the user is required to give a valid mail or telephone address.

**Postconditions:**

- User account is created and stored in the database.
- Registered users are verified and allowed to access the dashboard or access allowed functions.
- The system keeps a safe log of the attempts to log in and the registration.

**Main Flow (Basic Flow):**

i. Registration Flow:

- User gets to the registration page.
- User gives personal information: full name, email, phone number, and password.

- System checks the input data against completeness, inaccuracy, and originality.
- System records the user information within the database.
- System sends a confirmation mail or OTP to confirm identity of the user.
- OTP or confirmation link is used to verify the account by the user.
- System logs in the account and reroutes the user to the log in page.

ii. Login Flow:

- User accesses the login page.
- User provides credentials (email/ username and password).
- System compares the credentials with the data stored.
- Once authenticated successfully, the system will provide access and will redirect the user to the dashboard.
- In case of any failed authentication, the system will show an error and re-attempt will be prompted to the user.

iii. Forgot Password Flow:

- User clicks on the Forgot Password.
- System asks the user his/her registered email or phone number.
- System sends OTP or a password reset link.
- User authenticates the connection/OTP and offers a new password.
- Password is updated by System and successful reset is checked.

iv. Alternate Flows / Exceptions:

- Invalid Registration Input: The system will remind the user to make corrections in case he/she provides incomplete or invalid data.
- Duplicate Account: In case the email/phone number is registered the system displays it.
- Wrong email or password: In case of wrong email or password, the system denies the user access and shows an error message.
- Expired OTP / Confirmation Link: System should ask the user to reissue an OTP or a link.

v. Business Rules:

- The passwords should be of the minimum security standards (e.g., length, complexity).
- Customers are not allowed to open accounts using identical emails or phone numbers.
- Checking of accounts has to be done in a stipulated time.
- All login attempts and registration should be audited by logging them in the system.

vi. Assumptions:

- Users have internet access.
- Instruction in verification emails or OTP is used correctly by users.
- The database of the system is safe and accessible at all times.

vii. Notes:

- This is a basic use case that will be referenced by other modules within the system that will need authentication.
- This process is combined with security considerations, which include password encryption and safe delivery of OTP.

### 3.3.2.3. Activity Diagram

The activity diagram represents the movement process of the operations on the login and registration. It begins with the input of the user and then goes through checking and validation procedures and then results in permission or error messages. This diagram will assist the team to comprehend the chain of events and the possible decision-making stages of the user journey.

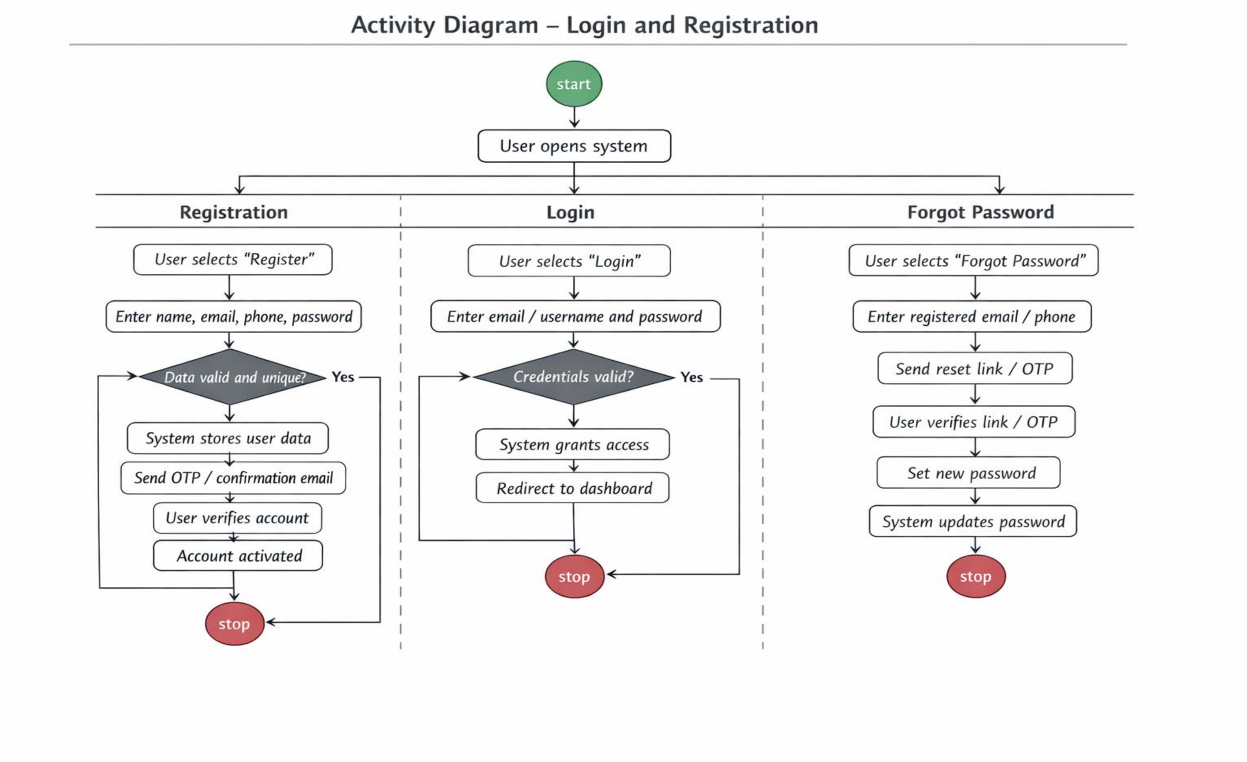



*Figure 5: Activity Diagram of sprint 1*

## 3.3.2.4. Sequence Diagram

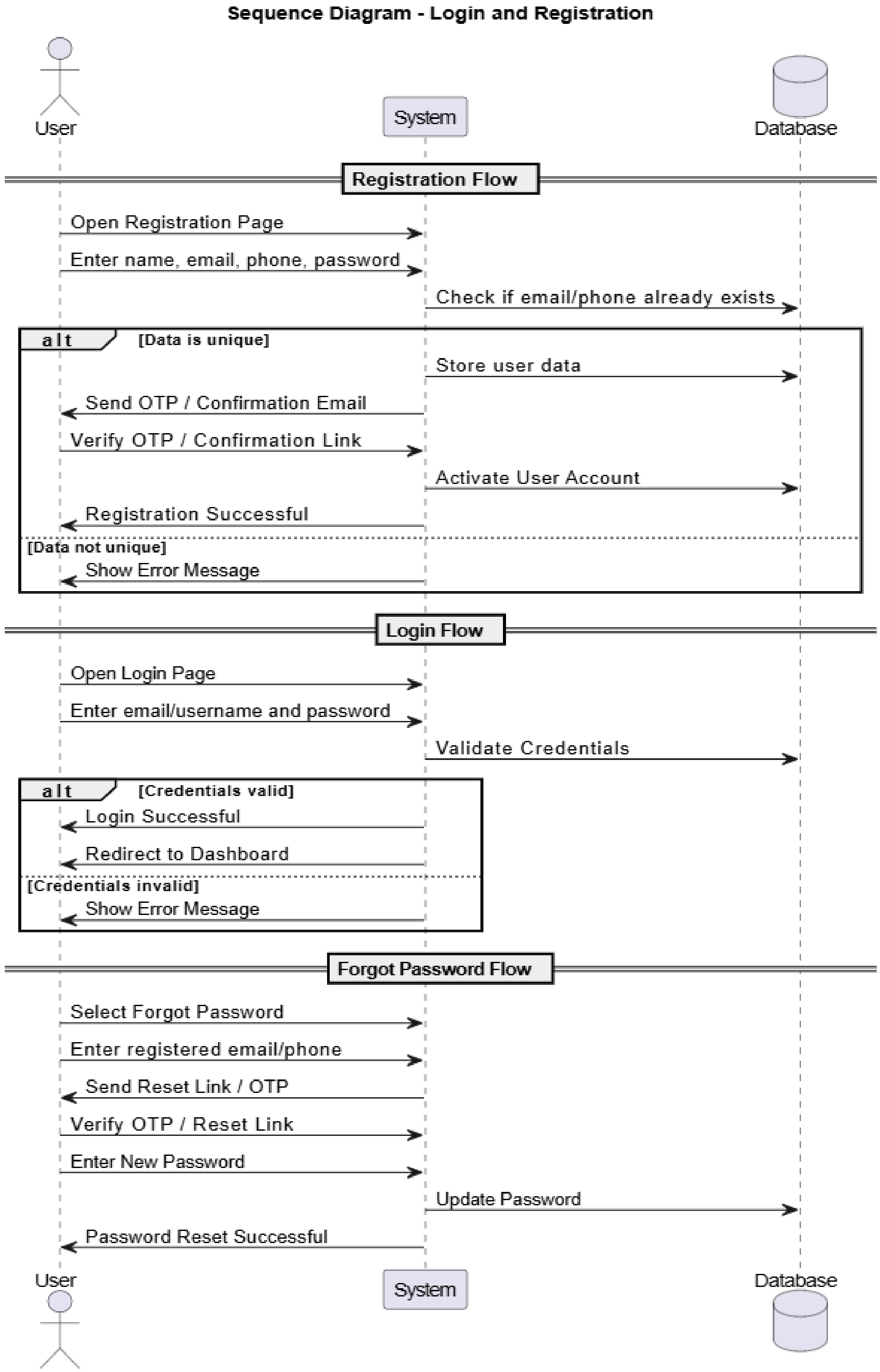


*Figure 6: Sequence diagram of sprint 1*

The sequence diagram shows the communication between the user interface (UI) and backend server and database when an individual gets authenticated. It brings out the sequence of requests and reply like entering the details of logins, validating the data in the database, and providing the response in authenticated outcomes.

### 3.3.2.5. DFD Level 1

The Level 1 Data Flow Diagram (DFD), gives the detail of how data is flowing in the system to authenticate the users. It presents user input, user processing by the backend services, databases, and user output in the form of success message or failure message. This guarantees clarity as far as the management of information is concerned.

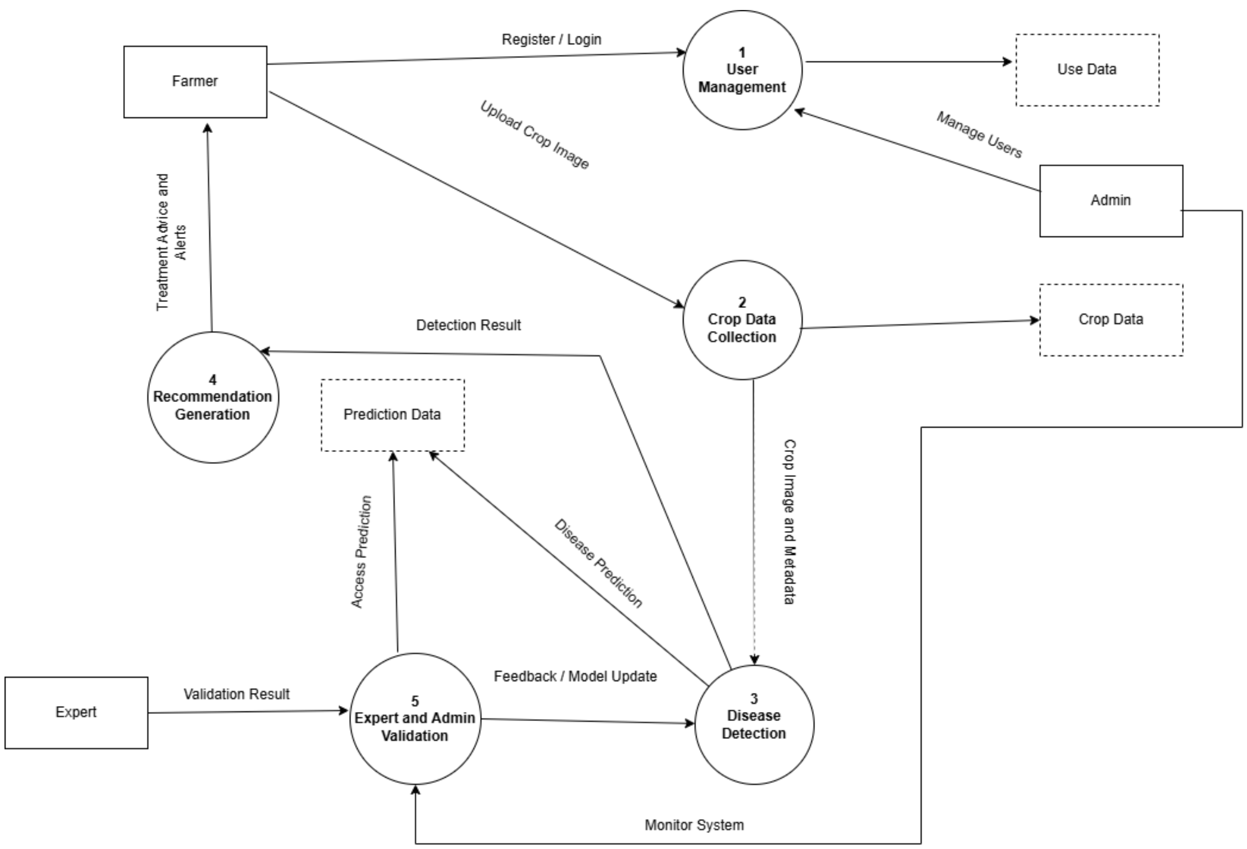



*Figure 7: DFD Level 1 of Sprint 1*

### 3.3.2.6. Other UML Diagrams

Flowcharts and mind maps were made to support the main diagrams and display the reasoning and brainstorming of authentication workflows. These visual tools assisted the team in explaining complicated processes and organizing task performance in an effective way.

### 3.3.2.7. Wireframe

The login and registration screens were created in low-fidelity wireframes. These wireframes were layout-centric, input-field centric, button centric, and bare navigation centric without going into the details of styling. This was aimed at prototyping the interface as soon as possible so that it could be tested by the initial user.

### 3.3.2.8. Mockup

This was followed by high fidelity mockups which included refined user interfaces using real colour schemes, typography as well as design components. These mocks were utilized to check visual design decisions and get ready to frontend development.

## 3.3.3. Development

### 3.3.3.1. Frontend

The frontend was developed with the help of React and interactive and responsive elements were built on the login and registration form. Form validation was done, error handling and user feedback were added to provide an easy user experience.

#### 3.3.3.2. Backend

APIs, which dealt with registration, login and authentication, were developed using Node.js and Express. Password hashing, JWT-based authentication, and input validation were used as security measures to ensure the protection of user data.

#### 3.3.3.3. Mobile

The design was responsive and PWA-compatible to allow users to access the platform without any issues on both web and mobile platforms and provide a similar user experience regardless of the screen size.

#### 3.3.3.4. AI

Though Sprint 1 did not focus on AI, the first dataset analysis and model planning were done in preparation of further sprints when image-based disease detection would be introduced.

### 3.3.4. Sprint Review

During the sprint review, the team had shown the working system of login and registration. Every task was completed on time and successfully, and the sprint objective to have secure user access was reached.

### 3.3.5. Sprint Retrospective

The retrospective found some small UI improvements in the form of improved placement of error messages and improvements in input field validation which would be implemented in subsequent sprints.

### 3.3.6. Velocity Chart

The speed of sprints was estimated using the number of points achieved on stories, and this will give the potential of the team to deal with further sprints.

### 3.3.7. Burndown Chart

The burndown chart has shown the timely execution of tasks and familiarized an understanding of the progress in achieving the sprints and all the scheduled items were provided within the time frame.

## 3.4. Sprint 2 / Prototype 2 Disease Detection.

### 3.4.1. Sprint Planning

The primary objective of Sprint 2 was to adopt the disease detection component with AI that will help farmers place an image of the crop and get precise forecasts. At the sprint planning event, the team was divided into front-end task of uploading images, backend of processing image data, integration of AI inference, displaying the result and logging predictions into the database. Each task had points allocated by how challenging the AI model integration would be, and this was due to providing a realistic schedule. The sprint was intended to show the presence of a functional disease detection besides ensuring that the user experience remained smooth.

## 3.4.2. Design

### 3.4.2.1. High-Level Use Case

On the upper level, the application of this sprint was disease detection through images. The participants consist of the farmers who post pictures of their crops and are given forecasts and treatment plans. The AI model is also presented as a virtual actor in the system that works on the images and makes predictions.

**Name:** Disease Detection and Recommendation Use Case.

**Actors:**

i. Primary Actor: Farmer/User – posts the images or crop data of the crop to identify disease.

ii. Secondary Actor: System (AI Crop Disease Detection Platform), Expert (optional validation).

**Description:**

This application scenario is aimed at empowering users to detect the diseases of crops through the AI system. Users post crop pictures or take input data and the system uses an AI-based analysis to diagnose the disease and prescribe suitable treatment. Predictions can be checked and verified by experts or administrators to be correct. This is the main functional module of the system and it directly helps farmers to manage the diseases.

**Preconditions:**

i. The user is authorized and gets into the system.

ii. The system is connected to the trained AI model and the databases of crops.

**Postconditions:**

- The system gives the predicted disease of the uploaded data of the crop.
- Treatments or remedial actions are provided to the user based on recommendations.
- History of prediction is saved to be referred to and analysed.

**High-Level Flow (Summary):**

- The module of detecting the disease is accessed by the user.
- Crop data and pictures are uploaded by users.
- The system checks input information to ensure that it is complete and qualitative.
- The data are processed by AI which predicts the disease.
- System produces treatment prescriptions (organic, chemical, and fertilizer prescriptions).
- The user is provided with predictions and recommendations.
- Predictions can also be checked by experts or administrators.

**Alternate Flows:**

- In the event of low quality uploaded data/image, the system would remind the user to re-upload.
- In the event that the AI model is not good enough to make a prediction of a disease, it will raise the red flag so that it can be looked into by the expert.
- The system shows an error message and gives an option of a second attempt in the event of network errors or server errors.

**Business Rules:**

- Disease detection features can only be accessed by authenticated users.
- Posted data on crops should be of good quality (e.g. resolution, format).
- The predictions and recommendations should be recorded and audited and learnt by the system.

**Notes:**

- The identified use case will form the core of the value proposition of the system, which will offer AI-based and real-time crop disease diagnostics.
- The recommendation module is integrated to make the farmers have actionable insights.

### 3.4.2.2 Expanded Use Case

The extended use case describes every stage of the disease detection procedure:

- The farmer posts a picture of a crop leaf either through the web or mobile platform.
- The system authenticates the image nature and dimension.
- The picture is uploaded to the backend, and it is preprocessed by the AI model (e.g. resizing, normalization).
- With the help of the AI model, the disease is predicted and a confidence score is calculated.
- This prediction is recorded in the database with the ID of the farmer and the time of this prediction.
- The system provides the user with the result showing disease name, severity and recommended treatments.
- This use case makes the system strong and easy to use as all the interactions, such as uploading images and showing results, are considered.

**Name of Use Case**: Disease Detection and Recommendation.

**Actors:**

i. Primary Actor: Farmer/User- uploads the images or data of the crops to identify disease.
ii. Secondary Actor: System -AI Crop Disease Detection Platform which processes crop data and produces forecasts.
iii. Option Actor: Expert – will review predictions to validate in case of low AI confidence.

**Description:**

In this use case, the system allows farmers to identify the disease of the crop in question and get recommendations on what should be done. Crop images or data are entered by the user and processed by the system on the basis of AI models. According to the forecast, the system proposes treatments or warns the user against further expert intervention. This application case will mean that farmers will be able to make the right decision at the right time to avoid losing yields and preserving crop health.

**Preconditions:**

i. The user has been logged into the system.
ii. The AI model and database of crops are functional.
iii. User can upload images or feed crop data.

**Postconditions:**

- Prediction of disease is acquired to the uploaded data of crops.
- The user is given treatment prescriptions (organic, chemical, fertilizer guidance).
- The system contains prediction and recommendation history.
- Predictions are marked to be reviewed by experts in case of need.

**Main Flow :**

- User navigates onto the Disease Detection module.
- Crops are uploaded by users or crop data is typed in by users.
- System checks the input against completeness, format and quality.
- The data is processed with an AI, which forecasts possible disease(s).
- System comes up with treatment recommendations.
- System provides the user with prediction and recommendations.
- System records the transaction and a history of prediction is kept.
- When the confidence of AI is low, prediction is flagged to be reviewed by an expert.
- Predictions are flagged through expert reviews and it offers feedback about the validation.

**Alternate Flows / Exceptions:**

- Bad Quality Image/Data: System asks user to re-upload good data.
- Uncertain Prediction: The AI model is not able to give a confident diagnosis→ system flags to be reviewed by an expert.
- System Error / Network Issue: System gives an error message and gives the user an option to retry.
- Unauthorized Access: Disease detection is not accessible to unlogged in or unverified users.

**Business Rules:**

- Disease detection can be accessed by only authenticated users.
- Crop data needs to be of a minimum quality to be processed by AI (e.g. resolution, file format).
- The audit and future model enhancement are recorded on all predictions, recommendations, and expert validations.
- A critical evaluation by experts will be required in the event that the AI confidence is low relative to an established threshold.

**Assumptions:**

- The users are guided by system instructions on how to submit images/data correctly.

- System has a trained and updated AI model with the capability of identifying crop diseases.
- When the AI predictions are not clear, the expert can be used to verify them.

**Notes:**

- This is one of the central use cases of the system and the key value proposal.
- The combination with recommendation and notification modules areas will provide actionable insights in a manner that is useful.
- The system is also used in gathering data which will be used in future training of AI and model enhancement.

### 3.4.2.3 Activity Diagram

Activity diagram Sprint 2 represents the flow of the disease detection process. It begins with the posting of a picture by the farmer, proceeds to the validation stage, preprocessing, AI inference and finally the results are presented and the history of the prediction is stored. Error-prone areas like the invalidity of the image format or mistakes in the deception of the AI are well indicated to respond to the errors.

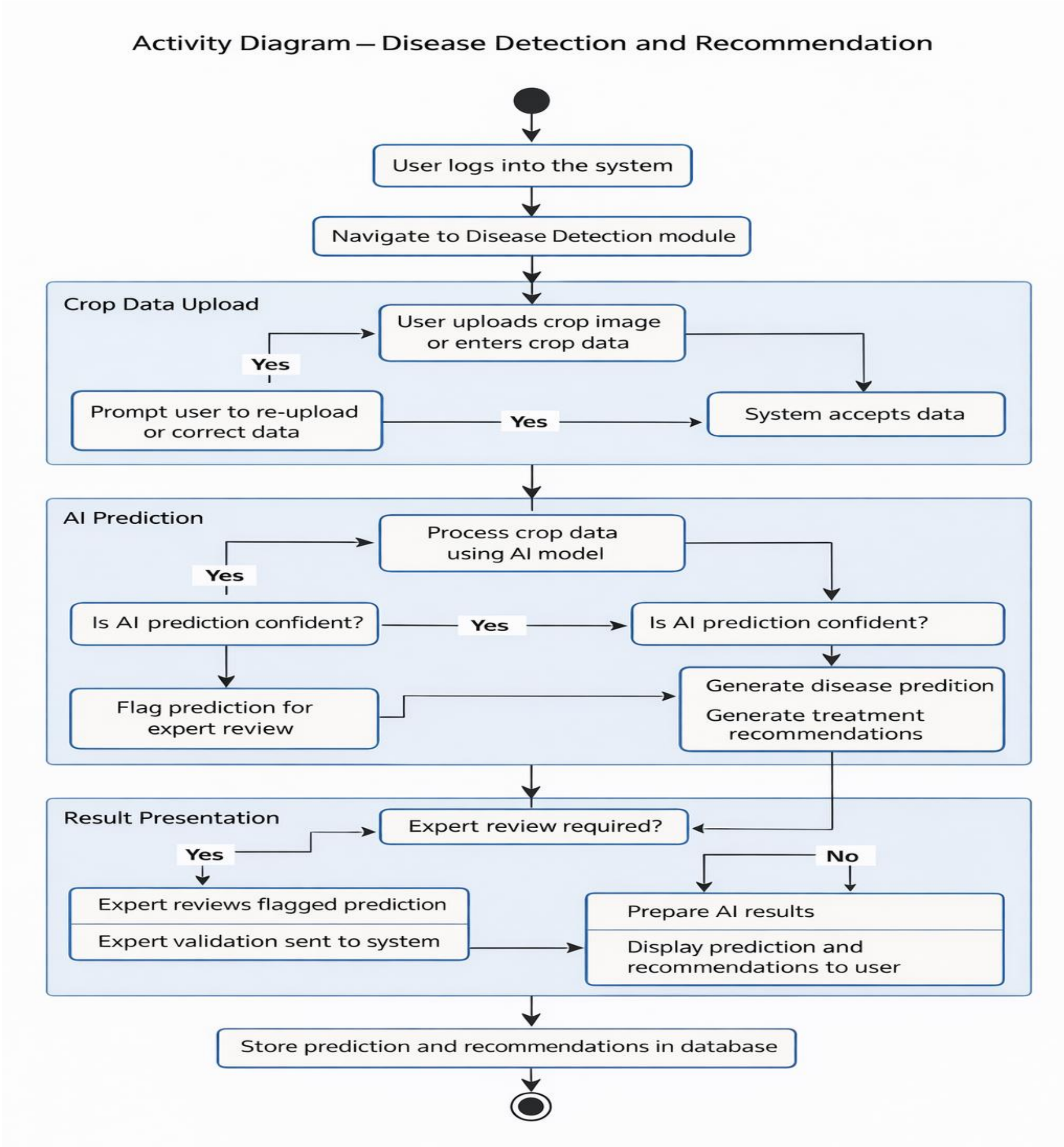



*Figure 8: Activity Diagram of sprint 2*

### 3.4.2.4 Sequence Diagram

The sequence diagram depicts the interaction of the user interface, the backend server, AI module and the database in a chronological manner. It documents the major processes involved in sending the image to the backend, pre-processing, using AI model, prediction, and updating database. This assists in the knowledge of data flow and integration among components.

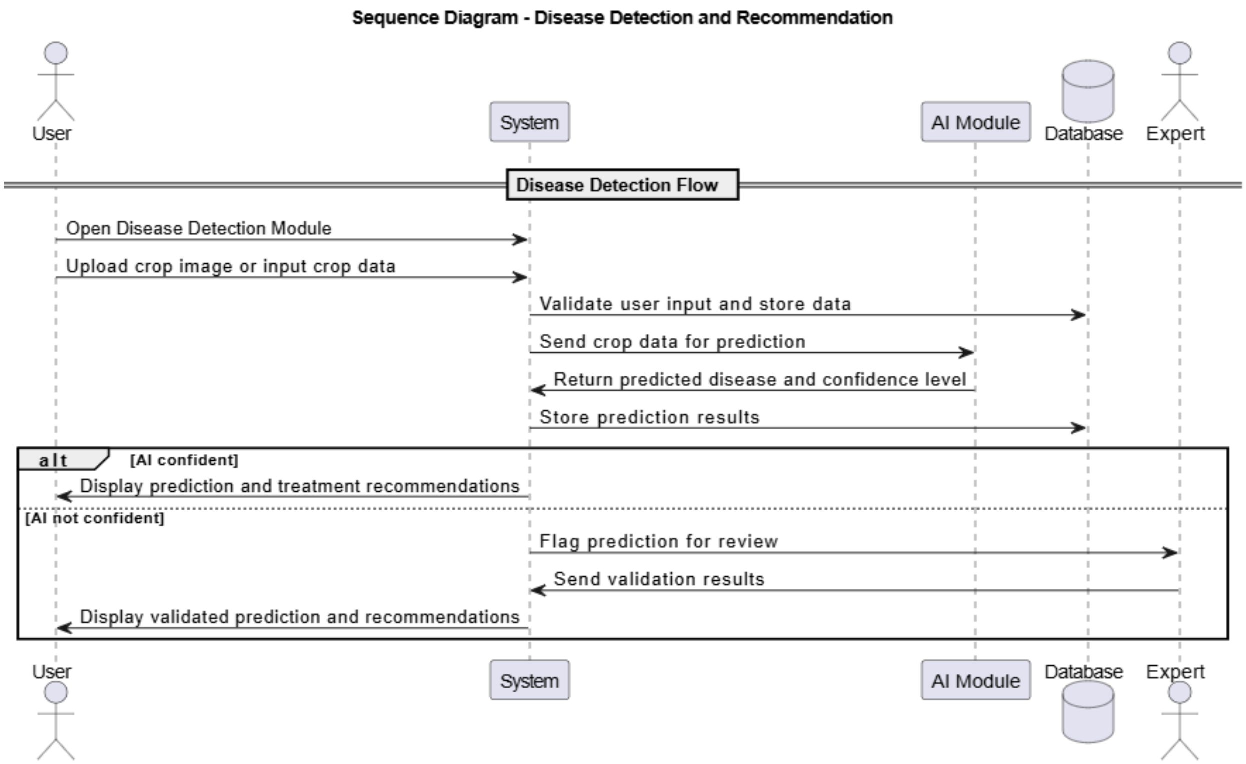



*Figure 9: Sequence diagram of sprint 2*

## 3.4.2.5 DFD Level 1

The Level 1 Data Flow Diagram gives a more detailed picture of the flow of data through the system as it is detected in the system during disease detection. It depicts the user input, processing, and storage in the database and output of predictions and recommendations. This will ascertain that there is clear insight on how sensitive information (images and results) is managed safely.

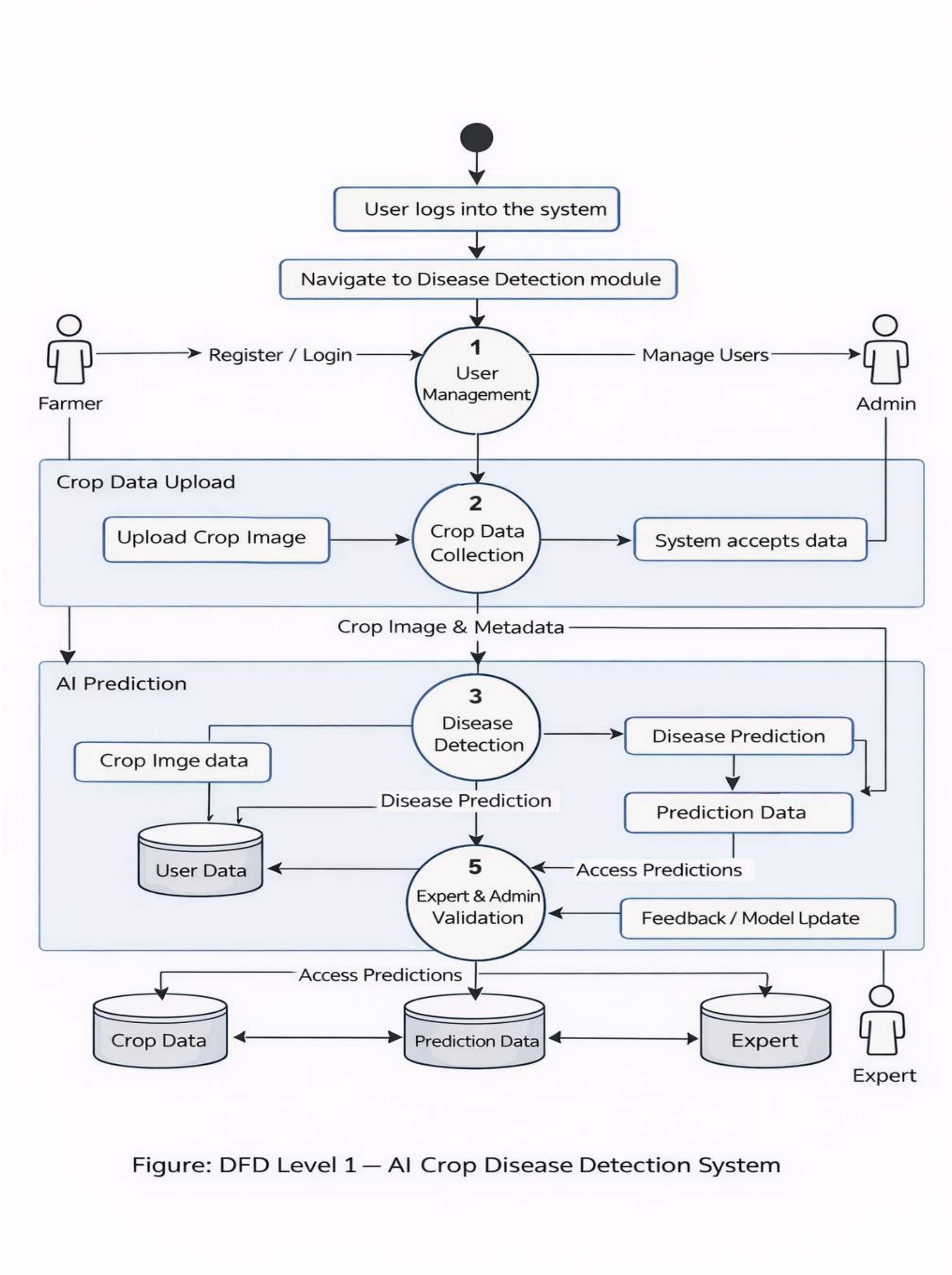



*Figure 10: DFD level 1 of Sprint 2*

### 3.4.2.6. Other UML Diagrams

Other flowcharts and mind maps were developed to illustrate AI preprocessing, model inference, and workflows of error-handling. They were useful in optimizing model integration and creating a seamless interaction between AI and the backend.

### 3.4.2.7. Wireframe

The image upload interface and the page with prediction results were created using low-fidelity wireframes. It was usability-oriented and did not confuse the farmers with the process of choosing pictures, uploading them, and seeing the results.

### 3.4.2.8. Mockup

The mockups were high-fidelity and offered smooth UI designs of the image upload screen and the result display. They provided visual representations of the confidence of the prediction, suggested treatments and past prediction records that they could easily refer to.

### 3.4.3. Development

### 3.4.3.1. Frontend

React was used to add image upload controls, drop functionality, and dynamic prediction results display to the frontend. Checking of validation It was introduced to make sure only supported image formats are uploaded.

#### 3.4.3.2. Backend

The APIs to get received images, preprocess them, call the AI model, and provide predictions were developed using Node.js and Express. The security system was safeguarded by rate limiting and input validation in order to prevent the system abuse.

#### 3.4.3.3. Mobile

The mobile interface was also responsive and PWA-enabled so that farmers could post pictures with their smartphones or tablets. The interface was performance optimized and it was fast even on slower connections.

#### 3.4.3.4. AI

To process images and analyze their content to predict crops diseases, the AI module was installed based on the Convolutional Neural Network (CNN) model. Preprocessing of the data such as resizing, normalization, and data augmentation was done. The first training and validation findings were considered, and the model was tuned to precise predictions with high confidence.

### 3.4.4. Sprint Review

At the sprint review, the team also displayed a prototype of a disease detector that was fully operational. Farmers were able to post leaf pictures, get proper forecasts and see recommended remedies. The sprint was able to achieve its target of introducing AI-based disease detection into the system.

### 3.4.5. Sprint Retrospective

The retrospective revealed the places of work improvement, including making the models more responsive, including more user-friendly error messages, and supporting more image formats. These were to be dealt with in subsequent sprints.

### 3.4.6. Velocity Chart

Story points were used to monitor velocity and assist the team to have a more accurate view of progress and plan future sprints.

### 3.4.7. Burndown Chart

Burndown chart captured the workflow as all the tasks were completed in a timely manner, and it ensured that the team remained on track.

# 4. Progress Analysis

Project management analysis is a basic tool of project management since it allows orderly monitoring and assessment of project activities alongside the set plans, schedules, and objectives. It makes sure that development is being in check, it is measurable and it is also on track with what the stakeholders have expected. The progress analysis, especially, is relevant in software and AI-based projects as the development process is rather iterative, and technical risks are linked to complex system elements (Sommerville, 2016).

Structured planning and tracking tools in this project were the Gantt chart and progress table, which were used to provide progress analysis. These tools give elevated and close-up views of the project progress and make informed decisions across the lifetime of development.

## 4.1. GanttChart

The Gantt chart was created at the stage of project planning to establish project milestones, timeframes of tasks, and dependencies. It is a visual representation of the sequence of actions since the analysis of the requirements to the system implementation and the integration of AI. Kerzner (2017) argues that Gantt charts are useful planning tools since they will provide a clear picture of the sequencing, and time allocation of tasks, which allows planning project activities to be coordinated much better.

The Gantt chart has been the tool of constant monitoring during the project and not a onetime planning document. This has been done to compare actual progress with planned timelines in order to detect deviations at an early stage. Such a proactive strategy helps take corrective measures on time, including redistributing efforts or rescheduling the main tasks, before delays affect key milestones (PMI, 2021).

Moreover, Gantt chart can be used to address task dependencies by ensuring that the prerequisite activities, i.e., system design and data preparation, are accomplished prior to the start of the dependent activity i.e. AI model training and system integration. This regulated planning eliminates development risks and helps in efficient management of workflow, which is critical in development projects of sophisticated systems.

## 4.2. Progress Table

Whereas the Gantt chart gives a macro level picture of the project time line, the progress table gives a micro level, task oriented evaluation of the development activities. The progress table records work completed, in progress and pending since most of it is normally classified into completed, in progress and pending. This method gives a chance to accurately monitor the development results and enhance reporting transparency (Schwalbe, 2019).

The progress table is also especially helpful when it comes to tracking parallel and iterative work, i.e. user interface refinement, backend development, and AI dataset preparation. Such activities do not necessarily have a simple representation in terms of timelines only, and they may have varying rates. Through a progress table which is always updated, project stakeholders can easily spot the bottlenecks, gauge workload allocation, and also ensure that, all the planned features are coming as scheduled.

Moreover, the progress table offers quantifiable facts of the project progress, which is essential to academic assessment and official project reviews. It promotes accountability since the deliverables are well defined and the tasks left are also well indicated hence leading to good project governance.

## 4.3. Analysis

The Gantt chart and the progress table analysis revealed that the project is on track as per the initial plan. The basic initial phases such as requirement gathering, system architecture design, database modelling, and initial prototype development have been successfully done. The stages provide a stable technical base on which other stages of development can occur (Pressman and Maxim, 2020).

The project is currently at the AI integration stage, and the tasks involved in this phase are model creation, training, validation, and integration with the core system. This step adds more technical complexity and this phase must be tested thoroughly to guarantee accuracy, reliability and performance of the system. Nevertheless, the development continues workably, and critical risks and deviation of the schedule have not been observed.

Continuous progress analysis means that risks are spotted in time, resources are utilized effectively, and project goals could be met. However, overall, the results show that there was proper planning, execution and discipline in the project management, which puts the project in a good position to be successful.

# 5. Further Work

Despite the fact that the modern stage of the project has already managed to create a working system base and resulted in the integration of AI, there are a number of significant aspects that can be developed in the future. They need to be improved to achieve better system intelligence, usability, scalability, reliability, and applicability to the real world. The next steps of work are the continuation of the improvement of the AI model, reinforcing the data analytics, and user engagement, as well as the further improvement of automation and system performance and reaching the status of full deployment. In sum, all these developments are intended to convert the system into an efficient prototype to a robust and production-oriented solution.

## 5.1. Elaborated Model Training and Lifelong Learning.

The first possible direction of work in the future is the continued training and optimisation of the AI model. Though the given model gives the first predictions with the acceptable level of accuracy, it can be improved considerably with the exposure to bigger and more different data. Heavy methods of model training like hyperparameter optimisation, ensemble learning, and transfer learning can be implemented to enhance prediction accuracy and generalisation (Goodfellow, Bengio and Courville, 2016).

The ability to include data that will be obtained in the real world in the process of using the system will allow learning continuously and making improvements. Since regions and seasons have different agricultural conditions, disease distribution, and other environmental factors, retraining on a regular basis will enable the model to be relevant and true with time. The iterative learning technique has been highly suggested to artificial intelligence-based decision-support systems to guarantee the effectiveness in the long-term (Mitchell, 1997).

The further analysis of the model will also incorporate the advanced performance measurements, e.g. precision, recall, F1-score, receiver operating characteristic (ROC) curves. These measures give more insight into the performance of classification in addition to the simple accuracy, especially in imbalanced datasets which are typical in agricultural disease detection (Provost and Fawcett, 2013).

## 5.2. Improved Analytics Dashboards and Data Visualisation.

The other significant direction of work in the future is the creation of all-encompassing analytics dashboards. These dashboards will inform the stakeholders with valuable data based on the system information such as trends in the occurrence of diseases, accuracy of prediction, and user activities, and the performance of the system. Simplification, intuitive interpretation of complex data may be supported with the help of visual analytics tools, including bar charts, line graphs, heatmaps, and geographical mappings (Few, 2013).

To administrators, analytics dashboards will facilitate successful monitoring of the system, evaluation of its performance, and making decisions that are based on data. Trend analysis can assist with early detection of disease outbreaks and seasonal trends in the case of agricultural experts and researchers. Such are the data-oriented insights that add superior practical value of the system as they convert raw data into usable knowledge (Han, Kamber and Pei, 2012).

## 5.3. Notification systems and Intelligent Alert Systems.

The combination of intelligent notification and alert mechanisms is also planned in the future work. These systems will automatically notify the user about the crucial events like disease diagnosis responses, treatment prescriptions, unusual levels of risk, or updates of the system. The notifications can be sent in various formats, such as email, text message, or even in-app notifications, which will make them accessible and timely.

The Early warning systems are exceptionally useful in farming areas where there is a significant crop loss reduction when early intervention is performed. Proactive decision-making can also be facilitated by intelligent alerts with their level of prediction confidence or environmental threshold, and help the user develop trust in the system (Sommerville, 2016). The option to customise notification settings will also enhance user experience, as it will enable a person to regulate the nature and volume of received alerts.

## 5.4. Local, Teamwork, and Sharing of Knowledge.

The next step in the evolution of work is the addition of community and collaboration capabilities to the system to encourage the involvement and intention to use it by the user. These can be discussion groups, professional consultation courses, question and answer sections and peer-to-peer interaction areas. These capabilities offer users an incentive to exchange experiences, consult with professional advice and communicating best practices in respect to crop management and disease prevention.

Community-based systems improve both collaborative learning and building trust because they integrate AI-based recommendations with expert knowledge (Wenger, 1998). The measurement of moderation and the validation of specialists will be crucial to guarantee the accuracy of information and eliminate the dissemination of fake news. Through facilitating collaboration, the platform is capable of changing into a holistic knowledge ecosystem, instead of being a single diagnostic tool.

## 5.5. Scalability, Optimisation of performance, and Infrastructure Enhancement.

With the increased use of a system, the work in the future should be done in the context of scalability and performance optimisation. These comprise enhancing the backend architecture, optimising the database queries, increasing the efficiency of APIs and caching to minimize response time. There will be the need of load balancing and horizontal scaling policies to help mitigate the augmented user traffic without damaging system reliability (Bass, Clements and Kazman, 2013).

The implementation of cloud-native infrastructure and containerisation technology could also contribute to the increased scalability and flexibility of deployment. These methods allow being very resource management effective, fault tolerant and able to scale cost-effectively, which is essential in a real-world system deployment (Buyya et al., 2019).

## 5.6. Security, Privacy, and Regulatory Compliance.

Securing the systems and privacy of data will also be the priority of the future improvements. Since the platform will deal with user-generated and potentially sensitive farming information, it is important that strong security protocols are in place. These are secure authentication system, role based access control, encryptions of data and periodical security audits.

It is important to ensure that the appropriate data protection regulations are observed so as to ensure that user confidence in the system and credibility of the system is preserved. The principles of privacy-by-design must be implemented in further development in order to reduce the risks of data exposure and guarantee the ethical treatment of data (ISO/IEC, 2011). Enhancing security will also ensure that the system is secured against unauthorised access, data breach, and cyber threats.

## 5.7. Complete Implementation, testing and verification.

The last step in future employment is deployment of the entire system and large-scale real-life testing. This is comprised of rolling out the system into a production server, pilot testing on real users and getting systematic feedback that can be used to improve further. The real operating conditions will contribute to the evaluation of the system usability, reliability, and effectiveness, and the user acceptance testing and monitoring of the system will assist in that (Pressman and Maxim, 2020).

The evaluation of any system after deployment will be significant in detecting the limitations and tweaking the system features. An iterative improvement will be promoted with the help of continuous feedback loops, which will guarantee the system will develop as user requirements and environmental conditions are changing.

## 5.8. Summary

To sum up, the next wave of work will be to upgrade the system into a mature, scalable, and intelligent platform with an ability to produce long-term value. The system can be trained with high levels of AI, provide analytical visualizations, smart notifications, collaboration, robust infrastructure, high security, and complete deployment to attain realistic impact and sustainability. These intended developments indicate an evident path of constant enhancement and show the best practices in the creation of software and AI systems.

# 6. References

The references list is very essential in sustaining the quality and credibility of this report. All materials used and cited during the study are cited using the Harvard Anglia referencing style that is commonly used by UK institutions of higher learning. This ensures that there is uniformity in referencing format, and readers find it easy to trace and confirm the sources of original work.

References used are a mix of scholarly and professional literature i.e. textbooks, peer-reviewed journal articles, conference papers, industry reports and respectable online sources. All of the sources have been critically chosen to reinforce theoretical underpinnings, methodological, system design, and analysis used in the report. There is an in-text citation in which the original authors of ideas, models, frameworks, and technical concepts that are being discussed are mentioned.

The use of the Harvard Anglia style also builds the academic integrity as all the borrowed ideas, quotes, and paraphrased materials are referenced. The practice prevents plagiarism and can be considered good research behaviour. Moreover, proper referencing supports validity of the report because it indicates that the conclusions and design choices are based on existing research and best practices in software engineering, artificial intelligence as well as project management fields.

On the whole, the references list gives a clear account of intellectual sources that were used to develop the project and justify the confidence and intellectual quality of the work.

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- Sommerville, I., 2016. Software Engineering. 10th ed. Harlow: Pearson Education Limited.


# 6. Appendix

The supplementary material presented in the appendix does not interfere with the flow of the main body of the report. It also consists of meticulous evidence of the process of creation, technical artefacts, and documentation created during project lifecycle. Such materials have been added in order to make it more transparent, show the practical implementation and give verifiable evidence about the progress and result attained.

Iteration documents and evidences are documents that are typically acquired through iterative processes.

## 7.1. Iteration Documents and Evidence

These are documents that are usually obtained using iterative processes.
This subsection contains full supporting material which shows the development and evaluation that was done in the project in an iterative manner. The appendix presents verifiable statements on planning, implementation, and review actions performed in several cycles of implementation, which is an expression of agile development approach.

The documentation contains sprint planning and sprint review documents, and they contain iteration objectives, allocated tasks, and anticipated deliverables. These reports show the way project requirements were converted into development goals to be implemented systematically as well as how such an evaluation at the end of every iteration was carried out. Iteration logs also capture the development of what can be seen as improvements carried out with regard to technical issues, stakeholder feedback, and changing project requirements.

Furthermore, pre-survey tools and feedbacks are also given to show how preliminary user needs and expectations were determined and confirmed. The survey tools such as structured questionnaires were aimed at collecting information about user requirements, the challenges, and features preferences to the proposed system. The survey responses to the same provide empirical data to substantiate major design and functionality changes, which ought to be made at the initial stages of development. The fact that both the instruments and their responses are included makes the system transparent and it helps to make sure that the system design was based on user-centred research and not assumptions.

Other materials produced in the development process like configuration files, testing evidence, validation output, and iteration summaries are added where necessary. The sum of the iteration documents and supporting evidence gives a full audit trail of the lifecycle of project development, which allows the evaluators assess the development process and the results of the project development in a systematic and transparent way.

**Online Prey-Survey Form**

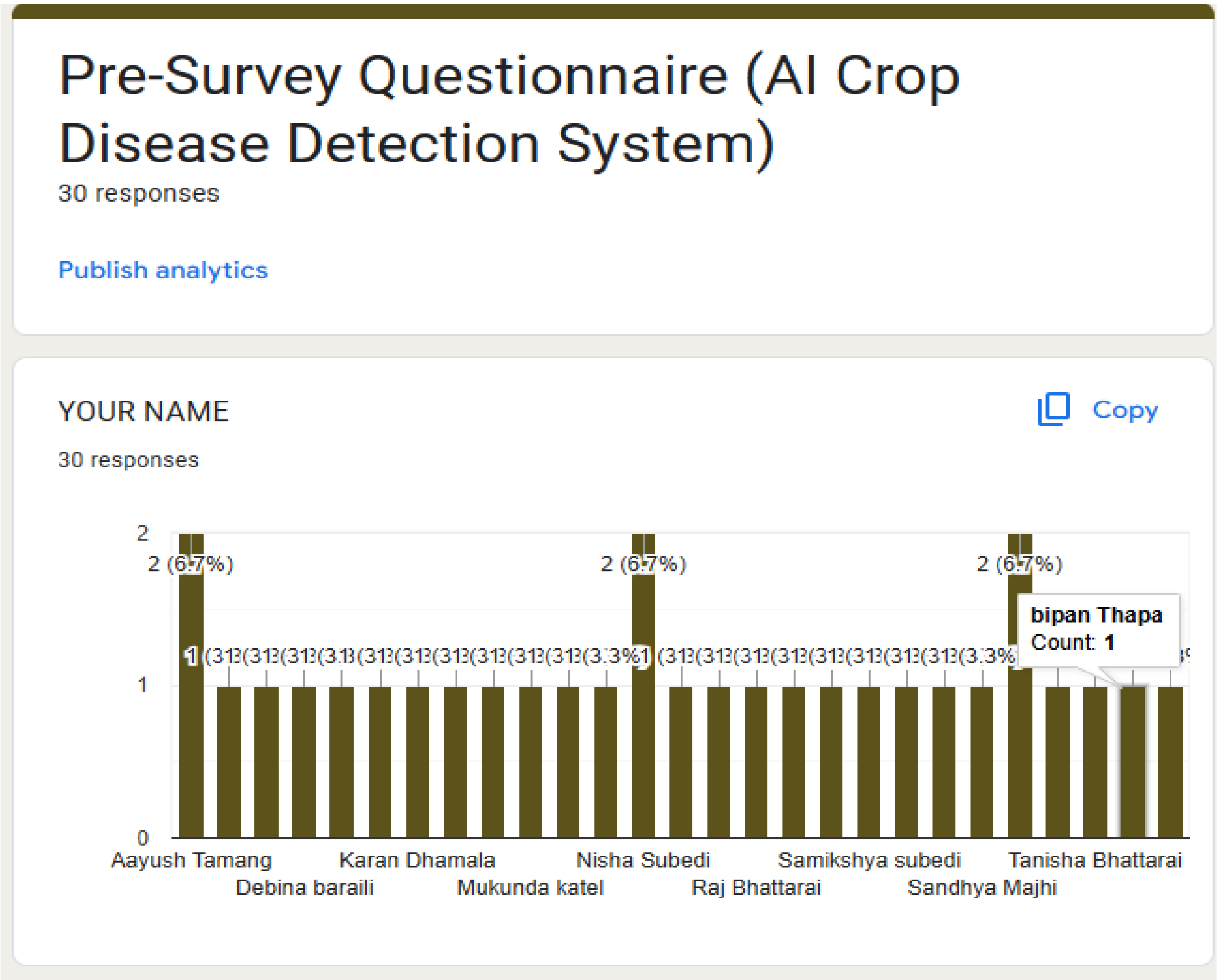
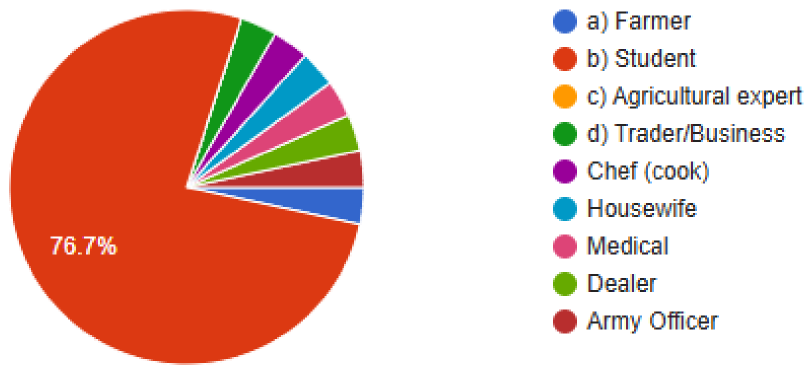



*Figure 11: Online Pre-Survey Form*

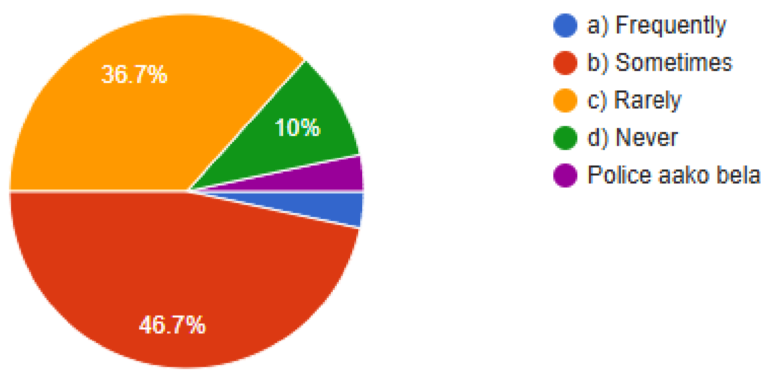
1. What is your occupation ?

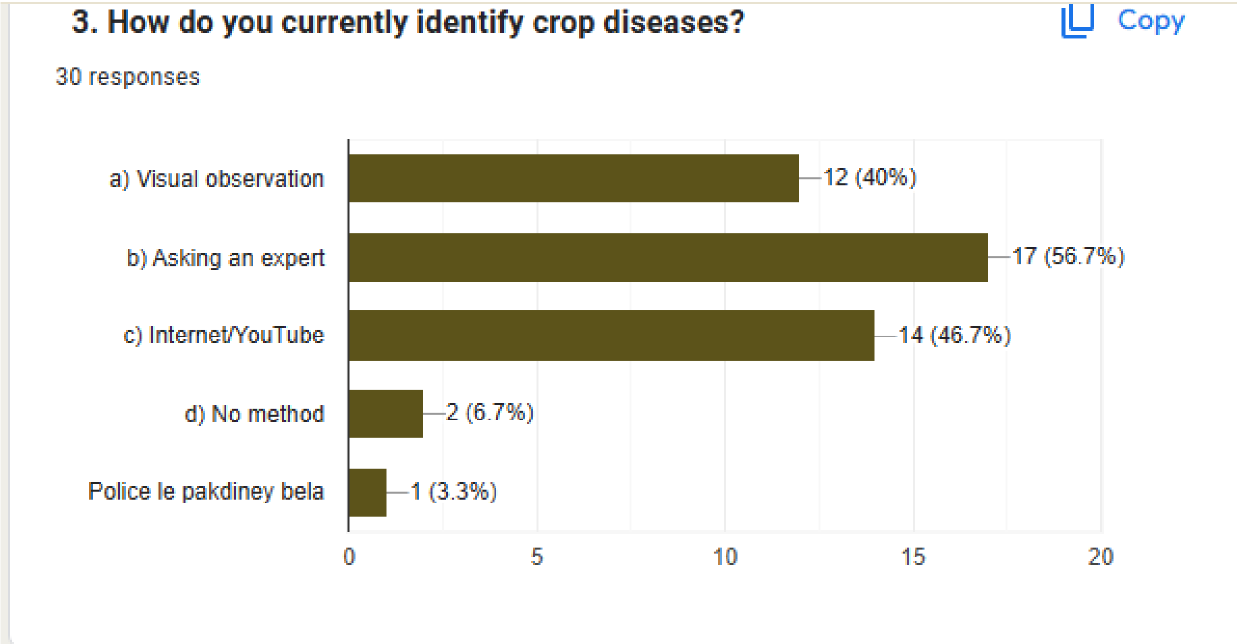
30 responses

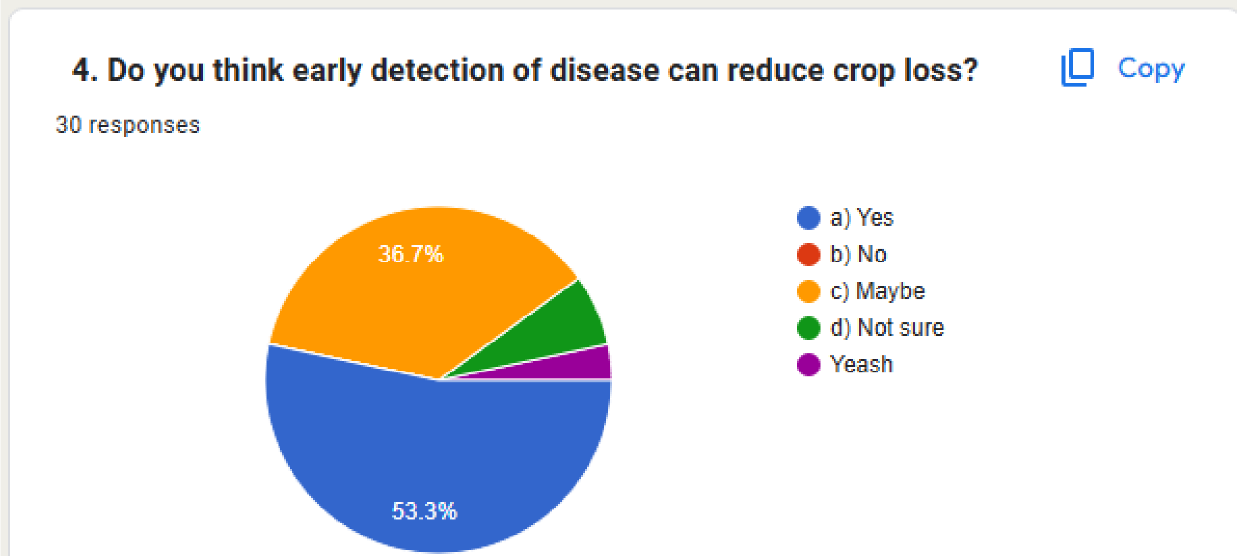


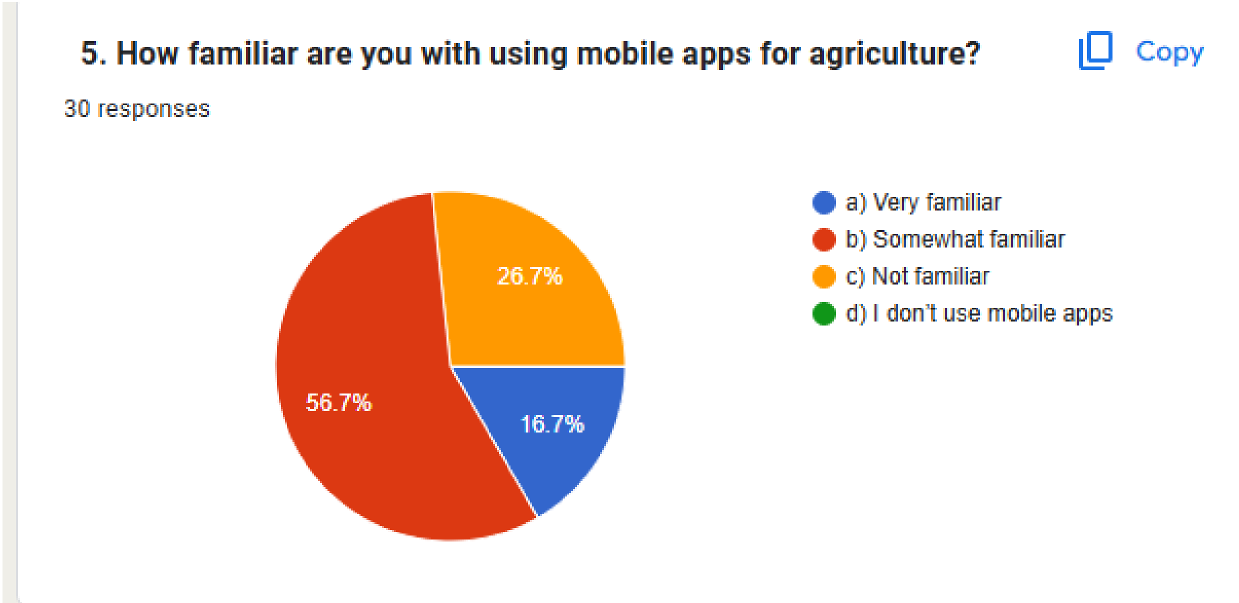

2. How often do you face crop disease problems?

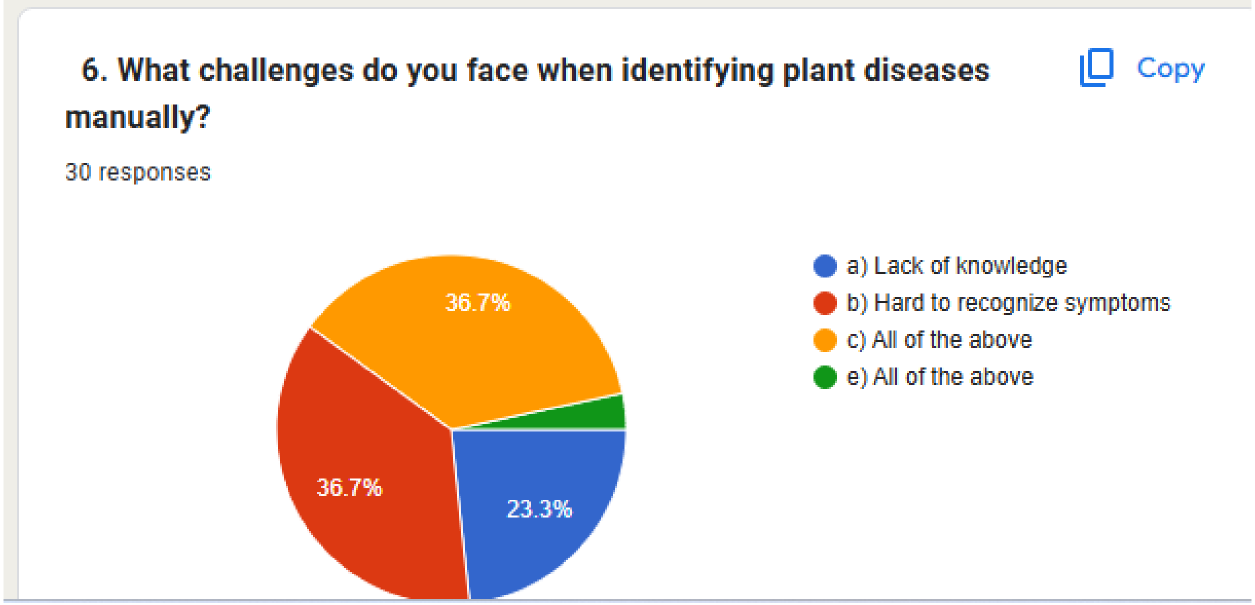
30 responses




3. How do you currently identify crop diseases?
Copy
30 responses
a) Visual observation
b) Asking an expert
c) Internet/YouTube
d) No method
Police le pakdiney bela
12 (40%)
17 (56.7%)
14 (46.7%)
2 (6.7%)
1 (3.3%)
0
5
10
15
20


4. Do you think early detection of disease can reduce crop loss?
Copy
30 responses
36.7%
53.3%
a) Yes
b) No
c) Maybe
d) Not sure
Yeash

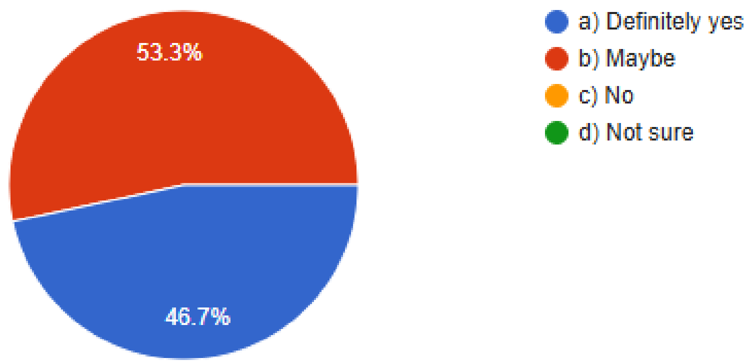

5. How familiar are you with using mobile apps for agriculture?
Copy
30 responses
a) Very familiar
b) Somewhat familiar
c) Not familiar
d) I don't use mobile apps
26.7%
56.7%
16.7%


6. What challenges do you face when identifying plant diseases manually?
Copy
30 responses
a) Lack of knowledge
b) Hard to recognize symptoms
c) All of the above
e) All of the above
36.7%
36.7%
23.3%

7. Would you use a mobile app that detects crop diseases using photos?

Copy

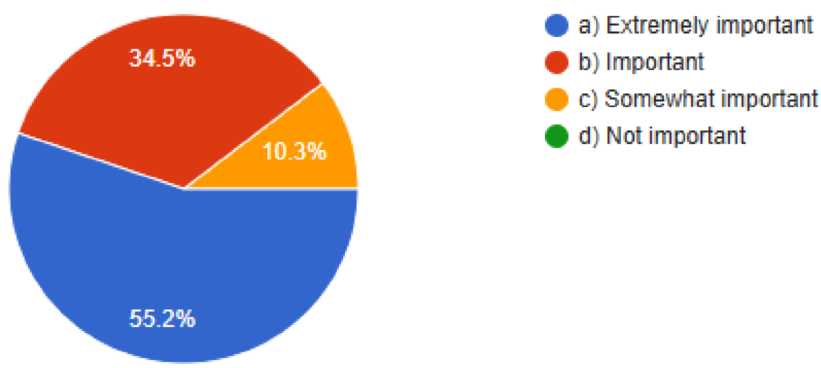
30 responses




8. How important is accuracy in a disease detection app?

Copy

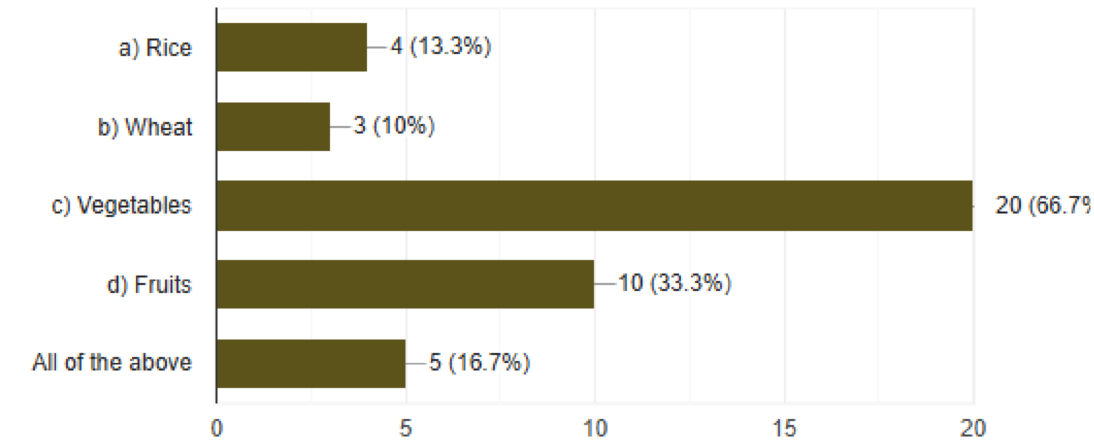
29 responses



**9. What type of crops do you grow the most?**

Copy

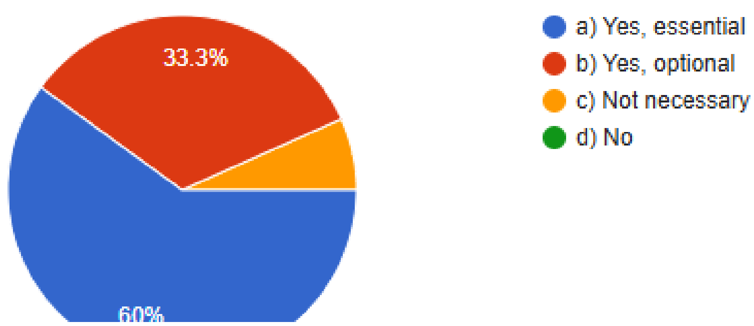
30 responses

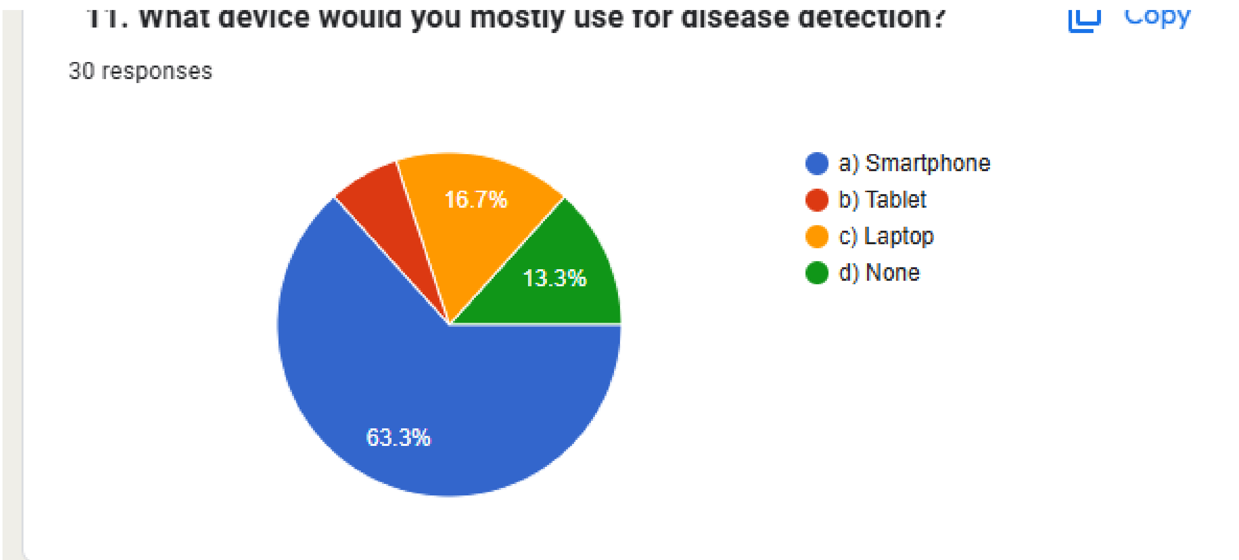


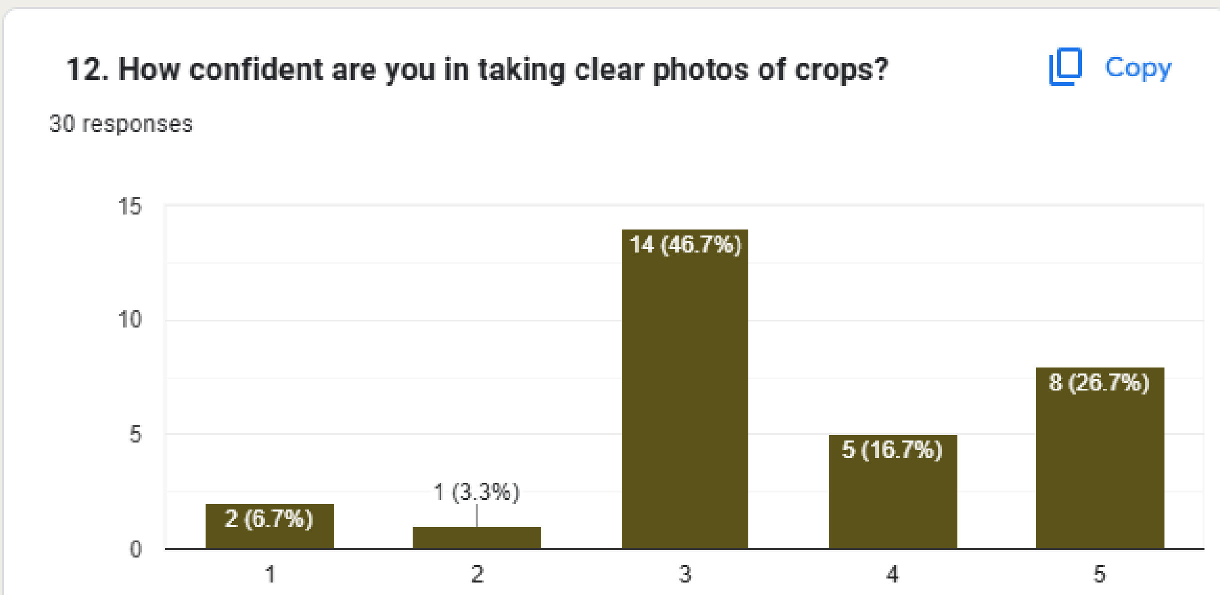

**10. Would you prefer an app that also gives treatment suggestions?**

Copy

30 responses

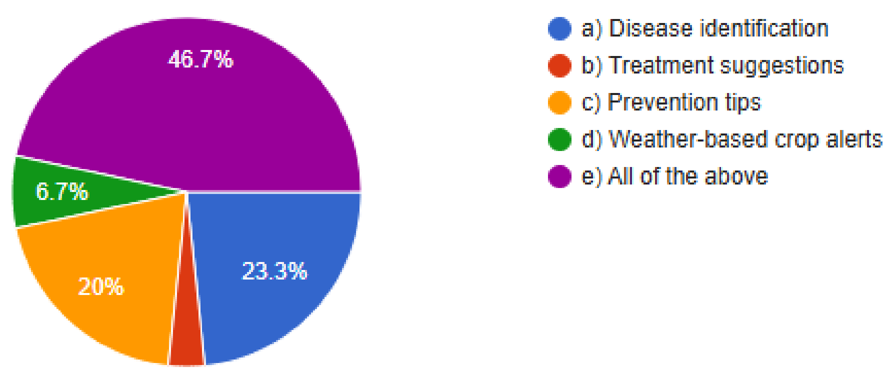



11. What device would you mostly use for disease detection?
Copy
30 responses
16.7%
13.3%
63.3%
a) Smartphone
b) Tablet
c) Laptop
d) None


12. How confident are you in taking clear photos of crops?
Copy
30 responses
15
10
5
0
2 (6.7%)
1 (3.3%)
14 (46.7%)
5 (16.7%)
8 (26.7%)
1
2
3
4
5

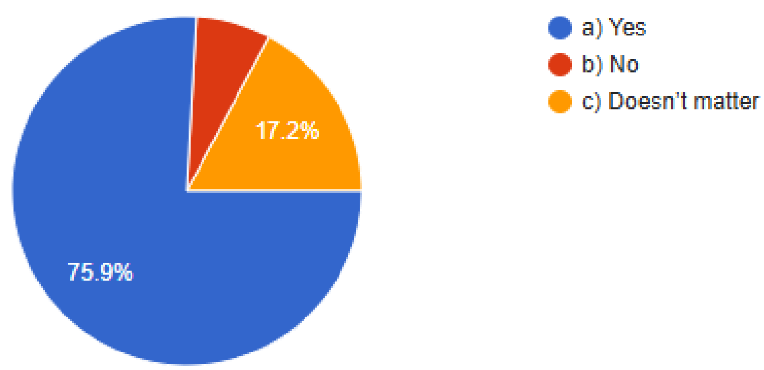
13. Which features would you find most useful in the app?

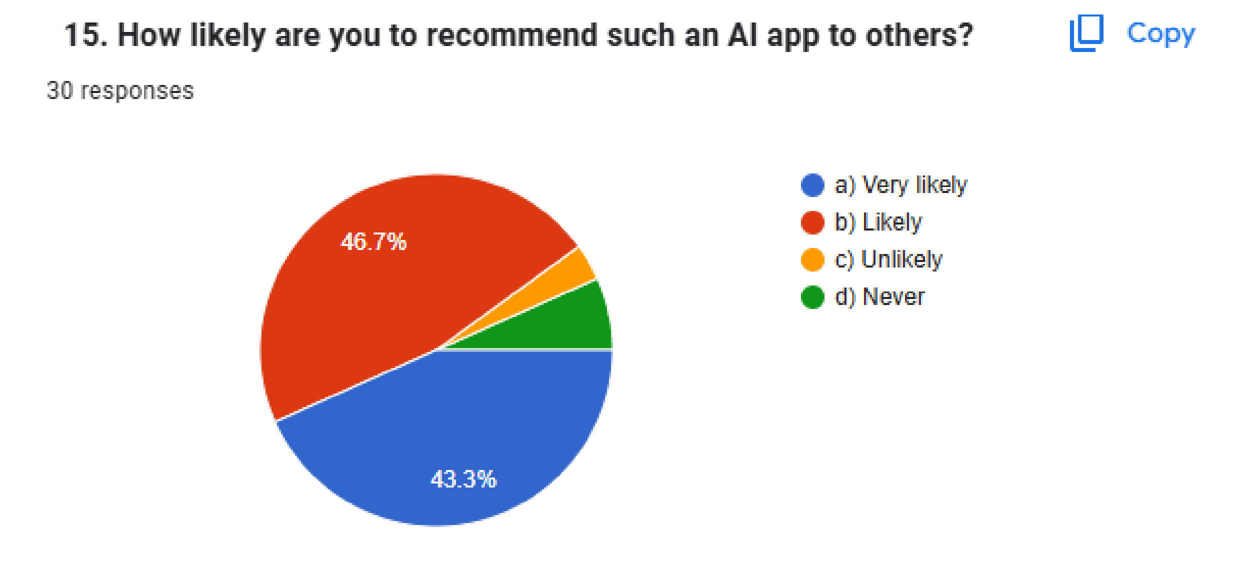
30 responses

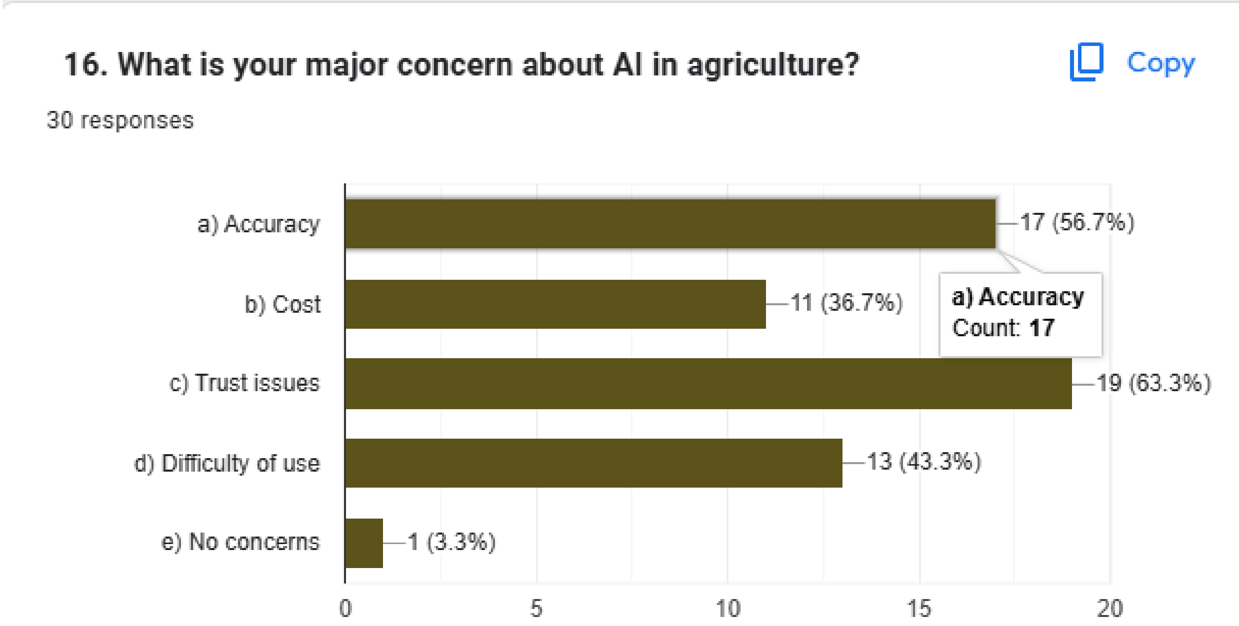



14. Do you prefer the app to work offline?

29 responses



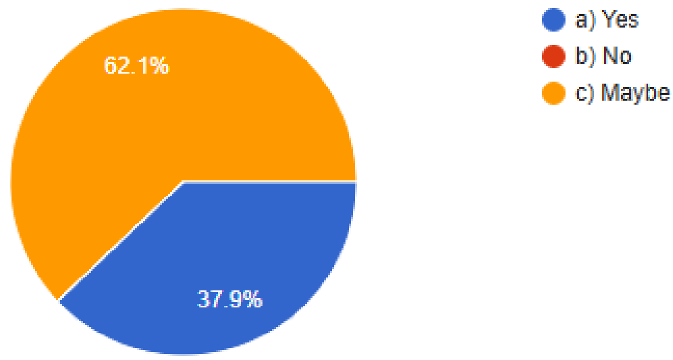

15. How likely are you to recommend such an AI app to others?
Copy
30 responses
46.7%
43.3%
a) Very likely
b) Likely
c) Unlikely
d) Never


16. What is your major concern about AI in agriculture?
Copy
30 responses
a) Accuracy
b) Cost
c) Trust issues
d) Difficulty of use
e) No concerns
17 (56.7%)
11 (36.7%)
19 (63.3%)
13 (43.3%)
1 (3.3%)
a) Accuracy
Count: 17
0
5
10
15
20

17. Do you believe AI can help increase agricultural productivity?

Copy

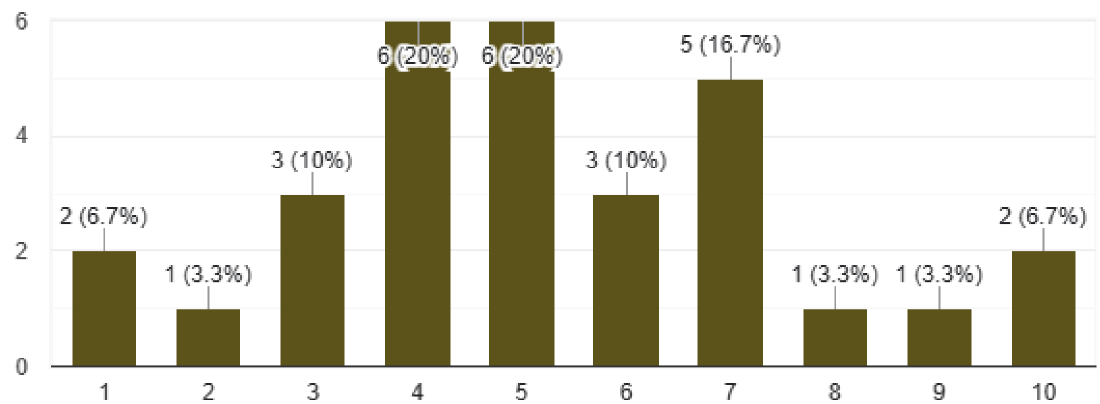
29 responses




18. How satisfied are you with current disease detection methods?

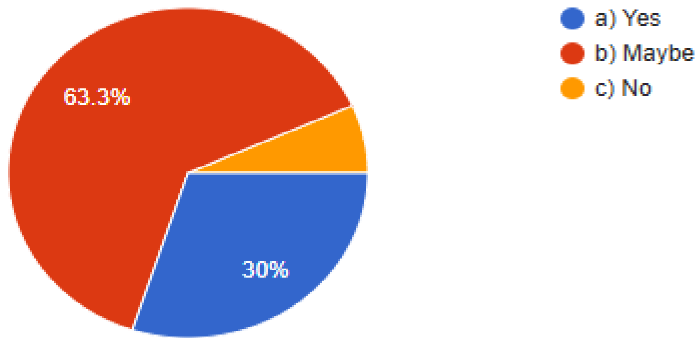
Copy

30 responses



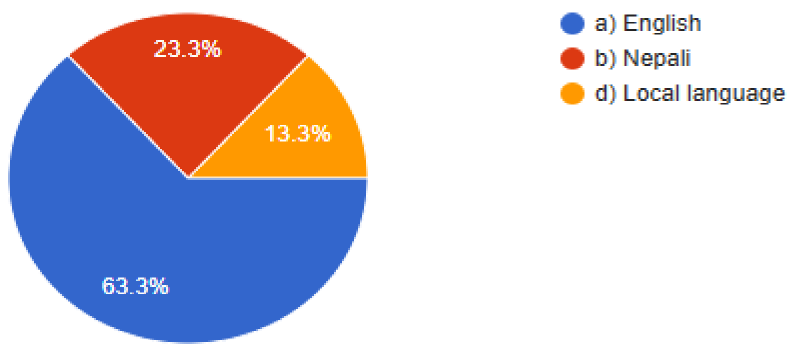
**19. Would you pay a small fee for accurate disease detection?** Copy

30 responses




**20. What language would you prefer the app to use?** Copy

30 responses




**"Please feel free to leave any feedback, comments, or suggestions for improvement.**
**Your thoughts will be very helpful in enhancing our AI Crop Disease Detection System."**

**Some Physical Pre-Survey Form:**

## Pre-Survey Questionnaire (AI Crop Disease Detection System)

This questionnaire is part of my Final Year Project titled AI Based Crop Disease Detection System.

Your feedback and responses will play an important role in guiding the development and improvement of this project. Your time and input are sincerely appreciated.
Thank you.

1. Email *

Achyut.Parajuli@gmail.com

2. YOUR NAME

Achyut Parajuli

3. 1. What is your occupation ?

Mark only one oval.

- a) Farmer
- ✓ b) Student
- c) Agricultural expert
- d) Trader/Business
- Other: ________

4. 2. How often do you face crop disease problems?

Mark only one oval.

- ✓ a) Frequently
- b) Sometimes
- c) Rarely
- d) Never
- Other: ________

5. 3. How do you currently identify crop diseases?

Check all that apply.

- a) Visual observation
- ✓ b) Asking an expert
- c) Internet/YouTube
- d) No method
- Other: ________

6. 4. Do you think early detection of disease can reduce crop loss?

Mark only one oval.

- a) Yes
- b) No
- ✓ c) Maybe
- d) Not sure
- Other: ________

7. 5. How familiar are you with using mobile apps for agriculture?

Mark only one oval.

- a) Very familiar
- ✓ b) Somewhat familiar
- c) Not familiar
- d) I don't use mobile apps

8. 6. What challenges do you face when identifying plant diseases manually?

Mark only one oval.

- a) Lack of knowledge
- b) Hard to recognize symptoms
- ✓ c) All of the above

9. 7. Would you use a mobile app that detects crop diseases using photos?

Mark only one oval.

- a) Definitely yes
- ✓ b) Maybe
- c) No
- d) Not sure

10. 8. How important is accuracy in a disease detection app?

Mark only one oval.

- a) Extremely important
- b) Important
- ✓ c) Somewhat important
- d) Not important

11. 9. What type of crops do you grow the most?

Check all that apply.

- a) Rice
- b) Wheat
- ✓ c) Vegetables
- ✓ d) Fruits
- All of the above

12. 10. Would you prefer an app that also gives treatment suggestions?

Mark only one oval.

- ✓ a) Yes, essential
- b) Yes, optional
- c) Not necessary
- d) No

13. 11. What device would you mostly use for disease detection?

Mark only one oval.

- ✓ a) Smartphone
- b) Tablet
- c) Laptop
- d) None

14. 12. How confident are you in taking clear photos of crops? 4 points

Mark only one oval.

| | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| conf | | | ✓ | | |

15. 13. Which features would you find most useful in the app?

Mark only one oval.

- a) Disease identification
- ✓ b) Treatment suggestions
- c) Prevention tips
- d) Weather-based crop alerts
- e) All of the above

16. 14. Do you prefer the app to work offline?

Mark only one oval.

- a) Yes
- b) No
- ✓ c) Doesn't matter

17. 15. How likely are you to recommend such an AI app to others?

Mark only one oval.

- a) Very likely
- ✓ b) Likely
- c) Unlikely
- d) Never

18. 16. What is your major concern about AI in agriculture?

Check all that apply.

- ✓ a) Accuracy
- ✓ b) Cost
- ✓ c) Trust issues
- d) Difficulty of use
- e) No concerns

19. 17. Do you believe AI can help increase agricultural productivity?

Mark only one oval.

- a) Yes
- b) No
- ✓ c) Maybe

20. 18. How satisfied are you with current disease detection methods?

Mark only one oval.

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Satis | | | | | | ✓ | | | | |

21. 19. Would you pay a small fee for accurate disease detection?

Mark only one oval.

- a) Yes
- ✓ b) Maybe
- c) No

22. 20. What language would you prefer the app to use?

Mark only one oval.

- a) English
- ✓ b) Nepali
- c) Local language
- Other: ________

23. "Please feel free to leave any feedback, comments, or suggestions for improvement.
Your thoughts will be very helpful in enhancing our AI Crop Disease Detection System."

It should be efficient to use.

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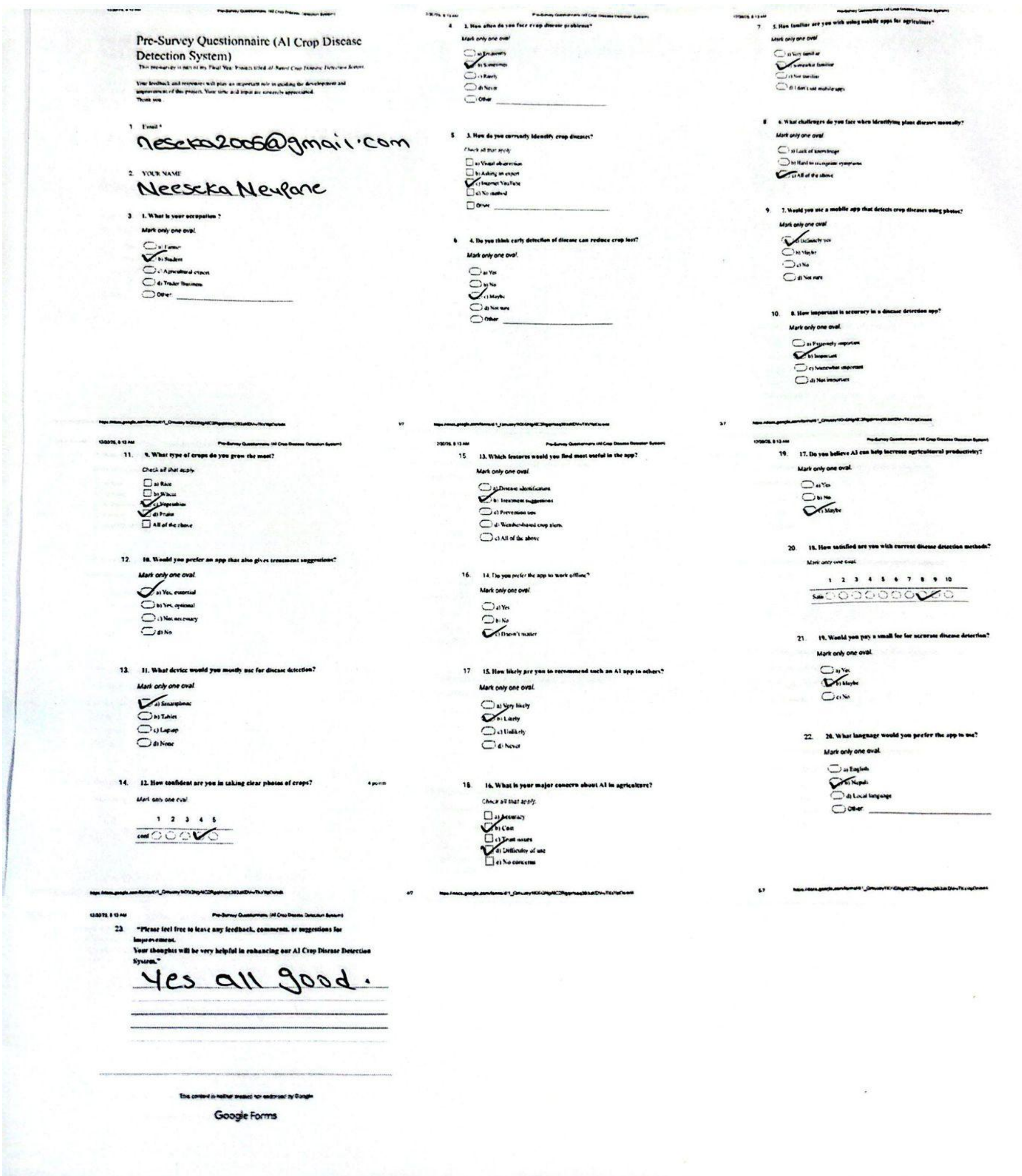
Google Forms

*Figure 12: Physical Pre-survey form*



Pre-Survey Questionnaire (AI Crop Disease Detection System)

1. Email *
neseka2006@gmail.com

2. YOUR NAME
Neeseka Neupane

3. 1. What is your occupation ?
Mark only one oval.
a) Farmer
b) Student
c) Agricultural expert
d) Trader Business
Other:

4. 2. How often do you face crop disease problems?
Mark only one oval.
a) Frequently
b) Sometimes
c) Rarely
d) Never
Other:

5. 3. How do you currently identify crop diseases?
Check all that apply.
a) Visual observation
b) Asking an expert
c) Internet YouTube
d) No method
Other:

6. 4. Do you think early detection of disease can reduce crop loss?
Mark only one oval.
a) Yes
b) No
c) Maybe
d) Not sure
Other:

7. 5. How familiar are you with using mobile apps for agriculture?
Mark only one oval.
a) Very familiar
b) Somewhat familiar
c) Not familiar
d) I don't use mobile apps

8. 6. What challenges do you face when identifying plant diseases manually?
Mark only one oval.
a) Lack of knowledge
b) Hard to recognize symptoms
c) All of the above

9. 7. Would you use a mobile app that detects crop diseases using photos?
Mark only one oval.
a) Definitely yes
b) Maybe
c) No
d) Not sure

10. 8. How important is accuracy in a disease detection app?
Mark only one oval.
a) Extremely important
b) Important
c) Somewhat important
d) Not important

11. 9. What type of crops do you grow the most?
Check all that apply.
a) Rice
b) Wheat
c) Vegetables
d) Fruits
All of the above

12. 10. Would you prefer an app that also gives treatment suggestions?
Mark only one oval.
a) Yes, essential
b) Yes, optional
c) Not necessary
d) No

13. 11. What device would you mostly use for disease detection?
Mark only one oval.
a) Smartphone
b) Tablet
c) Laptop
d) None

14. 12. How confident are you in taking clear photos of crops?
Mark only one oval.
1 2 3 4 5

15. 13. Which features would you find most useful in the app?
Mark only one oval.
a) Disease identification
b) Treatment suggestions
c) Prevention tips
d) Weather-based crop alerts
e) All of the above

16. 14. Do you prefer the app to work offline?
Mark only one oval.
a) Yes
b) No
c) Doesn't matter

17. 15. How likely are you to recommend such an AI app to others?
Mark only one oval.
a) Very likely
b) Likely
c) Unlikely
d) Never

18. 16. What is your major concern about AI in agriculture?
Check all that apply.
a) Accuracy
b) Cost
c) Trust issues
d) Difficulty of use
e) No concerns

19. 17. Do you believe AI can help increase agricultural productivity?
Mark only one oval.
a) Yes
b) No
c) Maybe

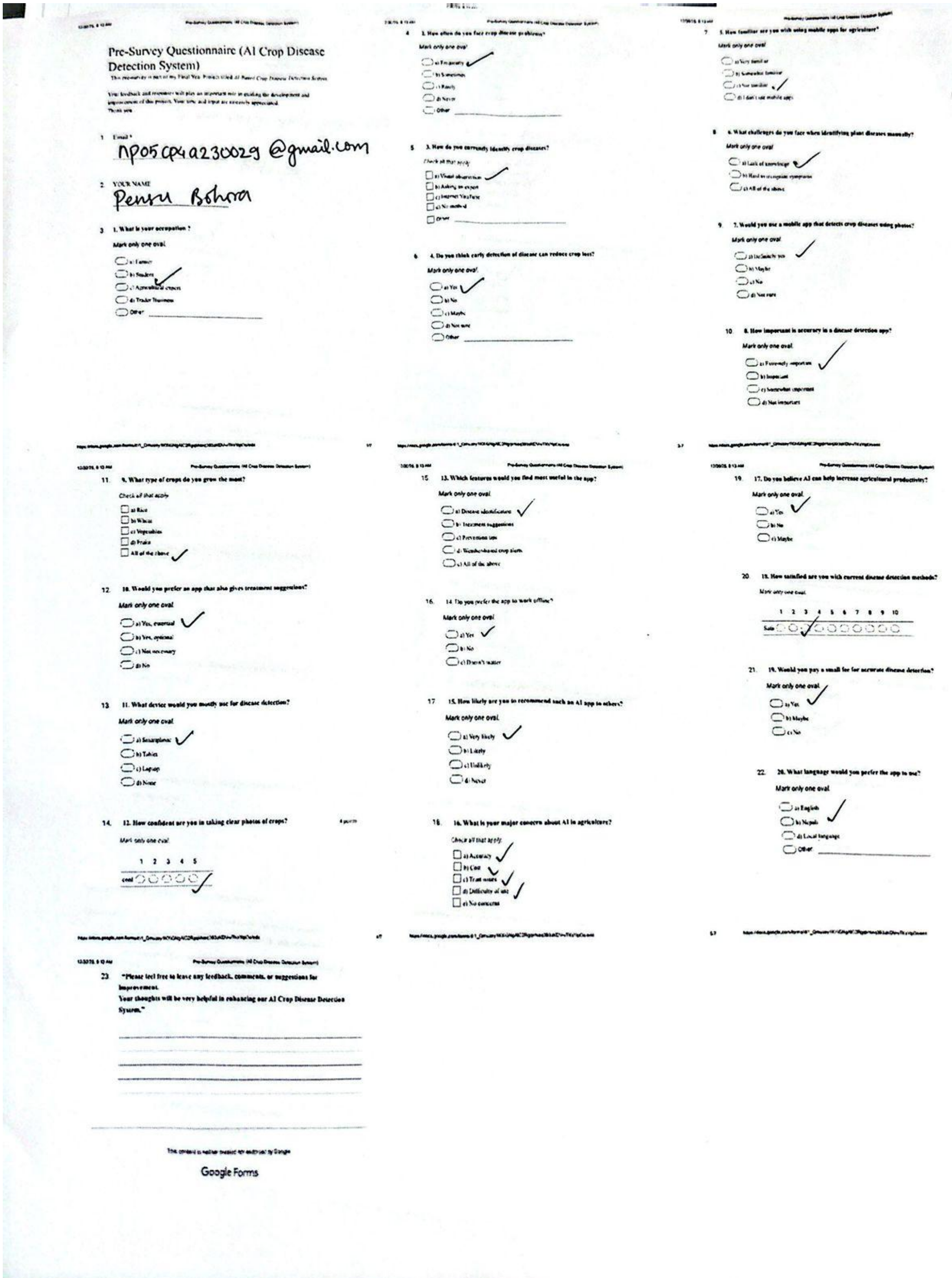
20. 18. How satisfied are you with current disease detection methods?
Mark only one oval.
1 2 3 4 5 6 7 8 9 10

21. 19. Would you pay a small fee for accurate disease detection?
Mark only one oval.
a) Yes
b) Maybe
c) No

22. 20. What language would you prefer the app to use?
Mark only one oval.
a) English
b) Nepali
d) Local language
Other:

23. "Please feel free to leave any feedback, comments, or suggestions for improvement.
Your thoughts will be very helpful in enhancing our AI Crop Disease Detection System."
Yes all good.

This content is neither created nor endorsed by Google.
Google Forms



Pre-Survey Questionnaire (AI Crop Disease Detection System)

1 Email *

np05cp4a230029 @gmail.com

2 YOUR NAME

Pensu Bohora

3 1. What is your occupation ?

Mark only one oval.

- a) Farmer
- b) Student
- c) Agricultural expert
- d) Trader/Business
- Other

4 2. How often do you face crop disease problems?

Mark only one oval.

- a) Frequently
- b) Sometimes
- c) Rarely
- d) Never
- Other

5 3. How do you currently identify crop diseases?

Check all that apply.

- a) Visual observation
- b) Asking an expert
- c) Internet/YouTube
- d) No method
- Other

6 4. Do you think early detection of disease can reduce crop loss?

Mark only one oval.

- a) Yes
- b) No
- c) Maybe
- d) Not sure
- Other

7 5. How familiar are you with using mobile apps for agriculture?

Mark only one oval.

- a) Very familiar
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11 9. What type of crops do you grow the most?

Check all that apply.

- a) Rice
- b) Wheat
- c) Vegetables
- d) Fruits
- All of the above

12 10. Would you prefer an app that also gives treatment suggestions?

Mark only one oval.

- a) Yes, essential
- b) Yes, optional
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Mark only one oval.

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- e) No concerns

19 17. Do you believe AI can help increase agricultural productivity?

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Mark only one oval.

1 2 3 4 5 6 7 8 9 10

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- c) No

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Mark only one oval.

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- b) Nepali
- d) Local language
- Other

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Koiralavinay@gmail.com

2 YOUR NAME

Binay Koirala

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Mark only one oval.

- a) Farmer
- b) Student
- c) Agricultural expert
- d) Trader/Business
- ✓ Other: Lecturer

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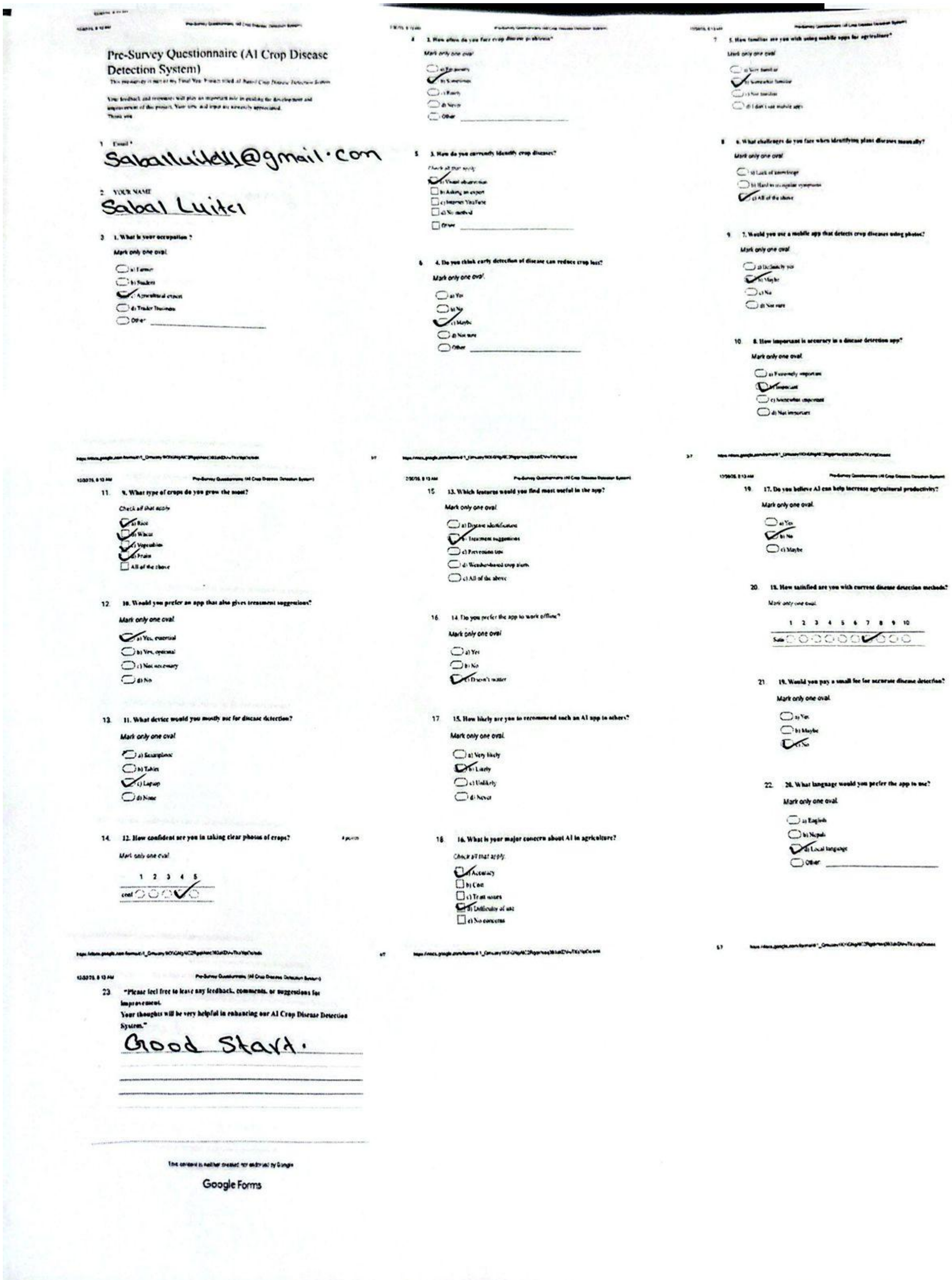

Pre-Survey Questionnaire (AI Crop Disease Detection System)

This questionnaire is part of my Final Year Project titled AI Based Crop Disease Detection System.

Your feedback and responses will play an important role in guiding the development and improvement of this project. Your time and input are sincerely appreciated. Thank you

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Saballuitel11@gmail.com

2 YOUR NAME

Sabal Luitel

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Good Start.

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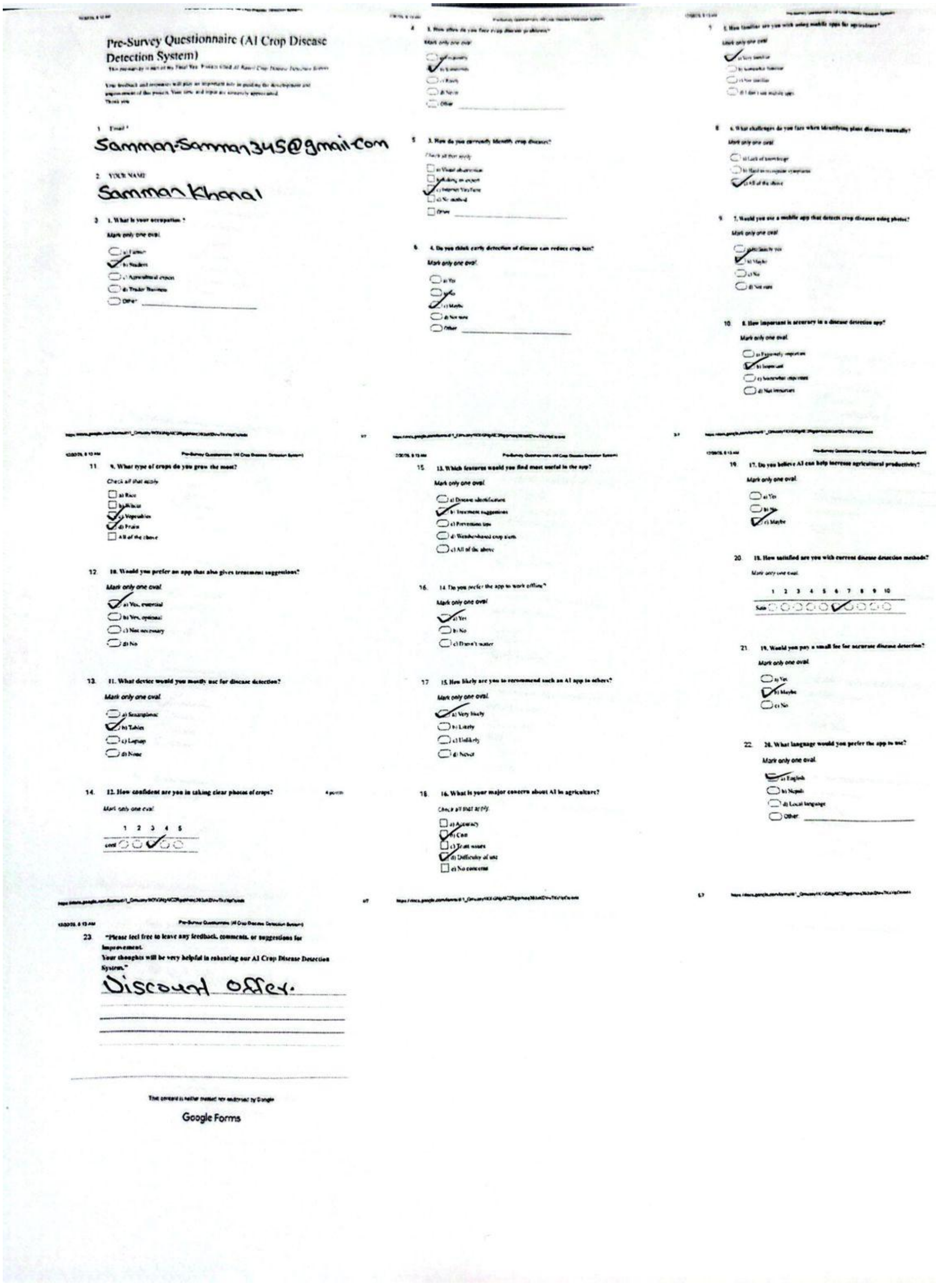

Pre-Survey Questionnaire (AI Crop Disease Detection System)

1. Email *

Samman-Samman345@gmail.com

2. YOUR NAME

Samman Khanal

3. 1. What is your occupation ?

Mark only one oval.

4. 2. How often do you face crop disease problems?

5. 3. How do you currently identify crop diseases?

Check all that apply.

6. 4. Do you think early detection of disease can reduce crop loss?

Mark only one oval.

7. 5. How familiar are you with using mobile apps for agriculture?

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Discount Offer.

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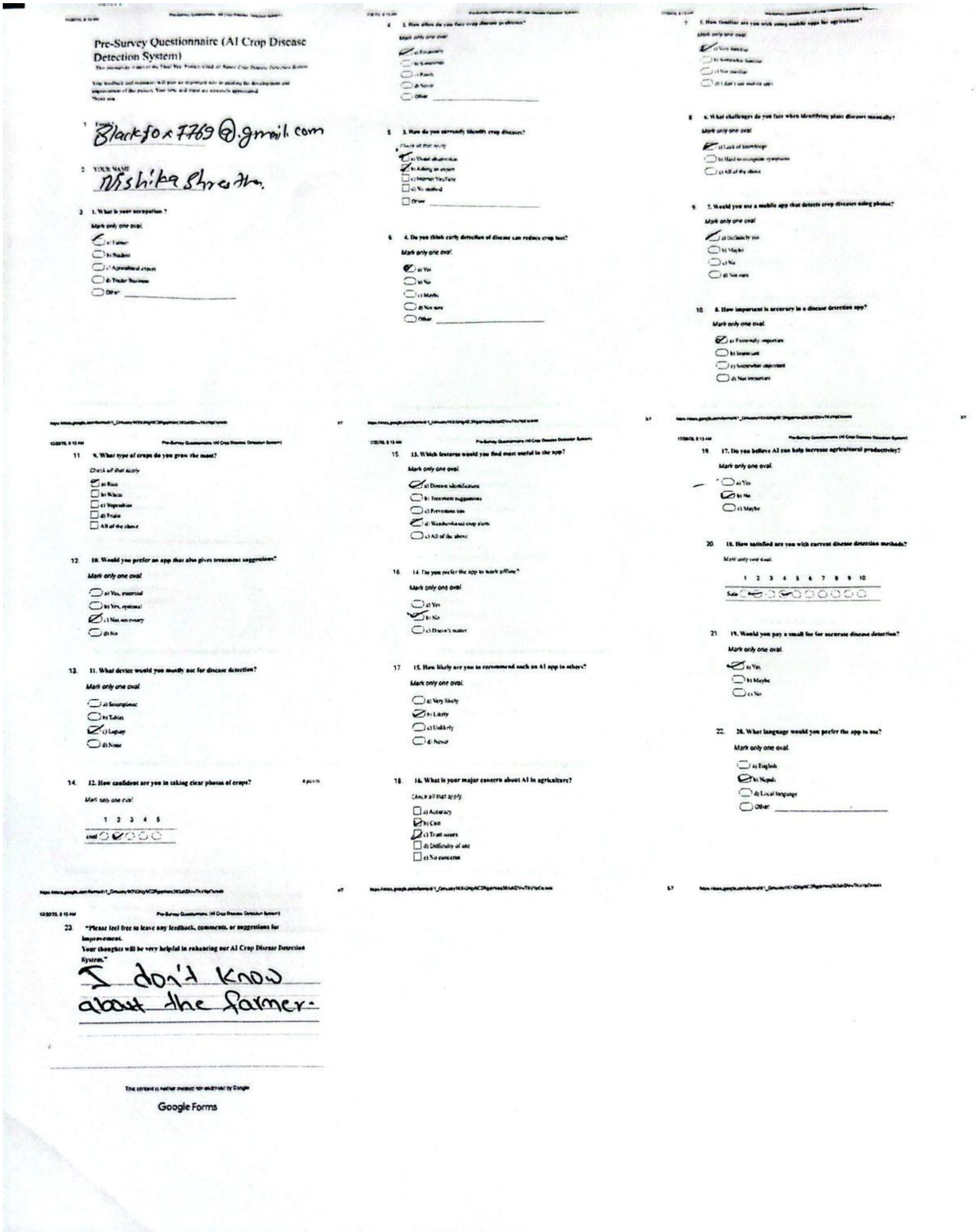
Pre-Survey Questionnaire (AI Crop Disease Detection System)

1 Email: Blackfox7769@gmail.com

2 YOUR NAME: Nishika Shrestha.

3 1. What is your occupation ?

Mark only one oval.

4 3. How often do you face crop disease problems?

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I don't know about the farmer.

Google Forms

# Pre-Survey Questionnaire (AI Crop Disease Detection System)

1 Email *

NikeshBhattarai963@gmail.com

2 YOUR NAME

Nikesh Bhattarai

3 1. What is your occupation ?

Mark only one oval.

- a) Farmer
- b) Student
- c) Agricultural expert
- d) Trader/ Business
- Other

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